

Mulvane EMS Protocols



The members of Mulvane EMS have given a commitment of themselves to become the best professional men and women possible in the quest for efficient emergency medical care for the citizens and visitors of Mulvane and the surrounding area.

Effective May 1, 2022

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Introduction:

These protocols have been designed to maximize the interventions appropriate for each level of care, decrease potential risks to the patient, and improve patient outcomes.

The color-coded format will serve to allow each EMS professional to easily follow the potential interventions for their level of care.

EMR

- EMR, EMT, AEMT, and Paramedic standing orders.

EMT

- EMT, AEMT, and Paramedic standing orders.

AEMT

- AEMT and Paramedic standing orders.

PARAMEDIC

- Paramedic standing orders.

PHYSICIAN OPTIONS

- Any order within the level of care for the provider.

Key Points/Considerations

- Additional points specific to patients that fall within the protocol.

BLS should be done before ALS, and advanced providers are responsible for all appropriate basic interventions. At all provider levels, the standing orders are highlighted.

EMRs and EMTs are only authorized to administer medications listed in the KBEMS approved medication list of April 5, 2019 if they have received authorized training from Mulvane EMS and have successfully completed credentialing process.

It is the provider's responsibility and duty to inform the physician of their level of training, and to decline any order that they feel is inappropriate for that level of training.

Hopefully, these protocols will make your job easier and provide greater assistance with patient care.

PRACTICES

Patient Confidentiality:

Patient is defined as an individual who requests or has the potential of needing medical evaluation and/or treatment.

104-191 Health Insurance Portability and Accountability Act of 1996 regulates the protection of information obtained when responding, evaluating, treating, and transporting patients. It is imperative that we maintain the confidence of patient information and ensure its security. Mulvane EMS prohibits the release of any patient information to anyone outside the organization unless required for purposes of treatment, payment, or health care operations.

Every effort should be made to ensure patient information is not left open in public view or unattended.

Legal Consent:

- **Informed Consent:** Originates from the legal and ethical right the patient has to direct what happens to his/her body. A conscious patient must be told in a manner he/she can understand, the nature and extent of the procedure to be performed and the possible risks involved. If consent is refused by a conscious and mentally competent adult for any treatment procedure following explanation, that person cannot be treated.
- **Implied Consent:** Happens when you are unable to communicate with the patient. Most commonly, this is because the patient is unconscious. When a patient cannot express his or her wishes, there is an assumption that he/she would ask to be treated at the scene and transported to a medical facility where emergency life-saving treatment may be given.
- **Minor Consent:** The age of majority in Kansas is 18 years of age. To administer care to a minor requires parental or guardian consent (either in person or by telephone). If a responsible person is not available, emergency treatment to sustain life and alleviate pain and suffering may be undertaken without consent.

Refusal of Treatment:

It is the policy of Mulvane EMS to give the competent adult patient the right to refuse examination, treatment and/or transport for any illness or injury.

The following guidelines must be followed to allow a patient to be released:

- The patient **MUST** be competent and rational to make the decision to refuse service.
- The patient **MUST** be 18 years of age or have written documentation of emancipation.
- The patient must be aware of complications that could arise from the decision to not be transported.

If the patient is able to answer basic questions (ie. age/name/etc.) but still seems confused or altered or if there is continued concern regarding the patient's decision-making capability, administer the Quick Confusion Scale to aid in determination of transport necessity.

The Quick Confusion Scale				
Item	Scoring System			
	# Correct	X weight	=	Total
What year is it now?	0 or 1 (score 1 if correct and 0 if incorrect)	2	=	
What month is it?	0 or 1	2	=	
Present memory phrase: "Repeat this phrase after me and remember it: <u>John Brown 42 Market Street, New York.</u> "				
About what time is it?	0 or 1	2	=	
Count backward from 20 to 1.	0, 1, or 2	1	=	
Say the months in reverse.	0, 1, or 2	1	=	
Repeat the memory phrase. (each underlined portion correct is worth 1 point)	0, 1, 2, 3, 4 or 5	1	=	
Final score is the sum of the totals:			=	

Explanation of Scoring for Quick Confusion Scale	
The highest number in category indicates correct response; decreased scoring indicates increased number of errors	
What year is it now?	Score 1 if answered correctly, 0 if incorrect.
What month is it?	Score 1 if answered correctly, 0 if incorrect.
About what time is it?	Answer considered correct if within one hour: score 1 if correct, 0 if incorrect
Count backward from 20 to 1.	Score 2 if correctly performed; score 1 if one error, score 0 if two or more errors
Say the months in reverse.	Score 2 if correctly performed; score 1 if one error, score 0 if two or more errors
Repeat the memory phrase: <u>John Brown 42 Market Street, New York.</u> "	Each underlined portion correctly recalled is worth 1 point in scoring; score 5 if correctly performed; each error drops score by one.
Final Score is sum of the weighted totals; items one, two, and three are multiplied by 2 and summed with the other item scores to yield the final score.	

Max score = 15. Score ≤ 11 likely cognitive impairment; score ≤ 7 = substantial impairment.

- ❖ **Note:** If the patient is not competent to make the decision and refuses treatment and transport, the next of kin should be notified to persuade the patient. If necessary contact the Police Department to place the patient under protective custody to provide legal treatment and transport.
- ❖ **Note:** If the patient is combative the safety of the crew is paramount and restraints should be used, refer to restraint procedure for guidelines.

Procedure when a Physician is on the Scene:

For situations in which a physician is on the scene, the following procedure should be followed; such situations are divided into two types for the purpose of this procedure.

Routine, Normal Situations

- Conduct yourself in a professional demeanor and respectful attitude at all times.
- Inquire if the physician is the patient's regular physician. If so, consult the physician as to the patient's history, prior treatment, etc.
- Advise the physician that we have medical standing protocols established by our Medical Director and request that we are allowed to continue under these standing orders.
- Follow any reasonable request by the physician.
- Verbal orders received from in-town physicians and/or practitioners may be followed without them riding in to the hospital with the proper documentation.

Problem or Crisis Situation

- Follow the steps outlined above.
- If the physician prefers to assume responsibility for the patient, the following requirements must be satisfied.
 - The Physician must provide proof of Kansas licensure.
 - The Physician must accompany the patient to the medical facility in our unit.
 - The physician must initiate any procedure that is contrary to our protocols.
- If, in the opinion of the charge technician, the patient's well-being is endangered or if the physician does not agree to our procedure in its entirety, the receiving Emergency Room Physician shall be contacted.

Do Not Resuscitate (DNR) Directives and Orders:

The following guidelines will help when a DNR situation is encountered.

- If the care provider believes a DNR order exists, attempt to locate the order while continuing with appropriate care.
 - If a valid DNR, Directive, or Order is produced at any point during a resuscitative effort, all resuscitative efforts should be stopped.
 - A Do Not Resuscitate (DNR) Directive or Order may be honored if ANY of the following conditions are met:
 - A valid pre-hospital directive or authorization by an attorney in fact named in a valid Durable Power of Attorney for Health Care Decisions.
 - A physician's order to "Do Not Resuscitate".
 - A DNR Medic Alert identifier (bracelet or necklace).

Terminally ill patients who have a Living Will or TPOPP/POLST (Transportable Physician Orders for Patient Preferences/Physician Orders for Life-Sustaining Treatment) may be treated per the provisions of these documents instead of Mulvane EMS Standing Orders or Protocols. In the absence of one of these valid documents, terminally ill patients may be transported without oxygen, IV fluids or medications at the request of a lucid patient or an attorney in fact named in a valid Durable Power of Attorney for Health Care Decisions.

Infection Control and Exposures:

It is everyone's duty to protect themselves, their crew, and the patients from infection and exposures. By simple measures all personnel and patients can be exposed to hazards. The following are precautions to help eliminate those exposures.

- At the beginning of every shift the crew will decontaminate the truck. This is to include the entire back and front passenger compartments and equipment.
- Immediately after every call:
 - Linens will be replaced.
 - Equipment wiped down.
 - Counters wiped down.
- Once each month each truck will have the cabinets emptied and thoroughly cleaned. This is to include the back, outside compartments, and the front passenger compartment.
- All sharps containers should be used for needles and sharp objects only. Do **not** fill past the recommended fill height. Any full containers will be properly disposed of and replaced with new ones.
- Do not eat, drink, apply makeup, or smoke as these will increase the risk of an exposure.
- Anytime patient contact is made, the provider should take universal precautions to reduce the risk of exposures. These key tips will help minimize exposures:
 - Wear non-latex gloves when there is a possibility of contact with any bodily fluids.
 - Wear eye protection and gown whenever there is a splash hazard.
 - Washing hands after every call is the easiest way to decrease exposure.
 - During vehicle extrication the crew will wear the provided protective clothing inside the work area established by the fire department.
 - A safety vest will be worn on all scenes on a roadway.
- If an exposure occurs contact one of the following:
 - EMS Lieutenant
 - EMS Captain
 - Public Safety Chief
- If the exposure is in the eyes or mucous membrane the area should be flushed for at least 15 minutes. All other exposures should be washed with water and a non-abrasive antibacterial soap. Lastly, an exposure report will be filled out and will remain on file for no less than 40 years.

Quality Assurance:

The role of Quality Assurance is to evaluate the quality of care provided to patients requesting service from Mulvane EMS. Quality Assurance is comprised of multiple quality control activities. These activities exist within our organization to provide consistent quality care, and to evaluate existing policies or procedures. Moreover, quality control activities exist to improve the quality of care provided and enhance the EMS system's responsiveness to the citizens and visitors of Mulvane and the surrounding area.

Quality Assurance is a structured and functional review of quality control activities and provides feedback to the involved individuals, which can effectuate desirable change. The approval of these Protocols and Standing Orders is contingent on the continued operation of a Quality Assurance Program and periodic (quarterly) reporting of the results of the process must be made to the Medical Director of Mulvane EMS.

Definitions and Components

- **Protocol:** Establishes the standards to be achieved under specific circumstances within the EMS system. Guidelines contained herein provide conditions, restrictions, or requirements when accomplishing medical procedures or treatment regimes. Protocols are useful in teaching and provide a standard by which care provided can be reviewed by the service Medical Director.
- **Standing Order:** Provides legal authority to carry out procedures on behalf of the physician(s) who has signed them. Additionally, they authorize treatment before radio contact is made. Usually these orders provide treatment of time urgent situations, i.e. cardiac arrest.
- **On-Line Medical Control:** Provides direct real-time physician to pre-hospital provider communication. Once contact is made, the pre-hospital provider becomes an agent of the on-line physician and carries out orders given.
- **Off-Line Medical Control:** Provides medical direction prior to or upon review of the actual real time emergency. Included in off-line medical control are: Protocols, Standing Orders and Call Critiques. Compliance by EMS personnel with Protocol and Standing Orders is evaluated by an audit and review process which is supervised by the Medical Director.
- **Medical Director:** A person licensed to practice medicine and surgery in the State of Kansas. The Medical Director is responsible for Educational Standards, Protocol Development, Critique, Planning and Quality Assurance. Active involvement in off-line medical control is essential.

Procedures for Quality Control

- **Off-Line Activities:**
 - **Chart Review:** Charts written by field personnel are audited to evaluate compliance with departmental Protocol and Standing Orders.
- **On-Line Activities:**
 - **Medical Control:** Each time medical orders, beyond Standing Orders are anticipated, a medical control physician is to be contacted. The seven local medical control facilities are:
 - 📍 Via-Christi St. Francis Regional Medical Center
 - 📍 Via-Christi St. Joseph Regional Medical Center
 - 📍 Via-Christ Wellington ER
 - 📍 Wesley Medical Center
 - 📍 Wesley Woodlawn Hospital
 - 📍 Wesley Derby
 - 📍 Rock Regional Medical Center
 - 📍 Robert J Dole Veteran's Hospital

These constitute the majority of receiving facilities for Mulvane emergency patients. The medical control physicians should be contacted during unusual occurrences or situations that require special attention. These orders require the following stipulations to be documented on the chart by the technician:

- 📍 The name of the Physician that gave the order; The verbatim content of the "Verbal Order."; The signature of the Physician giving the order.

Feedback and Evaluation:

Communication will be made to field personnel when a Protocol or Standing Order violation is suspected during the peer review process. Frequently, the communication will provide sufficient information to eliminate or confirm a suspicion. If Protocol and/or Standing Order violations are unclear, situations are referred to the Medical Director for determination. If a violation exists, remedial training and/or disciplinary action may be undertaken.

Triage/Transport

Patient Transport/Destination:

The following guidelines provide general rules for the establishment of hospital destinations for patients transported by Mulvane EMS.

Non-routine factors, such as poor road or weather conditions, diversion status of hospitals, number of patients requiring transport, etc., may alter how these guidelines are followed.

Approved Hospital Destinations

- Approved receiving hospital destinations for Mulvane EMS are:
- Rock Regional Hospital
- Wesley Derby ER
- Wesley Main
- Wesley Woodlawn
- Via Christi St. Joseph
- Via Christi St. Francis
- Via Christi Wellington (*Code Green and Orange only*)

Emergency Transports

- **Non-Critical Patients:** (Code Green) will be transported to either Rock Regional Hospital, Wesley Derby ER and Via Christi Wellington ER. The exception is VA patients, institutional calls or pre-arranged transfers.
- Workman's Compensation situations may allow the employer to determine hospital destination.
- Patients in police custody may be transported to a destination selected by the arresting officer.
- **Code Orange Patients:** Should be transported to the area hospital of the patient's preference unless a legal guardian, spouse, or law enforcement officer requests an alternate facility. Any ER can assess/treat patients with psychiatric/mental health complaints.
- **Emergency Critical Patients:** (Code Red or Blue) will be transported to the appropriate facility of the patient preference or as directed by a spouse or legal guardian, VA excluded. If a patient cannot or will not make such a choice, transport will be made to the closest appropriate Wichita hospital facility; VA excluded.
- **Deceased Patients (Code Black):** Will not be routinely transported by EMS. Only following a "coroner's release" will a patient be moved or transported to the morgue of the coroner's choice. Normally, transport of deceased patients is requested by law enforcement personnel at the scene when the body is in public view.
- **Isolated Carbon Monoxide Poisonings:** Transport to Wesley Medical Center for Hyperbaric Oxygen (HBO) treatment. If the etiology is uncertain, follow the most appropriate transport guideline.

Burn Alert Patients:

- **Burn Alert Patients:** Transport only to Via Christi St. Francis.
- ❖ ***Note:*** *Code Blue Burned OB (with or without trauma) shall be transported to Wesley Medical Center.*

STEMI Alert Patients:

- **STEMI Alerts:** Transport will be to Wesley Medical Center, Via Christi St. Francis, or Kansas Medical Center.
- **Code Blue/Post Code Blue:** Transport only to Via-Christi St. Francis or Wesley Medical Center.

Stroke Alert Patients:

- **Stroke Alert Patients (Code Red):** Transport only to Via Christi St. Francis or Wesley Medical Center.

Trauma Alert Patients:

- **General:**
 - EMS personnel may deviate from these protocols if conditions exist that would significantly affect transport time to the defined trauma center.
 - No more than four (4) trauma alert patients from a single incident will be transported to the same Trauma Center.
 - Obstetrical patients at or beyond 20 weeks gestation meeting any of the field triage criteria for a Code Yellow Rapid Fetal assessment or a Code Red or Code Blue Trauma Alert, will be transported to Wesley Medical Center, which must be notified they will be receiving an obstetrical trauma patient.
- **Trauma Alert Patients:** Transport will be patient preference or to the closest Level I Trauma Center (Via Christi St. Francis or Wesley Medical Center).
- **Pediatric Trauma:** Wesley Medical Center **ONLY**.

Non-Emergency Transport

- Routine transfer patients will be transported to any location the patient, patient's spouse, or legal guardian requests. If the patient's condition deteriorates during transport use the most appropriate emergency transport guide.

Type 1 Assists:

- **BLS Crew:** Sedgwick County EMS will be contacted for ALS assistance for any patient with a triage of Code Yellow, Code Red, or Code Blue. The intercept may take place at the scene or a safe place along the roadway. Procedurally, the paramedic should board our ambulance and treat the patient while they are en route to the receiving facility. It may be necessary to transfer the patient to the Type I unit; however, this should be limited to situations where lack of equipment, communication or vehicular problems exist with the Type II unit.

Guidelines for Patient Triage:

Code Green: (Non-Serious)

- No appearance of life-threatening injury or illness.
- Vital signs within normal limits.
- Closed fractures, minor lacerations, minor illness, isolated hip fracture (with stable vital signs).
- Patient is alert and oriented.
- Patients who are stable and are anticipated to remain stable.

Code Yellow: (Serious)

- Urgent Patients.
- Any Patient given any medication.
- Patient requires paramedic evaluation.
- IV and/or monitoring have been established.
- V or above on the AVPU scale.
- Initially stable but may deteriorate.

<i>Medical</i>	<i>Trauma</i>
CVA with stable vital signs (Outside of Stroke Alert)	Code Yellow Trauma Alerts
Cardiac/Chest Pain with stable vitals	Code Yellow Rapid Fetal Assessment
Respiratory Disease/Distress patient	
Medication intervention required	
Overdoses with stable vital signs and LOC of A or V	
Alterations in vital signs	
OB patients with impending delivery or field delivery without anticipated complications.	
Diabetic with LOC of V (AVPU)	
Other medical emergencies with anticipated alterations of vital signs.	
Postictal Seizures	
Newborn APGAR ≥ 8	
Reactive Airway Disease requiring medication intervention	

Code Red: (Critical)

- Requires ALS intervention
- Immediate Patients
- Unstable Vital Signs

<i>Medical</i>	<i>Trauma</i>
CVA/TIA within criteria (onset <6hr or wake-up stroke)	Code Red Trauma Alerts
LOC of P or U (AVPU)	Code Red Burn Alert
Hemodynamically unstable cardiac patients	
(Acute MI on 12 lead)	
Dysrhythmias or Cardiogenic Shock	
OB with complications	
Newborn with APGAR of < 7	
Overdose with LOC of P or U or unstable vital signs	
Code Red Sepsis Alert	

Code Blue: (Critical)

- Requires ALS
- Cardiac or Respiratory arrest

Code Orange: (Psychiatric)

- Patient demonstrates unusual, bizarre or socially aberrant action, ideas or moods and represents a threat to themselves or others. May require law enforcement.
- No medical/trauma issue/complaint is present (Must address/triage according to medical/trauma issue first)

Code Black: (Dead on Scene)

- Decapitation
- Blunt trauma without vital signs upon arrival (After performing basic airway maneuvers and bilateral needle chest decompression; it is at the discretion of the Paramedic to run the full cardiac arrest protocol.)
- Incineration
- Decomposition
- Rigor Mortis
- Dependent Lividity
- DNR Order
- Penetrating head trauma without vital signs
- Adult Only! Prolonged field resuscitation with an asystole or PEA rhythm 20 minutes or greater (in the absence of hypothermia)

* In rare circumstances ie body in public view the code black may be transported to facility of Sumner County Coroner's choice.

*If Code Black is in Sedgwick county do not move body unless requested by law enforcement or Sedgwick County Medical Examiner's Office.

General Assessment:

The General Assessment procedure will be used for all patients and should be completed for every patient that the provider encounters.

Patient Definition

- **Patient Criteria:** An assessment, history, and offer of transport will be performed and documented.
 - Request for EMS by him/herself.
 - Apparent illness or injury.
 - Has a Chief Complaint.
 - Vital signs are obtained. (Exception: walk-in request for blood pressure check without complaint of illness or injury.)
- **No Patient Criteria:**
 - Accidental medical alarm activation.
 - No person on scene.
 - MVC victim with no apparent injury or complaint.
- **Citizen Assist Criteria:**
 - Request for assistance with medical equipment who has no acute illness, injury, or complaint. (Example: changing an oxygen cylinder.)
 - Lift assistance with no injury or illness causing the fall, no complaint of injury, and no vital signs assessed.

Scene Size-up

- **Scene Safety:** Paramount. All providers will make a scene assessment upon arrival to determine any unsafe scene or patient. If the scene is unsafe, providers will standby until the scene is secured by proper personnel.
- **Body Substance Isolation (BSI):** All personnel will take proper precautionary measures when making contact with patients (i.e. gloves, glasses, gowns, and masks.) Providers should also wash hands after calls, as hand washing is one of the best ways to prevent spreading diseases.
- **Mechanism of injury (MOI) / Nature of illness (NOI):** MOI is the event that caused the injury. On-scene size up should indicate potential injuries. NOI is the patient's reason/chief complaint for calling EMS.
- **Number of patients and triage:** It is crucial that the number of patients is ascertained. If there are more than two patients, additional resources may be needed and/or a need to establish an Incident Command System. Triage is the condition of the patient.
- **Consider additional resources:** Additional resources should be requested as needed.

Initial Assessment

Complete as indicated. Scene time should be kept to a minimum. A Rapid Trauma Assessment and a detailed physical exam should be done on traumas. A focused history should be done on medicals.

- Determine level of consciousness.
- Assess circulation, airway, and breathing.
- Perform a Quick head-to-toe assessment.
- Identify any life-threatening injuries or illness and treat.
- Obtain and record vital signs, including: peripheral pulse (rate, regularity and quality), blood pressure, respirations (character and rate), skin (condition, temperature and color), breath sounds, motor function and pupils. This should be done a minimum of every fifteen (15) minutes or more frequently as indicated by patient condition.
- Obtain a SAMPLE history. **S**igns/symptoms, **A**llergies, **M**edications, **P**rior medical history, **L**ast oral intake, **E**vents leading up to, and **I**nterventions. This can be information from the patient, family and/or bystanders.

Detailed Physical Exam

A Detailed Physical Exam should be completed on all trauma patients. When performing a detailed physical exam, utilize DCAP-BTLS. **D**eformities, **C**ontusions, **A**brasions, **P**unctures/**P**enetrations, **B**urns, **T**enderness, **L**acerations, and **S**welling. A detailed physical exam should examine the following:

- **Head, Neck, and C-Spine:** Skull, face, eyes, mouth, ears, nose, and throat
- **Chest:** Breath sounds (Rate, rhythm, and quality), crepitus, and bruising
- **Abdomen:** Soft, firm, tender, distended, and guarding
- **Pelvis:** Stability, pain, and tenderness
- **Extremities:** Distal circulation, sensation, and motor
- **Back:** Stability and incontinence

Focused History

When warranted a focused history should be performed and included as a part of the SAMPLE history. OPQRSTI will assist with cardiac assessments. OPQRSTI is as follows:

- **Onset:** When did it start?
 - **Provokes:** What makes it better or worse?
 - **Quality:** What does the pain feel like?
 - **Radiates:** Does it go anywhere else on your body?
 - **Severity:** On a scale of 1-10, rate the pain?
 - **Timing:** Is pain constant or does it come and go?
 - **Interventions:** What have they done or taken to help?
- ❖ **Note:** If the patient is unresponsive a rapid assessment should be performed.

Field Treatments

Treat appropriately in order of priority.

Oxygen Administration

Oxygen administration may be applied by all levels of providers. Based on patient need and/or protocol, supplemental oxygen will be administered to maintain a saturation level above 94% with the appropriate delivery device determined by the technician.

Blood Glucose

Blood glucose will be assessed on all patients that have an altered level of consciousness, suspected hyper/hypoglycemia, unknown etiology, ETOH suspicion, overdoses, seizures and at the discretion of the technician. All patients that are being treated for diabetic emergencies need to have blood glucose levels taken every 15 minutes or whenever a treatment is done.

Communications

A patient report will be given to the hospital as early as possible during transport using the following format:

- Identify ambulance service, level of care (ALS/BLS), and yourself (John Doe, EMT).
- Patient's age, sex, vital signs, LOC, Doctor and triage.
- Patient's chief complaint and physical assessment findings.
- Pertinent medical history.
- Mechanism of injury (if applicable), or events preceding condition or problem.
- Treatment given, patients response, and present status.
- Estimated time of arrival (ETA) at hospital.
- Any other pertinent information requested.
- Advise emergency department of changes occurring in patient status during transport. Upon arrival, give a verbal report to the receiving nurse or physician.
- If radio communication with Sedgwick Co. 800MHz system is unavailable contact the receiving facility via cellular phone. All calls to facilities where orders are to be received should be placed through a dispatch center for recorded record of the orders.
- Run reports should be written and locked within two hours of returning to the station.

Inter-Facility Transfers:

Intra-facility transfer requests will be approved on an individual basis dependent upon availability of staff, equipment, and the urgency of the transfer. The patient will be transported to the requested facility.

- Primary Survey: CAB
- Secondary Survey
 - Head to Toe survey
 - Obtain pertinent medical history.
 - Obtain reason for transfer.
 - Obtain paperwork for receiving facility.
- Administer oxygen if indicated.
- Monitor vitals every 10-15 minutes.
- Monitor I.V. Solutions
 - EMT's may monitor the following IV Solutions:
 - D5W
 - D10W
 - D5LR
 - Lactated Ringers (LR)
 - 0.9% NaCL (NS)
 - 0.45% NaCL
 - 0.2% NaCl
- Place patient in a position of comfort.
- Re-evaluate patient condition for changing status.
- Treat patient appropriately for underlying causes.
- Follow treatment orders from sending facility and request to use standing orders for situations outside of current orders.

Alert Criteria:

All alerts must also have a complete radio report given to the receiving hospital during transport.

Burn Alert Criteria:

Burns

All burns will be assessed using the “Rule of Nines” to determine the total burn surface area (TBSA) of the patient and only partial (2nd degree) and full thickness (3rd degree) burns will determine burn alert codes. All burn patients will be transported to Via Christi St. Francis (VCSF) except OB. All obstetrical burn patients will be transported to Wesley Medical Center first before being transferred to Via Christi St. Francis.

Burn Alert Notification

All burn alerts need to be communicated with Dispatch as soon as possible and include the following:

- Triage
- Age
- Gender
- GCS
- Total Burn Surface Area, degree of burn, and location on patient’s body.
- Type of burn (thermal, chemical, electrical, inhalation).
- Any associated trauma.
- Destination.
- ETA.

Code Red Burn Alert Criteria

Any patient with any one of the following meets Code Red Burn alert:

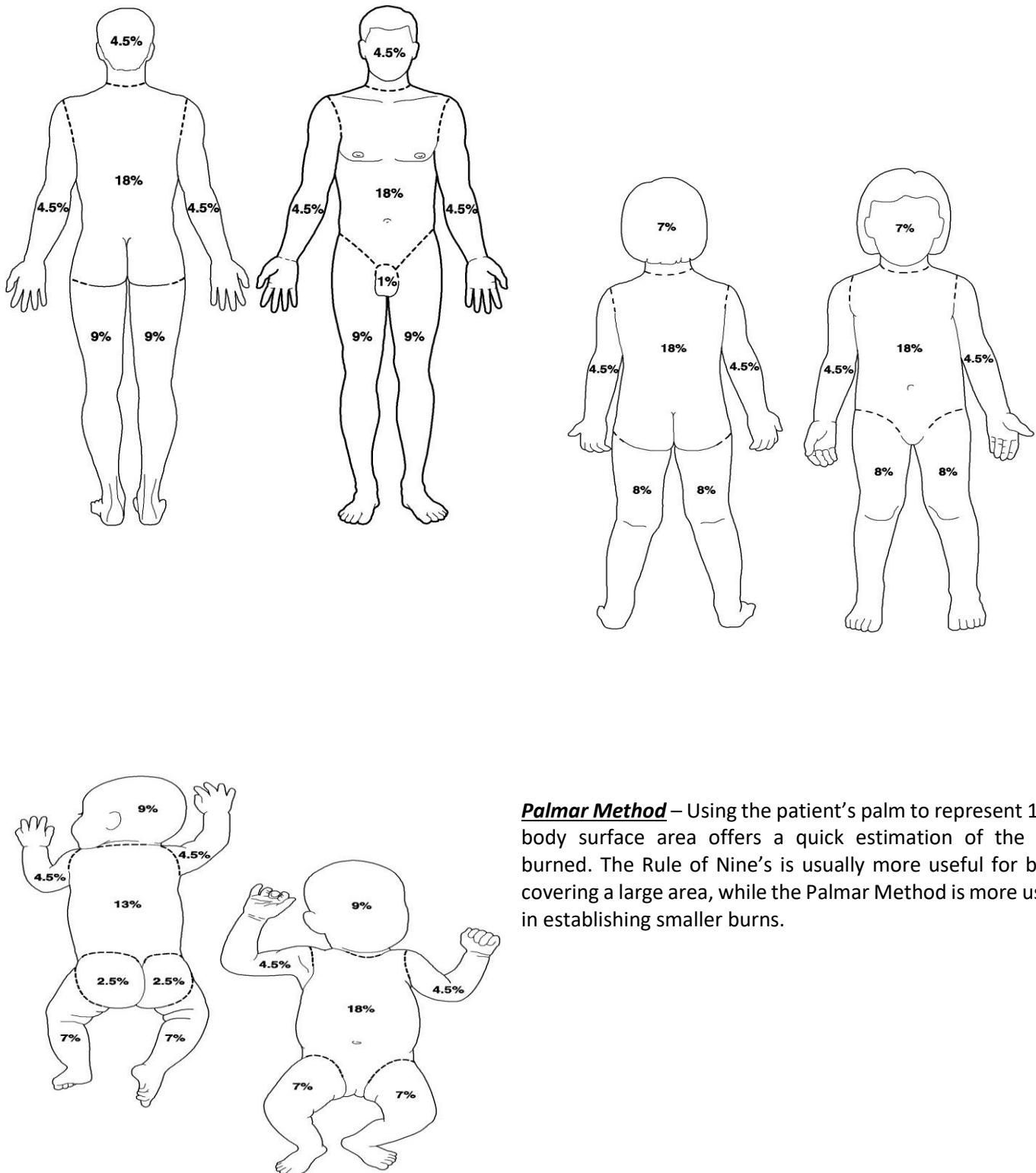
- 10% or greater Total Body Surface Area (TBSA) with partial thickness and/or full thickness burns.
- Airway and/or respiratory difficulties associated with inhalation of smoke (soot in nares/airway), chemicals, or thermal exposures.
- 440 volt or greater electrical burns.
- Lightning strikes.
- Circumferential burns to head, neck, and torso.
- Chemical burns to head, neck, torso, and face.
- Potential for inhalation injuries.
- Chemical burns to the hands, arms, legs, and feet.
- Circumferential burns to the hands, arms, legs, and feet.
- Burns to the peritoneal area, genitalia, and major joints.

Burn Alert Criteria:

Burns

All burns will be assessed using the “Rule of Nines” to determine the total burn surface area (TBSA) of the patient and only partial (2nd degree) and full thickness (3rd degree) burns will determine burn alert codes.

Rule of Nines



Palmar Method – Using the patient’s palm to represent 1% of body surface area offers a quick estimation of the area burned. The Rule of Nine’s is usually more useful for burns covering a large area, while the Palmar Method is more useful in establishing smaller burns.

Rapid Fetal Assessment Notification and Alert Criteria:

All obstetrical patients that are greater than 20 weeks gestation and meet the criteria for a Code Yellow Rapid Fetal Assessment, Code Yellow Trauma Alert, Code Red, or Code Blue will be transported to Wesley Medical Center.

Rapid Fetal Assessment (RFA)

Rapid fetal assessment alerts should be communicated to Dispatch as soon as possible and should include all of the following information:

- Triage
- Age
- GCS
- Mechanism of injury
- Estimated gestational age
- Presence of multiples
- Gravida and Para
- Destination
- ETA

Code Yellow Rapid Fetal Assessment Criteria:

Any OB patient with one of the following:

- OB patient greater than 20 weeks gestation.
- Blunt trauma with complaints of abdominal and /or back pain.
- Motor vehicle accidents where patient was unrestrained or has seatbelt abrasions to the iliac crests.

STEMI Alert Criteria:

STEMI:

ST-Elevation Myocardial Infarctions will be determined using a 12-lead (15-lead/posterior or right-sided 12-lead if necessary). It is defined as ST Elevation present at the J-point in at least two contiguous leads and >5mm. Presence of reciprocal changes helps confirm the diagnosis but is not required for STEMI activation. STEMI patients are to be transported to Via Christi St. Francis, Wesley Medical Center, or Kansas Medical Center.

STEMI Alert Notification:

All STEMI alerts need to be communicated with dispatch as soon as possible and include the following:

- Triage
- Age
- Gender
- GCS
- Patient's cardiologist (or specify if patient does not have a cardiologist)
- Destination
- ETA

Stroke Alert Notification and Criteria:

Stroke alerts should be communicated via dispatch as soon as possible and should include all of the following information:

- Triage
- Age
- Gender
- GCS
- Onset of symptoms/last seen normal
- Findings of Cincinnati Stroke Assessment
- Destination
- ETA

Stroke Scale Assessment:

The Cincinnati Pre-hospital Stroke Scale is a system used to diagnose the presence of a stroke in a patient. It tests three signs for abnormal findings which may indicate that the patient is having a stroke. If any one of the three tests shows abnormal findings, the patient may be having a stroke and should be transported to a hospital as soon as possible.

- **Facial droop:** Have the person smile or show his/her teeth. If one side doesn't move as well as the other and it seems to droop, that could be sign of a stroke.
 - Normal: Both sides of face move equally.
 - Abnormal: One side of face does not move as well as the other (or not at all).
- **Arm drift:** Have the person close his/her eyes and hold his/her arms straight out in front for about 10 seconds. If one arm does not move, or one arm winds up drifting down more than the other, that could be a sign of a stroke.
 - Normal: Both arms move equally or not at all.
 - Abnormal: One arm does not move or one arm drifts down compared with the other side.
- **Speech:** Have the person say, "You can't teach an old dog new tricks," or some other simple, familiar saying. If the person slurs the words, gets some words wrong, or is unable to speak, that could be a sign of stroke.
 - Normal: Patient uses correct words with no slurring.
 - Abnormal: Slurred or inappropriate words or mute.

NOTE: If a family member or the witnessing bystander does not accompany EMS to the hospital, get that person's phone number for the hospital to contact him/her with any further questions about the patient.

Trauma Alert Notification and Criteria:

Trauma Alert Notification

Trauma alerts should be communicated with Dispatch as soon as possible and should include all of the following information:

- Triage
- Age
- Gender
- GCS.
- Mechanism of injury.
- Major body system injured.
- Destination
- ETA

Trauma Alert Criteria

- **Code Red Trauma:**
 - GCS of less than 14.
 - Open or depressed skull fracture.
 - Systolic blood pressure < 90 mmHg.
 - Near respiratory and/or cardiac arrest. (Including hangings and traumatic drowning).
 - Respiratory distress: Adult rates of < 10 or > 29 breaths per minute (<20 in infant age < 1 year).
 - Penetrating injuries to chest, abdomen, head, neck, and extremities proximal to elbow and knee.
 - Total or partial amputation, crushed, mangled, or de-gloved extremity proximal to the wrist or ankle.
 - Two or more open or closed fractures to femur or humerus.
 - Pelvic fracture with unstable vital signs.
 - Paralysis.
 - Flail chest with paradoxical respirations.
 - Shock present.
- **Code Yellow Trauma:**
 - Loss of consciousness greater than 5 minutes.
 - Pelvic fractures with stable vital signs.
 - Total or partial amputations distal to the wrist or ankle.
 - Open or closed fractures with neurovascular compromise.
 - Falls:
 - Adults – greater than 20 feet
 - Pediatric under five (5) foot - use two times their height
 - Pediatric over five (5) foot - use 10 feet.
 - Motor vehicle accidents:
 - Auto vs. pedestrian, motorcycle, or bicyclist thrown, run over, or an impact greater than 20 mph.
 - Motorcycle crash greater than 20 mph.
 - Total or partial ejection from automobile.
 - Death of another occupant in same vehicle as patient.
 - Inside compartment intrusion greater than 12 inches on occupant side and 18 inches on any side.
 - Accidents greater than 35 mph with high index of suspicion for injury.
 - Co-morbidities and special considerations:
 - Age greater than 55 years old.
 - Bleeding disorders or patient on anticoagulants.
 - Renal disease requiring dialysis.

Triage Mass Casualty Incident:

Any situation that potentially overwhelms the local EMS resources should be declared a Multiple/Mass Casualty Incident. The first responding EMS unit should declare a possible major incident while responding to the scene when the dispatch information suggests the likelihood that one exists. As soon as possible upon arrival at the scene, the unit should verify that a major incident does or does not exist.

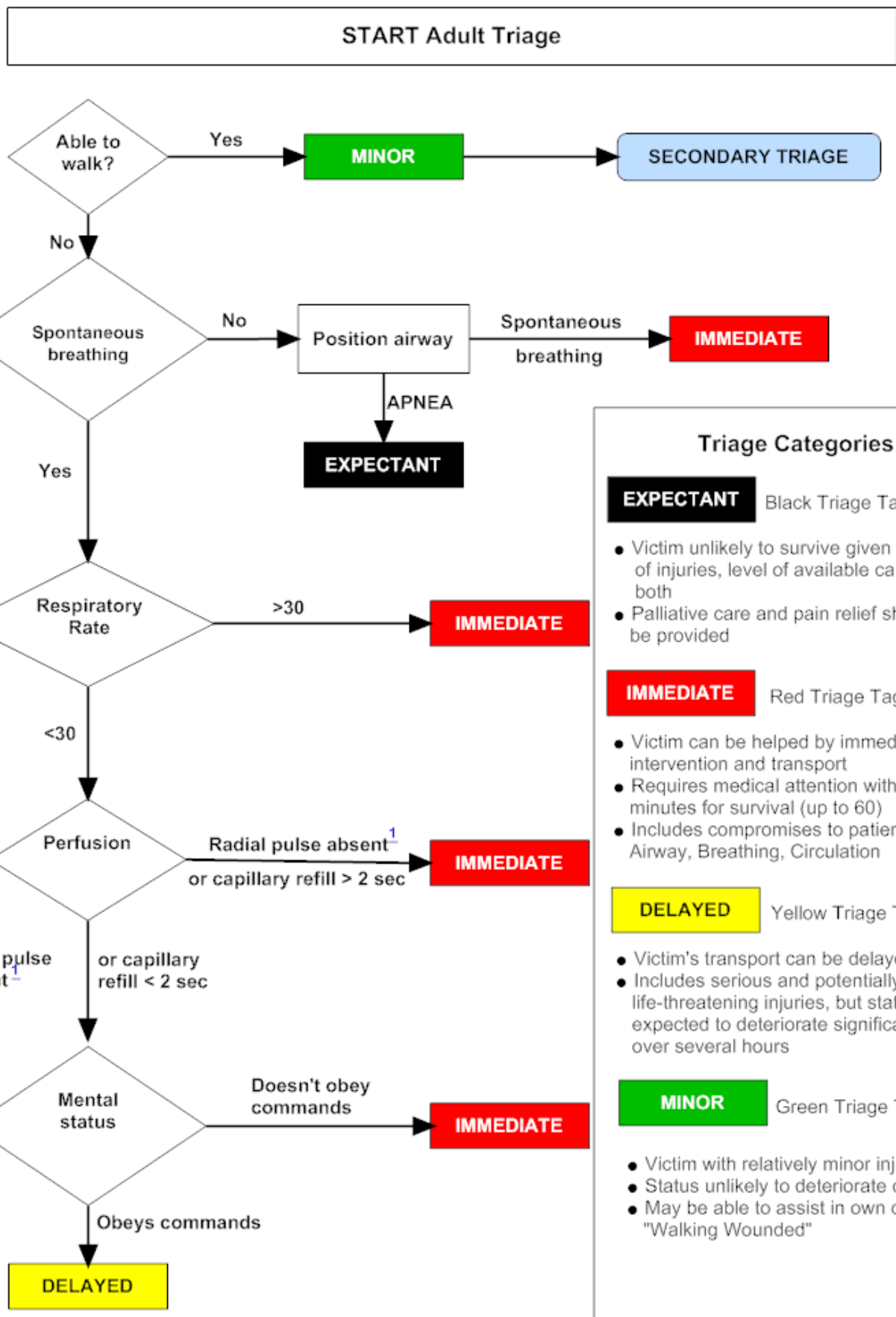
At a Multiple/Mass Casualty Incident involving mutual aid from several communities, a common EMS frequency should be established for ambulance communications.

Patients will be triaged according to START and JumpSTART triage criteria. During primary triage, providers should spend no more than 30 seconds with each patient. Patients may be treated on scene only after all patients have been triaged and staged. ALS providers should consider providing care at the BLS level in order to give care to as many patients as possible.

START Triage:

- Should be used for all adult patients.
- Walking wounded should be encouraged to congregate in a designated location under their own strength and triaged in the **GREEN** (minor) category.
- Patients with no respiratory effort should be triaged in the **BLACK** (deceased) category following an attempt to open the airway.
- Patients with difficulty in respirations, perfusion or mental status as specified below should be triaged in the **RED** (immediate) category.
 - Respirations >30/minute
 - Perfusion – No radial pulse or capillary refill times >2 seconds
 - Mental Status – Unable to follow simple commands

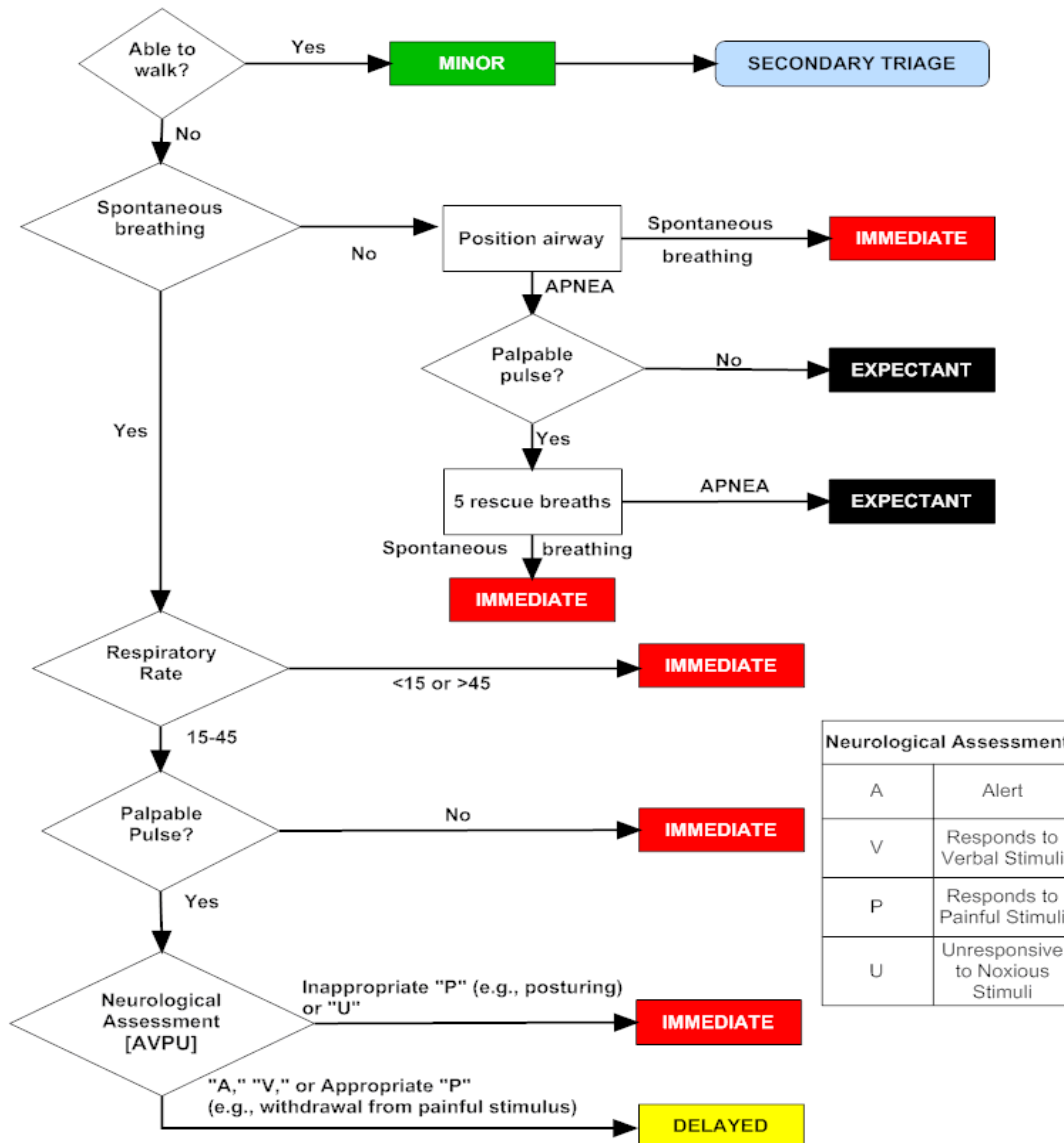
All patients who cannot walk, have respiratory effort, and do not meet criteria for the **RED** category should be triaged to the **YELLOW** (delayed) category.



JumpSTART Triage:

- Should be used for pediatric patients under 14 years of age.
- Walking wounded should be encouraged to congregate in a designated location under their own power and triaged in the **GREEN** (minor) category.
- Patients with a peripheral pulse but without respiratory effort should receive 5 ventilations then categorized as **RED** (immediate) if respiratory effort resumes or **BLACK** (deceased) if apnea continues.
- Position the airway for patients with no respiratory effort or peripheral pulse. No palpable pulse found the patient should be triaged in the **BLACK** (deceased) category.
- Patients with difficulty in respirations, perfusion or mental status as specified below should be triaged in the **RED** (immediate) category.
 - Respirations >45/min or <15/min
 - Perfusion – No peripheral pulse or capillary refill times >2 seconds
 - Mental Status – unresponsive or responsive to painful stimulus
- All patients who cannot walk, have respiratory effort, and do not meet criteria for the **RED** category should be triaged to the (delayed) category.

JumpSTART Pediatric Multiple Casualty Incident Triage



Use JumpSTART if the Patient appears to be a child.

Use an adult system, such as START, if the patient appears to be a young adult.

Triage Categories

EXPECTANT

Black Triage Tag Color

- Victim unlikely to survive given severity of injuries, level of available care, or both
- Palliative care and pain relief should be provided

DELAYED

Yellow Triage Tag Color

- Victim's transport can be delayed
- Includes serious and potentially life-threatening injuries, but status not expected to deteriorate significantly over several hours

IMMEDIATE

Red Triage Tag Color

- Victim can be helped by immediate intervention and transport
- Requires medical attention within minutes for survival (up to 60)
- Includes compromises to patient's Airway, Breathing, Circulation

MINOR

Green Triage Tag Color

- Victim with relatively minor injuries
- Status unlikely to deteriorate over days
- May be able to assist in own care: "Walking Wounded"

Glasgow Coma Scale:

The Glasgow Coma Scale provides a score in the range 3-15; patients with scores of 3-8 are usually said to be in a coma. The total score is the sum of the scores in three categories. For adults the scores are as follows:

Eye Opening Response	Spontaneous - open with blinking at baseline.	4 points
	Opens to verbal command, speech, or shout.	3 points
	Opens to pain, not applied to face.	2 points
	None	1 point
Verbal Response	Oriented	5 points
	Confused conversation, but able to answer questions.	4 points
	Inappropriate responses, words discernible.	3 points
	Incomprehensible speech	2 points
	None	1 point
Motor Response	Obeys commands for movement.	6 points
	Purposeful movement to painful stimulus.	5 points
	Withdraws from pain.	4 points
	Abnormal (spastic) flexion, decorticate posture.	3 points
	Extensor (rigid) response, decerebrate posture.	2 points
	None	1 point

For children under 5, the verbal response criteria are adjusted as follows:

SCORE	2 to 5 Years	0 TO 23 Months
5	Appropriate words or phrases	Smiles or coos appropriately
4	Inappropriate words	Cries and consolable
3	Persistent cries and/or screams	Persistent inappropriate crying &/or screaming
2	Grunts	Grunts or is agitated or restless
1	No response	No response

Firearms Security:

If during a response, EMS personnel find a firearm, either on the patient or with anyone who might accompany the patient, law enforcement officers, either on scene or in route shall be notified so they can take possession of, and be responsible for its disposition.

In the event that EMS needs to leave the scene prior to law enforcement arrival, the firearm shall be secured in a locking storage space in the ambulance with the gun lock secured and the muzzle pointed to the outside of the compartment.

In the case of a rifle or shotgun or any firearm being too large for the locking storage space it should be placed in the rearmost compartment on the passenger side in a vertical position with the butt down and barrel pointed upward and secured in place assuring that it will not fall against the door or other equipment in the compartment.

- ❖ **Note:** If the firearm is in its holster, leave it in the holster.
- ❖ **Note:** The firearm can be left at the patient's residence.

On arrival at the emergency facility, notify appropriate law enforcement officials to take possession of the firearm, noting the officer's name, badge number, and whether law enforcement took possession of the firearm on scene or at our destination, or if the firearm was left at the residence.

Air-Ambulance Response:

Criteria

- Patient's condition requires expeditious transport to a hospital capable of providing definitive care.
- The scene is in a remote location with an extended ground transport time.
- The patient's condition is a "life or limb" threatening situation demanding intensive multi-disciplinary treatment and care.
- Unstable trauma patients as defined by the vital signs and physical findings. (Code Red)
- Critical burn patients. (Code Red)
- Mass Casualty incidents.
- Sufficient area for helicopter landing (100' X 100' area) free of trees, power lines, and debris with approach avenues from both the north and south.
- Good clinical judgment should be used at all times during decision-making process.

Hazmat Response:

Always ensure crew safety during hazmat responses. EMS crews will standby at a safe location upwind and uphill of the scene. Once fire crews have established a safe, "green zone," the EMS crew may enter the area. Crews will take precautionary measures when making patient contact and ensure that the patient has been properly decontaminated prior to transport. Once patient transport has been completed the EMS crew will decontaminate the unit before returning to service.

Extrication:

During vehicle extrication the EMS crew will wear the provided protective clothing inside the work area established by the fire department.

Procedures

Medical Procedures Criteria:

OBJECTIVE

To ensure that considerations of medical procedures to be attempted are based upon or within:

- **DO NO HARM**

- Training given to a Paramedic, AEMT, EMT, and EMR.
- Established textbook guidelines.
- Boundaries established by the Medical Director of Mulvane EMS.
- Recommendations of the Emergency Room Physicians.
- Maintaining a consistently high level of care.
- Duty to Act.
- Standard of Care.

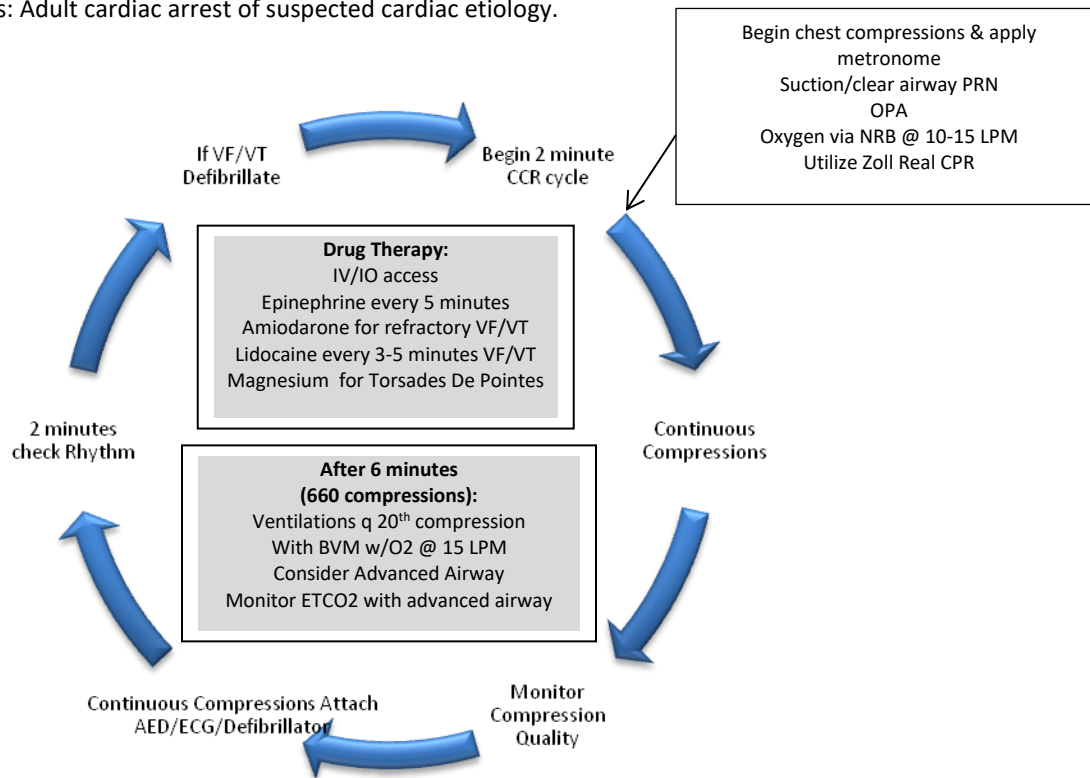
- **PROHIBITED ACTIVITIES:**

The following medical procedures will not be attempted by technicians in the field:

- Pericardiocentesis.
- Intracardiac medication injections.
- Arterial blood samples.
- Central venous catheterization of sub-clavian, internal jugular, or femoral. (External jugular catheterization may be attempted without specific physician's order.)
- Gastric lavage.

Adult Cardiac Arrest: Cardiocerebral Resuscitation (CCR)

Indications: Adult cardiac arrest of suspected cardiac etiology.



Treat reversible causes concurrently per appropriate protocol

CCR Quality

- Push hard (>2 inches), push fast (110/minute), allow complete recoil.
 - NOTE: Defibtech operates at 101/minute; Manual compression rate is to remain at 110/minute
- Minimize interruptions in chest compressions; interruptions should be infrequent and for no longer than 10 seconds.
- Avoid excessive ventilation. Ventilate at a rate of 6 breaths per minute or once every 20th compression.
- Rotate compressor every 220 or when compression quality decreases as indicated by Real CPR or capnography trending.

Electrical Therapy

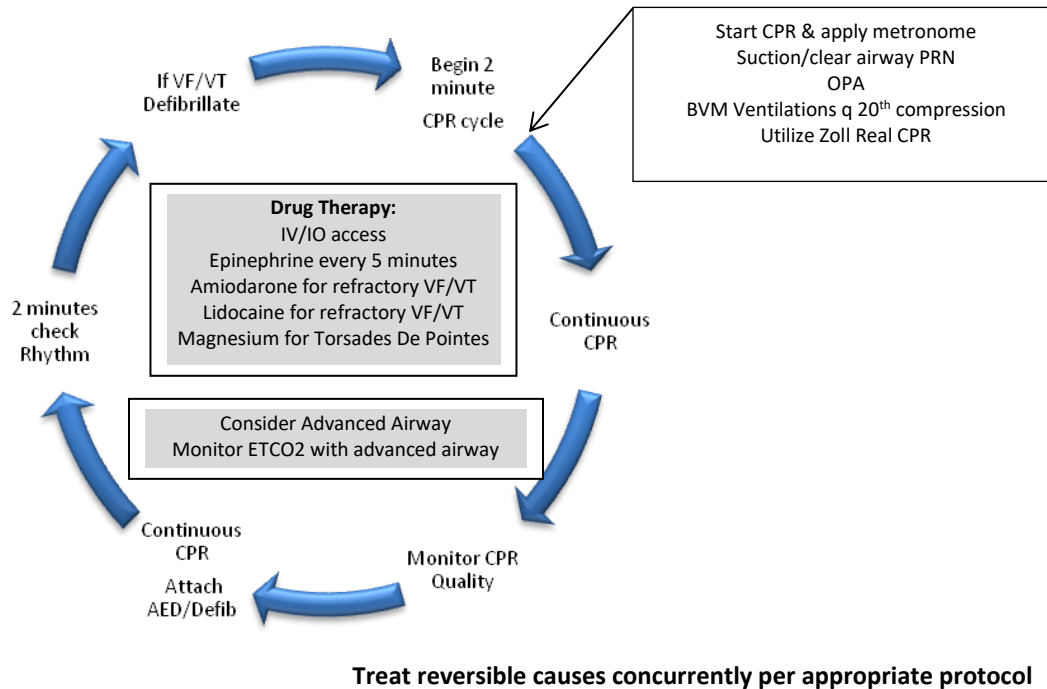
- Biphasic defibrillation: 200 joules.
- AED settings are preset by manufacturer.

Medication Therapy

- Normal Saline IV/IO TKO or bolus (if indicated) up to 2000 ml.
- Epinephrine 1:10,000 1 mg IV/IO repeat every 5 minutes up to 5 mg; **No more than 5 Epi total regardless of rhythm changes**
- Pulseless VF/VT antidysrhythmic options:
 - Amiodarone 300 mg in 20mL NS IV/IO; Repeat once in 3-5 minutes at 150 mg in 20mL NS IV/IO
 - OR**
 - Lidocaine 1.0-1.5 mg/kg IV/IO; Repeat every 3-5 minutes to maximum of 3 mg/kg.
 - Magnesium Sulfate 2 gm IV/IO bolus diluted in normal saline (for Torsades De Pointes only).
- Consider Sodium BiCarb 1mEq/kg after 10 minutes; Repeat once in 5 minutes at 0.5 mEq/kg (Asystole, PEA, VF, VT).
- **DO NOT ADMINISTER AMIODARONE AND SODIUM BICARBONATE IN THE SAME LINE**

Adult Cardiac Arrest: Cardiopulmonary Resuscitation (CPR)

Indications: Adult cardiac arrest of suspected non-cardiac etiology.



CPR Quality

- Push hard (>2 inches), push fast (110/minute), allow complete recoil.
 - NOTE: Defibtech operates at 101/minute; Manual compression rate is to remain at 110/minute
- Minimize interruptions in chest compressions; interruptions should be infrequent and for no longer than 10 seconds.
- Avoid excessive ventilation. Ventilate at a rate of 6 breaths per minute or once every 20th compression.
- One-person Adult CPR: 30:2 compression-ventilation ratio. Initiate continuous compressions with adequate resources.

Electrical Therapy

- Biphasic defibrillation: 200 joules.
- AED settings are preset by manufacturer.

Medication Therapy

- 🔪 Normal Saline IV/IO TKO or bolus if indicated up to 2000 ml.
- 🔪 Epinephrine 1:10,000 1 mg IV/IO, repeat every 5 minutes up to 5 mg; **No more than 5 Epi total regardless of rhythm changes**
- 🔪 Pulseless VF/VT antidysrhythmic options:
 - Amiodarone 300 mg in 20mL NS IV/IO; Repeat once in 3-5 minutes at 150 mg in 20mL NS IV/IO
 - OR**
 - Lidocaine 1.0-1.5 mg/kg IV/IO; Repeat every 3-5 minutes to maximum of 3 mg/kg.
 - Magnesium Sulfate 2 gm IV/IO bolus diluted in normal saline (for Torsades De Pointes only).
- 🔪 Consider Sodium BiCarb 1mEq/kg after 10 minutes; Repeat once in 5 minutes at 0.5 mEq/kg (Asystole, PEA, VF, VT).
- 🔪 **DO NOT ADMINISTER AMIODARONE AND SODIUM BICARBONATE IN THE SAME LINE**

Automatic External Defibrillator:

The Automatic External Defibrillator (AED) may ***only*** be used on an **unconscious, pulseless, and apneic patient.**

Indication

- The patient must weigh 90 pounds and/or have a large enough physical stature that the patches have ample space to be properly placed on the body.
- Insure that the patient is unconscious, pulseless, and apneic.
- Do not apply to patients that have a valid DNR, DPO, or Living Will that indicate that the patient does not want to be resuscitated.
- Follow the manufacturer's recommendations for adult and pediatric placement of patches and size of patches.

Precautions

- Do not use the AED in explosive or hazardous environments.
- Do not touch the patient or electrodes while the AED is analyzing or defibrillating.
- Do not place the patches directly over an implanted defibrillator or pacemaker.
- The patient must have a patent airway or the use of the AED will not be productive.

Procedure

- Determine that the patient is unconscious, pulseless, and apneic.
- If the arrest is unwitnessed CPR must be started for no less than two minutes.
- Apply the appropriately sized patches to the patient in the sterna and apex positions (shown on the patches).
- Turn on the AED and ANALYZE without touching the patient, AED, or the cables.
- If the AED advises "**SHOCK INDICATED**", at that time the patient ***must be cleared*** of all providers and only then can the **SHOCK** button be pressed.
- Resume CPR and ventilations with a BVM for two minutes and then the pulse should be checked.
- If the AED advises "**NO SHOCK INDICATED**", CPR must be immediately started.
- Resume CPR and ventilations with a BVM for two minutes and then the pulse should be checked.

Documentation

- The application, number of shocks, and the results of the patient must be properly recorded and documented in the patient care report.

Chest Decompression/Needle Thoracentesis:

A life-threatening situation may arise when a “one-way” valve is created that allows air to enter but not leave the pleural space. As the pressure in the pleural space exceeds the outside atmospheric pressure, the physiologic consequences of a simple pneumothorax are magnified to include tension pneumothorax. Increasing pressure within the pleural space further compromises pulmonary function/capacity and will force the mediastinum to the contra lateral side. This will result in increasing difficulty breathing (respirations) and a decrease in blood flow to the heart. As the pleural space increases, the mediastinum shifts and will result in a high venous pressure caused by the kinking of the Vena Cava. This causes a reduced blood flow into the heart and decreased cardiac output.

Signs and Symptoms

- Extreme anxiety, cyanosis, tachypnea, diminished or absent breath sounds to the injured hemisphere, bulging of the intercostal muscles, hyper-resonance, jugular vein distention, tachycardia, narrowing pulse pressures, hypotension, subcutaneous emphysema, and tracheal deviation (if seen - a late sign).
- Cyanosis and decreased breath sounds are the most common early findings with a pneumothorax. Other signs and symptoms develop later as pressure increases. Tachycardia and increased dyspnea despite treatment may be obvious later. JVD and tracheal deviation are among the latest of the signs and symptoms.

Authorized Use

- For relief of tension pneumothorax by standing order.
- For relief of pneumothorax by physician order.

Insertion site

- Second intercostal space
- Mid-clavicular line or axillary line if MCL site does not relieve pneumothorax, unable to access MCL, or if patient is obese.
- Same side as injury

Contraindications

- None

Technique

- Cleanse the site with betadine.
- Twist off the red cap and remove the 14 ga needle and catheter
- Insert at a 90-degree angle to the chest wall into the intercostal space over the superior border of the third rib, midclavicular line. If using anterior axillary site, insert at 90 degree angle to the chest wall into the 5th intercostal space over the superior border of the 6th rib, approximately a hands width from the axilla.
- Insert into the pleural space while listening for the sudden escape of air.
- Remove the needle portion and leave the catheter in place.
- Secure to the chest with a chest seal securing it adhesive side down with the valve directly over the catheter.
- Press dressing firmly in place to assure an occlusive seal.
- Re-evaluate breath sounds and vital signs.
- Monitor closely for recurrence of respiratory distress.

PEDIATRIC TECHNIQUE

- Insertion site is 2nd ICS mid-clavicular line ONLY
- Ages 0-1 years: use 22g IV needle
- Ages 1-4 years: use 20g IV needle

Continuous Positive Airway Pressure (CPAP):

Continuous Positive Airway Pressure has been shown to rapidly improve vital signs, gas exchange, reduce the work of breathing, decrease the sense of dyspnea, and decrease the need for endotracheal intubation in patients who suffer from shortness of breath from asthma, COPD, pulmonary edema, CHF, and pneumonia. In patients with CHF, CPAP improves hemodynamics by reducing left ventricular preload and afterload.

Indications

Any patient who is in respiratory distress with signs and symptoms consistent with asthma, COPD, pulmonary edema, CHF, or pneumonia and who is:

- Awake and able to follow commands.
- Is over 8 years old and is able to fit the CPAP mask.
- Has the ability to maintain an open airway.
- **And** exhibits two or more of the following:
 - A respiratory rate greater than 25 breaths per minute.
 - SPO² of less than 94% at any time.
 - Use of accessory muscles during respirations.

Contraindications

- Patient is in respiratory arrest/apneic.
- Patient is suspected of having a pneumothorax or has suffered chest trauma.
- Patient has a tracheostomy.
- Patient is actively vomiting or has upper GI bleeding.

Procedure

- **Explain the procedure to the patient.**
- Ensure adequate oxygen supply to ventilation device.
- Place the patient on continuous pulse oximetry.
- Place the patient on cardiac monitor and record rhythm strip.
- Place the patient on EtCO₂ monitoring via the nasal cannula before placing the mask but DO NOT break the seal to apply the cannula.
- Place the delivery device over the mouth and nose.
- Secure the mask with provided straps or other provided devices.
- For CHF/Pulmonary Edema, titrate to 10 cm/H₂O. For all other asthma & COPD patients titrate to 2-5 cm H₂O.
- Check for air leaks.
- Monitor and document the patient's respiratory response to treatment.
- Check and document vital signs every 5 minutes.
- Continue to coach patient to keep mask in place and readjust as needed.
- Contact medical control to advise them of CPAP initiation.
- If respiratory status deteriorates, remove device and consider intermittent positive pressure ventilation via BVM, and/or placement of non-visualized airway, or endotracheal intubation.

Removal Procedure

- CPAP therapy needs to be continuous and should not be removed unless the patient cannot tolerate the mask, experiences respiratory arrest, or begins to vomit.
- Intermittent positive pressure ventilation with a BVM, placement of a non-visualized airway, and/or endotracheal intubation should be considered if the patient is removed from CPAP therapy.

Special Notes

- Do not remove CPAP until hospital therapy is ready to be placed on patient.
- Watch patient for gastric distention which can result in vomiting.
- Due to changes in preload and afterload of the heart during CPAP therapy, a complete set of vital signs must be obtained every 5 minutes.
- Consider 0.5-1.0 mg Lorazepam IV/IO/IM/IN OR 2mg Midazolam IV/IO/IM/IN for severe anxiety to aid the patient in accepting the mask.

Defibtech Procedure:

Indication: For use during cardiac arrest to provide chest compressions

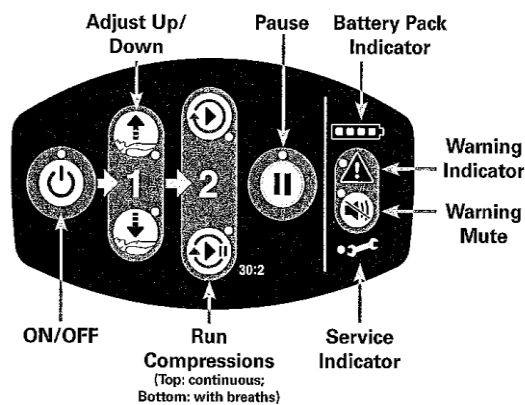
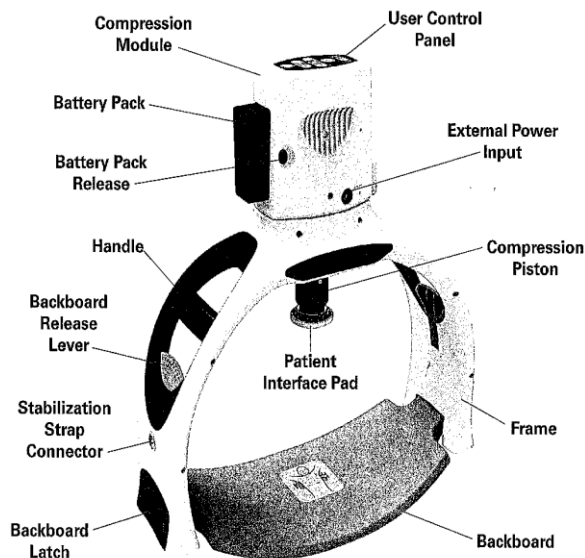
NOTE: The Defibtech compression rate is set at 101 compressions/min. However, manual compression rate is to remain at 110 compressions/min.

Contraindications:

- Defibtech cannot be placed safely or correctly on patient's chest
- Patient is too small for starting piston height to reach the patient's chest
- Patient is too large for the frame to attach to the backboard OR if the piston cannot be mounted without compressing the patient's chest

Procedure:

- Place soft cot under patient while also positioning the Defibtech backboard (grey curved piece) under the patient, lining the top of it up with patient's axillary creases.
- Roll patient onto soft cot and Defibtech backboard, re-positioning the backboard as needed for proper alignment. The center of the backboard should be in line with patient's nipple line.
- Attach compression module to frame, ensuring battery is placed properly in the compression module.
- Press ON/OFF button for one second to turn ON.
- Adjust piston height, ensuring proper positioning on patient's chest.
- Press Run button to perform compressions.
- PAUSE button is to be pressed prior to rhythm checks and pressed again to resume compressions.
- To ensure Defibtech remains appropriately positioned, apply the Stabilization Strap behind patient's neck and connect Strap to the frame, adjusting length as needed.
- Hands may be secured to the frame using the provided straps, however ensure IV access is not impeded.



External Cardiac Pacing:

The use of the External Transcutaneous Pacemaker is a temporary pacing device that may be used for rapid deployment in life threatening situations. The external pacemaker is to be used on symptomatic patients only or when all other means of treatment have been exhausted. The provider must provide basic life support measures first. Also, externally pacing a patient is most productive when in combination with drug therapy. Pediatric patients should have the appropriately sized patches applied.

Indications

- Symptomatic Bradycardia

Considerations

- Avoid hazardous environments, such as, flammable atmospheres, wet or moist environments, and conductive areas.
- Pacing a patient may be painful or discomforting. Consider pain management prior to pacing a patient if they have a low pain tolerance.

Application

- The application should follow the manufacturer's recommendations.
- Use appropriate size patches for the patient. The patient should have ample space between patches that they don't touch.
- The rate should be no less than 60 bpm for adults and children and 100 bpm for infants.

Procedure

- Ensure 4-lead is applied properly.
- Attach pacing pads, ensuring good contact with skin (i.e. remove clothing, hair, and moisture). Ensure pads are not covering any part of other electrodes.
- Select the lead (I, II, or III) providing the largest amplitude QRS complex.
- Verify R-waves are properly being detected.
- Press the PACER button. It will default to DEMAND mode.
- Set Pacer Rate.
- Turn On Pacer.
- Set Pacer Output.
- Determine presence of capture:
 - Electrical capture = presence of widened QRS complex, loss of any underlying intrinsic rhythm, and appearance of an extended, and sometimes enlarged, T-wave
 - Mechanical capture = presence of palpable peripheral pulse matching the pacer; USE ONLY femoral artery, right brachial artery, or right radial artery to check for capture

Contraindications

- Do not pace the following patients:
 - Trauma patients, except in electrocutions.
 - Hypothermic patients.
 - Patients on which the patches do not properly fit.

Fire Rehab:

Fire rehab is established so that personnel can seek rehabilitation from an incident and/or medical assistance when needed.

Indication

- Fire rehab will be established when requested by command for the following incidents:
 - Working fires.
 - Extended scene times on motor-vehicle accidents.
 - Large grass fires.
 - Full response Haz-Mat.
 - Any collapse or trench rescues with extended scene times
 - Extended scene times in inclement weather.
 - At the discretion of the Incident Commander.

Assignment

- All firefighters/personnel will report to rehab as a unit.
- One crew member will become Rehab Officer and notify command.
- All firefighters/personnel will remain in rehab for no less than 20 minutes.
- The unit assigned to rehab will not be released until all firefighters/personnel have been cleared.
- When a unit is assigned to rehab, the time is for them to relax. They will not be permitted to conduct duties on the incident grounds until released.

Assessment

- Anytime a formal rehab is established all firefighters/personnel assigned to rehab will need an initial set and exiting set of vitals.
 - Vitals will include a BP, Temp, HR, and LOC.
 - BP must be less than 160 mmHg systolic.
 - TEMP below 101°.
 - HR must be below 110bpm.
 - SpO² that sustains at >92%.
 - CO that sustains at 9% or lower.
 - All vitals must be below recommended limits prior to release from rehab.
 - The provider may use their discretion on limits due to patients PMHX.
 - At any time that an individual in rehab becomes symptomatic to medical problems or traumatic injuries that individual will become a patient and be treated as such.

Treatment

- Every unit in rehab should be able to relax in a relatively controlled environment.
- Water and/or Gatorade will be available to drink for rehydration.
- Cooling towels and/or blankets should be available.

Intramuscular Medication (IM) Administration:

Drugs administered by the intramuscular (IM) route are deposited into vascular muscle tissue, which allows for rapid absorption into the circulation.

Indications:

When medication administration is necessary, and the medication must be given via IM route or as an alternative route in selected medications.

Contraindications

- Active infection, cellulitis, or dermatitis at the site of administration
- Known allergy or hypersensitivity to the drug
- Acute myocardial infarction- the release of muscle enzymes may provide a confounding bias in making the diagnosis
- Thrombocytopenia
- Coagulation defects
- Hypovolemic shock- the absorption of the drug may be hampered owing to compromised vascularity to the muscle

Procedure

- Explain the procedure and gain consent.
- Before drug administration, check whether the patient has any allergies.
- Check the prescription is correct, following the 'five rights' of drug administration to reduce the risk of error.
- Assemble the syringe and needle and withdraw the required amount of drug from the ampoule. Some medicines are available in pre-filled syringes and manufacturer's instructions should be followed.
- Disperse air bubbles from the syringe.
- Expose the selected area and cleanse the injection site with alcohol.
- Insert the needle into the skin with a smooth, steady motion
- 90-degree angle skin pinched skin flattened
- Aspirate for blood
- Inject the medication.
- Withdraw the needle quickly and dispose of properly without recapping.
- Apply pressure to the site.
- Apply band aid
- Monitor the patient for the desired therapeutic effects as well as any possible side effects.
- Document the medication, dose, route, and time on/with the patient care report (PCR).

Intranasal Medication (IN) Administration:

IN medication administration is not intended to be a substitute for establishing IV access but is instead intended to be utilized in the **emergent situation** when IV access is not readily or quickly available.

Contraindications

- Severe nasal trauma
- Epistaxis
- Known prior nasal cocaine usage (relative)

Procedure

- Draw up medication in syringe.
- Attach MAD device to syringe.
- Place administration end of device gently against the patient's nare to form a good quality seal.
- Administer no more than 1 mL of solution into each nare per delivery.
 - Do not administer medication during patient exhalation through nose.

Intraosseous Access/Infusion:

The EZ IO device consists of a battery powered driver, a special needle assembly, a right-angle extension tubing ("EZ Connect"), and a wrist band.

Indications

- Patients where rapid, regular IV access is unavailable after 90 seconds with any of the following;
 - Cardiac arrest / Respiratory Failure / Respiratory Arrest
 - Multi-system Trauma with severe hypovolemia.
 - Severe dehydration with vascular collapse and/or loss of consciousness.
 - Severe Sepsis

Contraindications

- Fracture proximal to proposed insertion site
- History of Osteogenesis Imperfecta ("brittle bone disease")
- Current or prior infection at the proposed site
- Previous IO insertion or joint replacement at the proposed site
- Traumatized extremity
- Cellulitis
- Burns that may be infected by the technique used for insertion
- Congenital bone disease

Potential Complications

- **Technical:** Improper placement may result in sub-periosteal or soft tissue infusion; Penetration of posterior wall of medullary cavity resulting in soft tissue infusion and subsequent compartment syndrome; Slow infusion from clotting of marrow.
- **Systemic:** Osteomyelitis; Fat embolism; Slight periostitis at the injection site (usually short term); Bone fracture

Sites

- **Proximal Tibia:** Two finger widths below the patella or tibial tuberosity. Find the "flat spot" on the medial aspect of the tibial shaft. **Remember:** "Big Toe IO" means to look on the big toe side of the leg for the tibial plateau (the flat spot).
- **Distal Tibia:** Two finger widths above the ankle bone on the medial side. Again: "Big Toe IO".
- **Humeral Head:** The greater tubercle is located by having the patient place his hand on his belly button and relax his shoulder and elbow against the stretcher or turn the patient's arm with the elbow facing out and palm facing in. Slide your thumb up the anterior shaft of the humerus until you feel the greater tubercle; this is the surgical neck. Approximately 1 cm above the surgical neck is the insertion site for most adult patients. Vidacare recommends the 45 mm needle on patients >40kg (90lbs).

Insertion:

- Cleanse the site with betadine using aseptic cleansing techniques.
- Penetrate the skin to the bone at a 90-degree angle without running the driver.
 - Humeral site: point needle at 45 degrees to the horizontal plane, resulting in the needle hub lying perpendicular to the skin.
- Assure the 5mm mark is visible when the needle tip has come in contact with the bone. This assures there will be enough needle length to reach the marrow cavity. If the 5mm mark is not visible do not proceed, the needle is not long enough!
- Run the driver constantly with mild pressure for adult patients, no pressure with pediatric patients. With pediatrics, let the weight of the driver do the work. Stop the needle once it has become flush with the skin or the needle has entered the marrow.
- While holding the hub in place against the skin, unscrew the needle and remove stylet and attach supplied right angle lock.
- Aspirate a small amount of blood and marrow to confirm placement.
- *Prior to flushing with normal saline* administer 40 mg of 2% Lidocaine for conscious patients or 0.5mg/kg for pediatrics not to exceed 40 mg; flush should be easy. May repeat 20 mg X 1 for comfort of the conscious patient. Wait 60 seconds before flushing with normal saline.
- A second flush may be needed on larger patients; do the second flush with normal saline.
- Check for any leakage or extravasation or fluid gathering in extremity compartments.
- Put the arm band on the patient's wrist.
 - Humeral site: secure patients arm in place across the abdomen or to patient's side ensuring that the elbow remains facing out and the palm remains facing in.
- Attach standard IV set and pressure infuser and infuse fluids as specified.
- Secure needle with tape and/or gauze.
- Document.

Intravenous Access:

- IV access may be obtained on any patient deemed necessary by the AEMT or Paramedic for any type of medical or trauma complaint.
- Isolated orthopedic trauma patients may be triaged code green and still receive an IV.
- Code yellow patients or higher require IV access unless the patient refuses the procedure.
- Adult code red patients responsive or unresponsive require a minimum of two attempts before moving on to an IO access. Conscious patients require a physician order prior to starting an IO.
- Pediatric code red patients (responsive or unresponsive) require one attempt prior to starting an IO. Conscious patients require a physician order prior to starting an IO.
- Adult code blue patients without IV access established within 90 seconds should have IO access established ASAP.
- Pediatric code blue patients require one good faith attempt prior to IO access. (A good faith attempt means you looked and were unable to find and obtain IV access within 30 seconds)
- The approved method of obtaining IV access for medical or trauma patients is by using the IV lock and flush system. If the patient requires fluid resuscitation or the patient requires several IV medication administrations, then a drip set and bag of normal saline may be set up with the IV lock system to facilitate these treatments.
 - Normal Saline bags are to be used for fluid resuscitation or in the event of several medication administrations and medication drips only.

Needle Cricothyrotomy:

The only indication for a needle cricothyrotomy is to secure an airway when **ALL OTHER METHODS** have been unsuccessful. Needle cricothyrotomies may only be performed by full-time paramedics who have received training within in the last 3 years (either via MEMS or the Difficult Airway Course) AND have approval from the receiving physician through **on-line medical control**. Once an order for a needle cricothyrotomy has been approved, **only then** may it be performed.

Needle cricothyrotomy is to be performed on patients 11 years and younger. (If 12 years or older, surgical cricothyrotomy should be performed.)

EQUIPMENT:

- Iodine or other antiseptic solution
- Gloves and full-face protection
- Appropriate-sized IV catheter (14g, 16g, or 18g)
- 3mL syringe w/ 1mL of NS drawn up
- 10 drop set cut to approximately 18", keeping the hub adapter
- 2.5 or 3.0 ET tube connector
- BVM/Oxygen/Suction
- Tape

PROCEDURE:

- Have suction supplies available and ready.
- Locate the cricothyroid membrane utilizing anatomical landmarks.
- Use non-dominant hand to secure/stabilize membrane.
- Cleanse the area with iodine.
- Draw up 1mL of NS into a 3mL syringe and attach to the IV needle supplied in the cricothyrotomy kit.
- Insert the needle through the cricothyroid membrane at a 45-degree angle toward the feet.
- Aspirate for air with the syringe throughout the procedure of insertion.
- Once air bubbles return easily, stop advancing the needle.
- Advance the catheter into the trachea.
- Attach the modified drop set and secure with tape.
- Provide ventilation and verify placement with capnography, condensation in the tube, and breath sounds.

Restraint Procedures:

Procedure Statement

To ensure a safe and proper usage of restraint devices for medical and legal purposes.

- Restraints may be used at the discretion of the attending Tech if a patient shows to be a threat to the safety of themselves or others.
- Handcuffs can be used in place of restraints on any patient in the custody of law enforcement that is transported via EMS. The officer must be in the patient compartment during transport.
- Any patient restrained by EMS will be monitored continuously by visual means until another competent health care provider is physically in attendance and assumes medical responsibility for the patient.
- Anytime restraints are used, the need for their use must be documented in the patient narrative. Include in the narrative the time restraints were applied, how the patient was restrained, and any injuries the patient may have sustained while restrained.
- DO NOT USE restraint positions that could impede the patient's airway (i.e., choke holds, hogtie, prone, etc.)
- Restraints should not be routinely used. Assess each patient carefully, as a unique individual, to determine if restraints are truly necessary to ensure the safety of you and the patient.
- Properly apply and adjust restraints to maintain proper body alignment and patient comfort. Monitor and re-evaluate the restraints at least every 10 minutes. The patient's condition should be monitored regularly. Make sure the restraints are not cutting off circulation.
- Consider chemical restraint early to prevent overexertion of patient.
- **BLUE GOES ON WRISTS**
- **RED GOES ON ANKLES**

Spinal Motion Restriction:

Spinal Motion Restriction is defined as the application of a cervical collar and maintaining the spine in neutral alignment.

Indication

- Point tenderness on palpation of entire length of spinous processes.
- Focal deficit (motor or sensory) on neuro exam.
- High risk patients:
 - Vehicle ejection.
 - Motorcycle crash > 20 mph.
 - Auto vs. Pedestrian or bicycle at > 20 mph.
 - Axial load to head.
 - Fall from 3x patient's height.
- Low risk patients:
 - Patient not reliable and competent.
 - Below baseline level of alertness.
 - Evidence of clinical intoxication.
 - Distracting injury.
 - Unable to communicate adequately.

Procedure

- Perform appropriate trauma assessment.
- Determine if required by mechanism of injury.
- Criteria for spinal motion restriction met, patient should be transported flat and supine on the cot reinforcing the importance of limiting movement of cervical spine and encourage neutral positioning.
 - Perform a brief neurologic assessment to assess presence of movement and sensation to light touch in all extremities.
 - Evaluate:
 - Cervical spine for point tenderness and step-off
 - Neck for other injuries or findings
 - Place an appropriately sized c-collar.
- **Ambulatory** or able to self-extricate, assist to supine position on cot while limiting spinal movement.
- **Extrication** needed:
 - A long spine-board with a cervical collar in place may be used - **only** as a tool - and should be removed once extrication is complete and patient is supine on the cot maintaining neutral spine alignment throughout the process.
 - KED may be used if no other satisfactory method is available. It is not to be used for neuro intact patients seated in vehicles or when rapid extrication is required. If used, it should be unbuckled and fully released (including head straps) once the patient is laid flat on the cot.
- **Special Situations:** (Monitor and document neuro status in these alternate positions)
 - Vomiting: It is acceptable to place patient on their side to facilitate airway protection with repeated vomiting.
 - Dyspnea: If dyspneic with the cervical collar in place while laying completely flat, the head of the cot may be slightly elevated **no more** than 20 degrees.
 - Pregnancy (third trimester): To facilitate adequate venous return the patient may be placed on their left side or elevate the head of the cot **no more** than 20 degrees.

Surgical Cricothyrotomy:

The only indication for a surgical cricothyrotomy is to secure an airway when **ALL OTHER METHODS** have been unsuccessful. Surgical cricothyrotomies may only be performed by full-time paramedics who have received training within in the last 3 years (either via MEMS or the Difficult Airway Course) AND have approval from the receiving physician through **on-line medical control**. Once an order for a surgical cricothyrotomy has been approved, **only then** may it be performed.

Pt must be 12 years of age or older to have surgical cricothyrotomy performed. (If under age 12, needle cricothyrotomy must be performed.)

Equipment:

- Iodine or other antiseptic solution
- Scalpel
- Gloves and full-face protection
- 10mL syringe for ETT cuff
- ET tube 6mm cuffed
- Bougie
- Sterile 4x4
- Tape
- BVM/Oxygen
- Suction

Procedure:

- Have suction supplies available and ready
- Position patient in the neutral recumbent position. Maintain C-spine precautions for suspected cervical injury.
- Locate the cricothyroid membrane utilizing anatomical landmarks.
- Use non-dominant hand to secure/stabilize membrane.
- Cleanse the area with iodine.
- Make an approximately 2cm incision vertically over the larynx, then re-identify the cricothyroid membrane. Make a horizontal incision through the cricothyroid membrane until the widest part of the blade is inserted through the membrane.
- Place a finger in the hole to maintain patency.
- Place bougie in the hole, using finger as a guide and insert until resistance is felt. You might feel “clicks” along the way from the cricoid rings and the tip of the bougie should be in the carina.
- Slide cuffed ETT 6.0 over bougie and place in trachea. Inflate bulb and secure tube.
- Provide ventilation and verify placement with capnography, condensation in the tube, and bilateral breath sounds.
- Secure tube with tape.
- Make sure patients is reassessed often and tube is secure.

Synchronized Cardioversion:

Synchronized cardioversion is the preferred treatment for symptomatic tachycardic dysrhythmias. These patients must be symptomatic and show signs of cardiovascular compromise.

Indications & Settings

- Unstable V-Tach: 100 joules, 150 joules, and 200 joules.
- Unstable Supraventricular Tach: 50 joules, 100 joules, 150 joules, and 200 joules.
- Unstable new onset A-Fib: 100 joules, 150 joules, and 200 joules.
- Unstable A-Flutter: 50 joules, 100 joules, 150 joules, and 200 joules.

Application

- First provide basic life support measures.
- Obtain orders if indicated by protocols.
- Consider sedation prior to cardioversion if indicated.

Procedure

- Select a lead on the monitor that detects an “R” wave.
- Set the monitor to “Sync.” Ensure that the “R” wave is getting capture.
- Once capture is indicated, cardio-vert the patient to the specified energy setting stated in the protocols.
- If the rhythm changes follow appropriate protocol for treatment.

Taser Dart Removal and Care:

A taser works by firing two wire-attached darts that can strike a suspect from up to 15 feet or more. It delivers up to 50,000 volts of electricity to the recipient. This jolt of electricity is not harmful to vital body functions such as heart rhythm, pacemaker function or respirations. The taser dart usually penetrates the skin only a few millimeters. Technicians can safely remove a dart simply by pulling it out. The only exception is involvement of the eye, face, neck, breast, groin, spinal column or joint. In this case, leave the dart in place and transport the patient to the hospital for dart removal. Strongly consider scene safety and measures to protect yourself and other rescuers from a potentially violent patient in situations when a taser gun has been used. You do not need to transport a person to the hospital based solely on taser dart exposure. If a patient has no need for further medical evaluation, they can be left in police custody.

- Scene Safety.
- Confirm that the taser has been shut off and the probe is no longer connected to the taser gun.
- Obtain vital signs at the earliest opportunity. Violent and combative behavior may be secondary to intoxication, psychosis, hypoxia, hypoglycemia, OD or CNS infection. Treat trauma and seizure as applicable.
- Evaluate the anatomical location of the probe(s) puncture zone(s). High-risk/sensitive zones will require transport to a medical facility for removal. They Include:
 - Eyes, ears, nose, mouth and neck. (Darts to scalp, and low risk areas of forehead and cheek, can be removed in the field, but these wounds may require assessment by a physician).
 - Breasts (female)
 - Genitalia
 - Hands or Feet
 - Joints
 - Spinal column

Removal Procedure

- Utilize PPE.
- Prior to probe removal inform all caregivers that you are about to remove the contaminated sharp.
- Place hand in the form of a “V” around the taser dart in order to stabilize the surrounding skin and to keep loose skin from coming up with the dart. Firmly grasp the probe and with one smooth hard jerk, remove probe from skin. (remove probes one at a time)
- Examine the probe and the patient closely in an effort to make sure the probe tip did not break off during removal. Accordingly, it is important that the person removing the barb visually inspect it to make sure that the tip is fully intact. If the barb remains in the subject, the patient will need to be transported to a medical facility for barb removal.
- Be careful to avoid accidental needle sticks when removing probes.
- Removed probes should be handled like contaminated sharps and should be placed in the specimen container provided by the PD. They will likely log the probes into evidence and 6 inches of the wire must remain intact with the taser buds if the patient is transported.
- Provide wound care by cleansing the affected area with saline and apply a band aid.
- Inform patient of basic wound care and the need to seek additional care in the event that signs of infection occur (redness-fever-drainage-swelling etc.)
- Clear and thorough documentation is required in the body of the report narrative whether or not EMS transports the patient.
- If transport is necessary, transport to the closest appropriate hospital.

TOPICAL HEMOSTATIC AGENT APPLICATION, ADULT & PEDIATRIC

Indications:

- Life-threatening hemorrhage to junctional areas of the neck, axilla, and/or groin
- Severe scalp lacerations w/ uncontrolled bleeding
- Life-threatening hemorrhage to extremities unable to be controlled by direct pressure and/or tourniquet usage

Contraindications:

- Open thoracic, abdominal, and/or penetrating head wounds

Background:

QuikClot Combat Gauze is a sterile gauze (rolled or in 4x4 pads) impregnated with kaolin, a compound shown to increase clotting in severe hemorrhage by activating factor XII of the clotting cascade. Kaolin is a naturally occurring inorganic compound that has not been found to cause any allergic reactions.

Application:

1. Apply direct pressure to the wound and a tourniquet as applicable.
2. Remove excess pooled blood from around the wound while preserving any clots already in the wound.
3. For junctional and extremity injuries, pack as much rolled QuikClot gauze into the wound as possible and use it to apply pressure directly to the bleeding source until the bleeding stops. More than one package of hemostatic agent may be needed.
 - a. For severe scalp lacerations: apply the 4x4 QuikClot gauze over the wound, **DO NOT PACK**, and apply direct pressure to the bleeding source until bleeding stops. More than one package may be needed.
 - b. For adjunct therapy for extremity wounds uncontrolled by direct pressure and/or tourniquet, pack as much rolled QuikClot gauze into the wound as possible and use it to apply pressure directly to the bleeding source until the bleeding stops.
4. Continue applying direct pressure to the wound for 3-5 minutes.
5. Leave the hemostatic agent in place. Place empty package over wound to ensure it is transported w/ the patient to the destination facility.
6. Dress the wound with a pressure dressing to secure the hemostatic agent.
 - a. In the case of severe wounds to the neck, direct pressure may have to be continuously applied throughout transport.

NOTE: It is extremely important that direct pressure is provided immediately upon application of the hemostatic agent and held for *at least* 3-5 minutes.

Adult Cardiac

Adult Cardiac Arrest:

EMR

EMT

- Determine use of CCR versus CPR by suspect etiology.
- Begin CPR/CCR
- Attach AED
- Airway management and appropriate oxygen therapy via BVM if performing CPR **OR** NRB if performing CCR.



EMR and EMT STOP

AEMT

- Determine and document rhythm in two leads.
- Biphasic defibrillation 200 joules. (VF/VT)
 - Normal Saline IV/IO TKO or bolus if indicated up to 2000 ml.
 - Epinephrine 1mg IV/IO 1:10,000 (repeat 3-5 minutes) up to 5 mg; **No more than 5 Epi total regardless of rhythm changes**
- Pulseless VF/VT anti-dysrhythmic options:
 - Amiodarone 300 mg in 20mL NS IV/IO; Repeat once in 3-5 minutes at 150 mg in 20mL NS IV/IO.
 - Lidocaine 1.0-1.5 mg/kg IV/IO; Repeat every 3-5 minutes to maximum of 3 mg/kg IV/IO.



AEMT STOP

PARAMEDIC

- Magnesium Sulfate 2 gm IV/IO diluted in 20mL NS and administered IVSP over 1-2 minutes (for Torsades De Pointes only).
 - **MED MATH: supplied as 5Gm/10mL -> Draw up 2Gm Mag w/ 16mL NS in 60mL syringe for 10% solution**
- Consider Sodium BiCarb 1mEq/kg (max 100mEq) after 10 minutes of downtime; Repeat once in 5 minutes at 0.5mEq/kg (Asystole, PEA, VF, VT).



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Additional medications.
- Consider termination of resuscitation efforts.

Key Points/Considerations

- Do not interrupt compressions for placement of an advanced airway. Avoid excessive ventilation.
- Charge defibrillator without stopping chest compressions. Give everyone clear command. If a shockable rhythm presents ensure everyone is clear and shock. Otherwise continue compressions and dump charged defibrillator by changing energy settings. **BE SURE TO CHANGE BACK TO CORRECT ENERGY SETTING FOR NEXT DOSE.**
- Capnography to confirm and monitor ET tube placement and to measure effectiveness of chest compressions and watch for ROSC.
- Treat reversible causes concurrently per appropriate protocol.

➤ Hypoxemia	(hyperventilate)
➤ Hypovolemia	(fluid challenge up to 2000 ml)
➤ Hypoglycemia (<60 mg/dL)	(Dextrose)
➤ Hypothermia	(warm)
➤ Hyperkalemia/Hydrogen Ion (acidosis)	(Sodium Bi-Carb 1 mEq/kg with a maximum 100 mEq)
➤ Pulmonary Embolism	(ventilate & early advanced airway placement)
➤ Tension Pneumothorax	(needle decompression)
➤ Cardiac Tamponade	(fluid challenge up to 2000 ml)
➤ Overdose (Opiate)	(2 mg Narcan as necessary. May repeat up to 6 mg)

Adult Cardiac Arrest: Management/Post Resuscitation

EMR

- CAB and vital signs.
- Continue high flow O2 via non-rebreather mask or BVM as indicated.
- Acquire 4 lead and print (no interpretation).



EMR STOP

EMT

- Acquire 12-lead (no interpretation) while en route to intercept location.
- Capnography monitoring with an advanced airway in place.



EMT STOP

AEMT

- Acquire 12-lead.
 - Normal Saline IV/IO bolus up to 2000 mL to maintain blood pressure of 90 systolic (monitor lung sounds)



AEMT STOP

PARAMEDIC

- Treat dysrhythmias per appropriate protocol
- For hypotension not responsive to fluid:
 - Dopamine 2-20 mcg/kg/min IV/IO infusion (pre-mixed bag or via buretrol w/ added Dopamine)
 - **MED MATH:** DOPAMINE: Add one vial (200mg/5mL) Dopamine to 120mL NS in a buretrol for a total of 200mg/125mL = 1600mcg/mL; Drip rate is figured same as with pre-mixed bag



PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional Fluids.

Key Points/Considerations

- Contact Medical Control to tailor the treatment to the patient's specific needs.
- Vital signs should be recorded every 5 minutes on critical patients.
- Even mild hypotension should be avoided because it can impair recovery of cerebral function. It is appropriate to administer a fluid bolus unless it is known with certainty that the patient has volume overload.
- Symptomatic Bradycardia – reassess ventilations and airway, if profound bradycardia persists after reassessment; consider Cardiac Dysrhythmia/Bradycardia Protocol.
- DO NOT ADMINISTER Amiodarone or Lidocaine if converted rhythm is a high degree heart block.
- **Paramedic Option:** Placement of NG or OG tube for gastric decompression.
- **EMT & AEMT Option:** Placement of OG tube (with any supraglottic airway in place) for gastric decompression purposes.

Adult Cardiac Arrest: Termination of Resuscitation

EMR

EMT

- Resuscitative efforts for patients in cardiac arrest should not be initiated if:
 - The patient presents with lividity, rigor mortis, decomposition and/or injuries incompatible with life (such as decapitation).
 - There is a signed DNR order present.
 - Living Will directs to withhold resuscitative efforts. (Must have documents on scene)
 - Medical DPOA directs to withhold resuscitative efforts. (Must have documents on scene)
- For all other patients in respiratory or cardiac arrest, the EMS provider MUST start BLS care, including defibrillation, and request a Type I assist if Type II.



EMR and EMT STOP

AEMT

PARAMEDIC

- Complete standing orders appropriate to presenting rhythm.
- Persistent Asystole – lasting 20 minutes (with successful ET tube or King Airway placement and IV/IO medications given.)



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



If patient is hypothermic due to inclement conditions or drowning, contact online medical control or consider transporting.

Key Points/Considerations

- Is scene appropriate for termination order?
- Attempt to obtain a copy of the DNR or advanced directive to be attached to the chart.

Adult Cardiac: Chest Pain

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose and treat per diabetic protocol if indicated.
- Acquire 4 lead and print (no interpretation).
- Aspirin: 324 mg PO



EMR STOP

EMT

- Nitro: 0.4 mg sublingual every 3-5 minutes (3 dose maximum maintaining a systolic B/P >100 mmHg).
- Acquire 12-lead (no interpretation).
 - (Note only in a BLS setting may nitro be given prior to a 12-Lead being performed)
- Ondansetron ODT: For nausea 8 mg PO. May repeat once in 10 minutes.



EMT STOP

AEMT

- Cardiac monitoring.
- IV/IO access Normal Saline TKO.
 - Consider Normal Saline bolus up to 1000 ml if BP < 90 mmHg (monitor lung sounds closely).
- 12-lead acquired before any medications.
 - Nitroglycerin: 0.4 mg sublingual every 3-5 minutes as necessary maintaining a systolic B/P > 90 mmHg.
 - Nitro-Paste: 1 inch of 2% ointment topically.
 - Consider: 500 ml normal saline bolus IV/IO prior to nitro (inferior wall MI).
 - Morphine Sulfate: For pain control 2 mg IV/IO/IM/IN every 5 minutes; maximum of 10 mgs.
 - Hydromorphone: If allergic to morphine 0.25-1 mg IV/IO/IM/IN. Repeat every 3-5 minutes as necessary. Maximum of 4 mg.
- **OR**
- Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5min PRN at ½ initial dose, up to a total dose of 3mcg/kg
- Ondansetron: For nausea 4 mg IV/IM/IO. May repeat once in 5 minutes.



AEMT STOP

PARAMEDIC

- For recurrent hypotension up to an additional 1000 ml normal saline remaining aware of fluid overload.
- Follow appropriate protocol for rhythm changes.
- **PARAMEDIC STOP**

PHYSICIAN OPTIONS

- Any treatment outside of protocol

Key Points/Considerations

- Notify receiving hospital as soon as possible of a STEMI alert.
- Monitor B/P closely when administering nitro to patients showing signs of an inferior wall MI.
- Do not delay transport.
- Serial 12-leads on all suspected active MI patients or any suspicious cardiac related chest pain patients.
- STEMI Alerts transported to Wesley, Via Christi St Francis, or Kansas Medical Center only.

Adult Cardiac: Dysrhythmia/Bradycardia

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via NC or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead and print (no interpretation).

 **EMR STOP**

EMT

- Acquire 12-Lead (no interpretation).

 **EMT STOP**

AEMT

- Cardiac monitoring.
 - Acquire 12-lead.
 - IV/IO access Normal Saline TKO.
 - Normal Saline 500 ml bolus if systolic BP is < 90 mmHg (monitor for fluid overload).
 - Morphine Sulfate: 2 mg every 5 minutes IV/IO or 5 mg IM/IN every 10 minutes, maximum 10 mg.
 - SBP must maintain > 100mmHg.
 - Dilaudid/Hydromorphone (if allergic to Morphine) 0.25-1 mg IV/IO/IM/IN every 5 minutes as necessary to a maximum of 4 mg.
- OR**
- Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5min PRN at ½ initial dose, up to a total dose of 3mcg/kg

 **AEMT STOP**

PARAMEDIC

- Repeat Normal Saline bolus (monitor for fluid overload).
- Atropine (if symptomatic and BP < 90 with a HR < 60): 0.5mg IV/ IO. May repeat Atropine as necessary to maximum total dose of 3 mg.
- Transcutaneous pacing starting at 60 bpm for symptomatic patients **or** those with high degree heart block **or** bradycardia refractory to atropine.
- Consider sedation prior to pacing:
 - Midazolam 2 mg IV/IO or 5 mg IM/IN titrated to effect. Repeat as necessary up to a maximum of 10 mg.
 - Lorazepam (if allergic to Midazolam) 1-4 mg IVSP/IO or 2 mg IM/IN. Repeat as necessary to maximum of 4 mg. IV/IO do not exceed 2 mg/min. IV/IO must be diluted w/ equal amount normal saline.
- Dopamine 2-20 mcg/kg/min IV/IO infusion (only if refractory to the above treatments)
 - **MED MATH:** Add one vial (200mg/5mL) Dopamine to 120mL NS in a buretrol for a total of 200mg/125mL = 1600mcg/mL; Drip rate is figured same as with pre-mixed bag
- Epinephrine push-dose pressor 10-20mcg every 2 min, titrated to MAP>65mmHg

 **MED MATH:** Prepare 10mcg/mL (1:100,000) Epi by adding 1mL of 1:10,000 Epi (0.1mg/mL) to 9mL NS; Administer 10-20mcg boluses (1-2mL) every 2 minutes

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- Early physician contact.
- With serious signs and symptoms (decreased LOC, hypotension, CHF/pulmonary edema) do not delay transcutaneous pacing.
- Use EXTREME caution with atropine in patients with bradycardia caused by AMI. Atropine can exacerbate the AMI or cause VF or VT.
- Do NOT use Atropine in high degree heart blocks.

Adult Cardiac: *Dysrhythmia / Tachycardia with a Pulse*

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor 4-Lead only may be applied and printed (No interpretation of ECG rhythm)

EMR STOP

EMT

- 12-lead (no interpretation)

EMT STOP

AEMT

-  Cardiac Monitor.
- Acquire 12-lead.
- IV/IO access Normal Saline TKO.
 -  Normal Saline 500 ml bolus if systolic BP is <90 mmHg and if no signs and symptoms of fluid overload exist.
- Identify and treat possible causes.
- Unstable rhythms require synchronized cardioversion:
 - Narrow, Regular QRS: 1st 50J, 2nd 100J, 3rd 150J, 4th 200J
 - Narrow, Irregular QRS: 1st 100J, 2nd 150J, 3rd 200J
 - Wide, Regular QRS: 1st 100J, 2nd 150J, 3rd 200J
 - Wide, Irregular QRS: 1st 100J, 2nd 150J, 3rd 200J

AEMT STOP

PARAMEDIC

- If patient is stable use medications first, then perform synchronized cardioversion if refractory to medications.
- Narrow, Regular QRS (e.g. SVT):
 -  Adenosine 6mg Fast IV/IO followed by 10mL NS flush; Repeat in 2min with 12mg PRN
- Narrow, Irregular QRS (e.g. A-Fib, A-Flutter, MAT)
 -  If symptomatic, synchronized cardiovert at appropriate dose
- Wide, Regular QRS (e.g. V-Tach, SVT/A-Fib/A-Flutter w/ aberrancy):
 -  Amiodarone 150mg IV/IO over 10 min; May repeat in 10 min PRN; **(See MED MATH below)**
 -  Lidocaine 1-1.5mg/kg IV/IO every 5min to max of 3mg/kg
- Wide, Irregular QRS (e.g. A-Fib w/ aberrancy, WPW w/ A-Fib, MAT, Polymorphic VT/Torsades de Pointes)
 -  Torsades: Magnesium 1-2g IV/IO of 50% solution diluted in 20mL NS (to make 10% solution) over 10min (2mL/min) **(See MED MATH below)**
 -  Amiodarone 150mg IV/IO over 10 min; May repeat in 10 min PRN; **(See MED MATH below)**
- Consider sedation prior to cardioversion:
 -  Midazolam 2 mg IV/IO or 5 mg IM/IN titrated to effect. Repeat as necessary up to a maximum of 10 mg.
 -  Lorazepam (if allergic to Midazolam) 1-4 mg IVSP/IO or 2 mg IM/IN. Repeat as necessary to a maximum of 4 mg. IV/IO do not exceed 2 mg/min. IV/IO must be diluted w/ equal amount normal saline.

PARAMEDIC STOP

PHYSICIAN OPTIONS

-  Repeat synchronized cardioversion after fourth attempt.

Key Points/Considerations

- **MEDICATION MATH: Amiodarone infusion:** Draw up 150mg Amiodarone into 10mL NS and administer at 1mL/min. **Hold onto the syringe plunger while infusing to prevent backflow; will need to be given with fluids to allow for flushing between doses.**
- **MEDICATION MATH: Magnesium sulfate:** Draw up 2Gr Mag w/ 16mL NS for 10% solution and give 2mL/min; will take 10 min. to give

- Wide, regular QRS?
 - If **YES**, is there a RBBB/LBBB morphology?
 - **YES:**
 - Consider **SVT** w/aberrancy. Ask about hx + adenosine
 - If V1 has Rsr' (with left rabbit ear taller than right) then rhythm is more likely **ventricular tachycardia**.
 - **NO:**
 - Positive QRS in aVR and/or negative QRS in aVF?
 - If **YES**, is there all positive or all negative QRS complexes in V1-6?
 - If **YES**, then rhythm is most likely **ventricular tachycardia**.
 - If **NO**, consider **antidromic AVRT**.
 - If wide, regular with **NO** RBBB/LBBB morphology:
 - Consider **ventricular tachycardia** until proven otherwise.
- Wide, irregular QRS?
 - If **YES**, is there a RBBB/LBBB morphology?
 - **YES:**
 - Consider **BBB + afib**. Ask about hx.
 - If RBBB morphology, V1 **must** have rSR' morphology (right rabbit ear taller than left rabbit ear)
 - If **BBB + afib**: Fluids & treat underlying cause of tachycardia (sepsis, dehydration, etc.)
 - **NO:**
 - Consider **WPW + afib**. Do **not** treat with *adenosine*.
 - If **WPW + afib**, *cardiovert* PRN as rate can be very fast.
- Narrow, regular QRS?
 - If **YES**, does the rate fluctuate or does it stay the same constantly?
 - *Fluctuates:*
 - Consider **sinus tachycardia**
 - *Constant rate:*
 - Consider **PSVT**
 - **AVNRT** most common (tx: *adenosine*)
 - **AVRT** requires accessory pathway (*WPW*) (tx: *adenosine*)
 - **Aflutter** (look at II for flutter waves, V1 for individual P waves)
- Narrow, irregular QRS?
 - Consider **atrial fibrillation** or (less likely) **multifocal atrial tachycardia (MAT)**.
 - Afib + RVR is usually due to secondary etiology (dehydration, sepsis, dyspnea). Treat underlying cause before administering rate control medication.

Adult Medical

Adult Medical: Abdominal Pain/Abdominal Discomfort

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.

 **EMR AND EMT STOP**

AEMT

- Cardiac Monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- Ondansetron (Zofran) for nausea as needed 4 mg IV/IO/IM; May repeat once in 5 minutes.
- Morphine 2 mg IV/IO as necessary every 5 minutes up to 10 mg **OR** IM/IN 5 mg; Repeat once in 10 minutes.
 - If Morphine is unavailable: Fentanyl 0.5-1 mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5 minutes PRN at ½ initial dose, up to total dose of 3mcg/kg
- Dilaudid 0.25 mg – 1 mg IV/IO/IM/IN every 5 minutes as necessary up to 4 mg
- Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5min PRN at ½ initial dose, up to a total dose of 3mcg/kg

 **AEMT STOP**

PARAMEDIC

- Toradol 30mg IV/IO/IN or 60mg IM (Geriatric age 65+ is ½ the dose for both)

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- Begin transport as soon as possible.
- Transport patient in a position of comfort if possible.
- Use caution palpating patients with possible cardiac involvement, i.e. aortic aneurysms.
- EMR and EMT may apply 4 – lead ECG if indicated. No Interpretation.
- SPO2 and cardiac monitoring required with any narcotic administration.

Adult Medical: Abuse / Neglect

EMR

EMT

AEMT

PARAMEDIC

- Ensure that scene is secure.
- CAB and vital signs.
- Initiate treatment as necessary
- If possible remove patient from scene, transporting to hospital even if there is no medical reason for transport.
- If patient is in immediate danger, let law enforcement handle scene.
- **AVOID ACCUSATIONS** as this may delay transport. Adult with child may not be the abuser.
- **Never** communicate via radio abuse suspicions.
- Carefully document findings and report to physicians at the hospital.
- Immediately contact the appropriate law enforcement agency to file a report. Make sure to include this in your documentation of the call.



EMR, EMT, AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- EMS care providers are MANDATORY REPORTERS by law with the duty to notify authorities of suspected child abuse.
- Abuse/neglect are widespread enough that nearly all emergency responders will see these problems at some time. The first step in recognizing abuse or neglect is to accept that they exist and to learn the signs and symptoms:
 - Noticeable injuries not mentioned in the history.
 - History does not explain physical findings of the injury.
 - Injuries with a pattern:
 - Bite marks
 - Cigarette Burns
 - Rope (circumferential or linear marks at the ankles or wrists)
 - Loop scars (doubled-over electrical wire or telephone cord)
 - Belt marks
 - Etc.
 - Injuries to genital or perineal areas.
 - Multiple repetitive unexplained injuries.

Adult Medical: Altered Mental Status/ Unresponsive

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if trauma suspected.
- Check blood glucose level. (See Diabetic Protocol)
- Acquire 4-lead (no interpretation).
- Determine and treat causes per appropriate protocol.



EMR STOP

EMT

- Acquire 12-lead (no interpretation).



EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor
- IV/IO access Normal Saline @ TKO.
- Up to 1000 ml Normal Saline bolus with systolic B/P 90 mmHg or less and no signs or symptoms of fluid overload exist.
- Determine and treat causes per appropriate protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Consult Medical Control as soon as possible.
- IO access should be attempted if no peripheral access can be obtained after 2 attempts or 90 seconds in patients with cardiovascular collapse.
- Potential causes of altered mental status:
 - A – Alcohol
 - E – Epilepsy, Endocrine, Electrolytes
 - I – Insulin
 - O – Opiates and other drugs
 - U – Uremia
 - T – Temperature, Trauma
 - I – Infection
 - P – Psychogenic, Poisoning
 - S – Shock, Stroke, Space-occupying lesion, Subarachnoid hemorrhage
- Questions to consider:
 - What was patient doing prior to this?
 - Has it happened before?

Anaphylaxis/ Allergic Reactions

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
 - Consider ventilator assistance for severe dyspnea or hypoventilation.
- Determine cause of reaction, time of exposure, and amount of substance patient was exposed to if possible.
 - Remove patient from source or source from patient if still present.
 - For Moderate Allergic Reaction (e.g., hives, itching, redness, stable vitals) administer:
 - ☞ Diphenhydramine 50 mg PO

For Anaphylaxis administer:

- ☞ Epinephrine 1:1,000 dose 0.3 mg IM used first for shock states; Repeat x1 in 5 minutes as necessary.

EMR STOP

EMT

- ☞ Albuterol 2.5 mg in 3 mL (unit dose) + Ipratropium 0.5 mg in 2.5 mL (unit dose) mixed together via nebulizer for wheezing. May repeat as necessary.

EMT STOP

AEMT

- Cardiac Monitor as indicated.
- IV/IO access Normal Saline TKO.
 - ☞ Normal Saline 500 mL fluid bolus. Repeat one time as necessary to maintain BP > 90.
 - ☞ Diphenhydramine 25 – 50 mg IV/IO/IM.

AEMT STOP

PARAMEDIC

- ☞ If Epinephrine IM is unsuccessful after 2 doses, administer 0.5mg 1:10,000 I/IO
- ☞ Methylprednisolone (Solu-Medrol) 125 mg IV/IO.

PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

-
- S/S of allergic reaction: gradual onset, mild flushing, rash, or hives, mild bronchoconstriction, and normal mental status.
- S/S of anaphylaxis: sudden onset, severe flushing, rash or hives, severe bronchoconstriction, laryngospasm, anxiety, sense of impending doom, may decrease to confusion then unconsciousness.
- Severe reactions may develop within 2-20 minutes following exposure.

Adult Medical: Dehydration

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).

 **EMR AND EMT STOP**

AEMT

PARAMEDIC

-  Cardiac Monitor
-  Acquire 12-Lead
- IV/IO access.
- With signs and symptoms of dehydration present:
 - Patient must have tachycardia >120 bpm and at least one additional sign or symptom:
 - Dry mucous membranes
 - Skin tinting
 - Hypotension
 - Decreased LOC related to suspected dehydration
 - High Fever
 - Severe vomiting or diarrhea
 - Suspected heat related emergency
 - Pointed and peaked T-waves
 - Pale or clammy skin
-  Normal Saline bolus 1000 mL; May repeat x1 if patient remains tachycardic.

 **AEMT AND PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- Monitor lung sounds closely throughout bolus.
- Monitor SPO2.
- Excessive rapid fluid replacement can cause an imbalance in electrolytes and other lab values. This may increase the patient's stay at the hospital. With this said, minor dehydration will not be treated in the field.
- If the patient is on a beta blocker medication, consider the heart rate may be less than 120 bpm; look for additional S/S of dehydration.

Adult Medical: Diabetic Emergencies

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy.
- Cardiac Monitor 4-lead only may be applied and printed (no interpretation).
- Check blood glucose level.
 - If blood glucose is known or suspected to be low and patient is able to self-administer and swallow on command, give oral glucose one unit dose (24 gms). Repeat one time as necessary.



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- Hyperglycemia: Blood glucose is above 250 mg/dL.
 - Normal Saline bolus up to 2000 mL IV/IO.
- Patients presenting with signs and symptoms of hypoglycemia:
 - Dextrose 25 gms IV/IO. May repeat in 10 minutes if signs and symptoms of hypoglycemia are present.
 - Follow Dextrose administration w/100mL normal saline infusion.
 - If unable to obtain IV access, administer Glucagon 1 mg IM.
 - IO administration only if IM Glucagon is unsuccessful.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- DKA has slow onset lasting 12-24 hours. S/S include polyuria, polydipsia, polyphagia, warm, dry skin and mucous membranes, nausea/vomiting abdominal pain, tachycardia, Kussmaul's respirations, fruity breath odor, decreased mental function, or coma.
- If the patient wishes to refuse transport:
 - Patient is alert.
 - Blood glucose greater than 60 mg/dl.
 - No other complaints or complications.
 - Instructed to eat complex carbohydrates, contact physician, and to recheck blood glucose regularly.

Adult Medical: Nausea and/or Vomiting

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead. (no interpretation)



EMR STOP

EMT

- Ondansetron (Zofran) ODT 8mg PO. Repeat once in 10 minutes if needed.



EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor if indicated.
- IV/IO access Normal Saline @ TKO
 - Ondansetron (Zofran) 4 mg IV/IO/IM if needed. Repeat once in 5 minutes.
- See Dehydration Protocol for suspected dehydration.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- See Dehydration protocol for severe cases.

Adult Medical: OB/GYN: Childbirth

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- If OB (>20 weeks gestation) and patient is a trauma patient place on left side, maintaining neutral spine.
- Assess for delivery, crowning.
- If childbirth is imminent, prepare for delivery.
 - Position patient in manageable position for delivery.
 - Support baby's head over the perineum. If membranes cover the baby's head as it is delivered, tear the sac as gently as possible to allow for amniotic fluid to escape. Make note of any meconium in amniotic fluid.
 - Once head has been fully presented release any nuchal cords and suction oropharynx and then nostrils.
 - If a nuchal cord is present after the head has delivered clamp the cord with two clamps and cut cord. A nuchal cord does not allow proper oxygenation to the baby.
 - Gently guide the head downward until the shoulder appears and the other shoulder should present with a little upward guidance.
 - Once baby has fully been delivered keep baby at same level as vagina and clamp the cord at 7 and 5 inches from base and cut between clamps.
 - Assess APGAR score.
- If childbirth is breech:
 - Support extremities or buttocks until back appears.
 - Once back has presented grasp iliac wings and apply gentle downward pressure, DOT NOT PULL. Gently swing infant in path of least resistance, anteriorly and posteriorly. Both shoulders should appear posteriorly.
 - Splint the humerus next to the body with your fingers and gently pull traction.
 - Once torso has delivered, swing legs upward until baby is vertical. This will assist in delivery of the head.
 - Follow normal delivery of infant.
- If there is a prolapsed cord or limb presentation:
 - Place mother in prone, head to knee, position.
 - Place a sterile gloved hand in the vagina and attempt to hold the baby's head away from the cord.
 - Keep the cord moist with a sterile dressing and sterile water.
 - Transport as soon as possible.



EMR AND EMT STOP

AEMT

PARAMEDIC

- Monitor EKG if indicated.
- IV/IO access Normal Saline @ TKO.
- Consider 1000 mL fluid bolus IV/IO post-delivery. May repeat x1 if patient remains hypoperfused
- Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes.
- Manage airway as appropriate for the infant. If meconium is present, use meconium aspirator to suction the airway.
- Treat OB patient seizures per Seizure protocol **OR** Pre-eclampsia/Eclampsia protocol as appropriate



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Contact medical control as soon as possible for OB complications.

Key Points/Considerations

- Transport as soon as possible.
- Assess OB patients for estimated date of delivery, Para (live births), Gravida (been pregnant), expected complications, multiple births, and pre-natal care.
- REMEMBER once baby has been delivered to dry, warm, suction, and stimulate. Newborns are very susceptible to hypothermia.
- When suctioning meconium use no more than 100 mmHg of suction.
- All afterbirth tissues need to be transported to the hospital with patient for exam.
- Gently massage the uterus after birth and allow infant to suckle as this will assist in placenta delivery.
- The APGAR should be done at 1 and 5 minute intervals.

- APGAR SCORE CARD:

	SIGN	0	1	2
A	Apperance	Blue, pale	Body pink, limbs blue	Completely pink
P	Pulse	Absent	Below 100	Above 100
G	Grimace	No Response	Grimace	Vigorous cry
A	Activity	Limp	Some Flexion	Active motion
R	Respirations	No effort	Weak, irregular	Strong cry

- If pre-term labor (24-27 weeks), do not delay transport. Transport as soon as possible and give supportive care.
- EMR and EMT may assist paramedic in applying 3 or 4 – lead ECG if indicated.

Adult Medical: OB/GYN: Pre-Eclampsia/Eclampsia

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor (4-lead) only may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

- Monitor EKG if indicated.
- IV/IO access Normal Saline @ TKO
 - Ondansetron (Zofran) for nausea IV/IO 4 mg. Repeat once in 5 minutes.



AEMT STOP

PARAMEDIC

- Severe features of **Pre-Eclampsia**:
 - SBP>140 **OR** DBP>90 and at least **one** of the following symptoms:
 - Headache
 - Mental confusion
 - Visual changes/disturbances
 - Right Upper Quadrant or Epigastric pain
 - Pulmonary edema
- **Treatment for Pre-Eclampsia**: Magnesium sulfate 4gm IV/IO (20% solution) administered over 20 minutes
 - **MED MATH: Using 60mL syringe, draw up 4gm/8mL Magnesium and then draw up 12mL NS and mix -> administer at 1mL/min**
- **Eclampsia** = pre-eclampsia symptoms plus seizures
 - **Treatment for Eclampsia**:
 - Magnesium Sulfate 4gm IV/IO (50% solution) administered over 10-20 minutes
 - **MED MATH: DO NOT DILUTE: Using a 10mL syringe, draw up 4gm/8mL in 10mL syringe and give 1mL every 2 minutes (will take 16 minutes total to administer)**
 - If seizures persist after 2-3 minutes since starting Magnesium infusion:
 - Lorazepam 1-4mg IVSP/IO or 2mg Deep IM/IN up to maximum dose of 4mg
 - IV/IO do not exceed 2mg/min
 - Must be diluted in equal amount of normal saline to give IV/IO; IM do not dilute, give deep into muscle mass
 - Midazolam 2-5mg IV/IO/IM/IN; max single dose 5mg; May repeat one time
 - **BE AWARE OF RESPIRATORY DEPRESSION**



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Additional Magnesium (if available) or for continued seizures despite administration of benzodiazepine
- If imminent delivery is possible, contact medical control **BEFORE** administration of magnesium as it is contraindicated 2 hours immediately preceding delivery

Key Points/Considerations

- Transport as soon as possible.
- Mag toxicity (progressive): hypotension -> loss of deep tendon reflexes (check patellar reflexes often) -> somnolence, slurred speech -> respiratory depression/paralysis -> cardiac arrest
- Treatment for Mag Toxicity: Stop magnesium infusion, support respiratory effort
- Placental delivery is the only definitive management for pre-eclampsia and eclampsia therefore must be aggressive and early with interventions/treatments

Adult Medical: Overdose or Toxic Exposure

EMR

- CAB and vital signs
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather
- Determine what was taken, when, and how much
- Check blood glucose level
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation)
- Suspected Opiate Overdose -
 - ☞ If symptomatic (unresponsive with respirations less than 10/min or signs of inadequate perfusion, pinpoint pupils): Naloxone (Narcan) 2mg IM repeat in 5 minutes as necessary to a maximum of 4 mg.
 - ☞ Cardiac Arrest Dose: 2mg IM repeat in 5 minutes as necessary to a maximum of 4 mg.

EMR STOP

EMT

- Apply capnography (normal reading 35-45 mmHg)
- Ingested non-caustic poisoning within 1 hour or less: (patient must be alert enough to control their own airway)
 - ☞ Activated Charcoal 50 gms PO
- May perform 12-lead (no interpretation)

EMT STOP

AEMT

- Cardiac Monitor
- Acquire 12-lead
- Monitor capnography
- IV/IO access Normal Saline @ TKO
- ☞ Symptomatic opiate overdose: Naloxone (Narcan) 0.5 – 1mg IV/IO or 2 mg IM/IN (if all other routes are unavailable) every 5 min as necessary to a maximum dose of 6 mg
- ☞ Cardiac Arrest Dose: 2 mg IV/IO

AEMT STOP

PARAMEDIC

- Organophosphate poisoning:
 - ☞ Atropine 2 mg IV/IO every 3 – 5 minutes until secretions dry
- Dystonic reaction:
 - ☞ Diphenhydramine (Benadryl) 25 – 50 mg IV/IO/IM
- TCA overdose:
 - ☞ Sodium Bicarbonate 1 mEq/kg IV/IO over 2 minutes
- Beta blocker OD:
 - ☞ Glucagon 1 mg IV/ IO
- Calcium channel blocker OD:
 - ☞ Glucagon 1 mg IV/ IO

PARAMEDIC STOP

PHYSICIAN OPTIONS

-  Additional antidotes as needed.

Key Points/Considerations

- If toxicity is uncertain or multiple substances are involved contact a physician prior to pharmacological intervention.
- Includes patients who are unconscious/unresponsive without suspected trauma or other causes, and patients with a brief loss of consciousness.
- Dystonic reaction is uncontrolled contractions of face, neck, or tongue.
- Monitor patients with TCA overdose closely for rapid changes and deterioration and changes in their condition.
- Obtain and give to ER staff all meds/containers found.
- Obtain what meds were taken (pill bottles), number of pills taken, time of ingestion and if vomiting occurred.
- S/S of Phenothiazine overdose: (Anti-psychotics including Compro, Prolixin, Thorazine, Compazine, and others)
 - Tardive Dyskinesia (involuntary movements of the facial muscles and tongue) and acute dystonic reactions.

• S/S of TCA overdose:

- Widened QRS >0.10
- Tachycardia >120 bpm
- Hypotension

• S/S of Organophosphate poisoning:

- Muscle fasciculation
- Diarrhea
- Wheezing
- Abdominal cramping
- Salivation
- Seizures
- Altered mental status
- Pinpoint pupils
- DUMBELS: Diarrhea, urination, miosis, bradycardia, bronchospasm, bronchorrhea, emesis, lacrimation, salivation, sweating, secretions, seizures.
- SLUDGE: Salivation, lacrimation, urination, diarrhea, GI cramps, emesis.

• S/S of atropinization:

- Pupillary dilation
- Tachycardia
- Flushing
- Diminished diaphoresis
- Drying of secretions

Calcium Blockers	TCA	Beta Blockers	Organophosphates
Verapamil	Amitriptyline	Inderal	Disulfoton
Cardiazem	Amoxapine	Lopressor	Tetraethyl Pyrophosphate
Nifedipine	Clomipramine	Tenormin	Parathion
Nicardipine	Desipramine	Corgard	Phosdrin
Nimodipine	Dothiepin	Blocadren	Phorate
Felodipine	Imipramine	Cartrol	Doumaphos
Amlodipine	Lofepamine	Visken	Chlorpytigos
Bepidil	Maprotiline	Sectral	Trichlorfon
Flunarizine	Nortriptyline	Normodyne	Ronnel
Isradipine	Protriptyline	Betapace	Diazinon
Nitrendipine	Trimipramine	Zebeta	Malathion

List is not all inclusive.

Adult Medical: Pain Management

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.

 **EMR STOP**

EMT

-  Acetaminophen 640mg/20mL oral solution PO (160mg/5mL) or two 325 mg (650mg) tablets PO

 **EMT STOP**

AEMT

- Cardiac Monitor.
- IV/IO access Normal Saline @ TKO.
 -  Ondansetron (Zofran) 4 mg IV/IM/IO for nausea. Repeat as necessary once in 5 minutes.
- Utilize one of the following as necessary:
 -  Morphine 2 mg IV/IO as necessary every 5 minutes maximum 10 mg (burns maximum dose is 20 mg.)
OR IM/IN 5 mg Repeat once in 10 minutes.
 -  Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5min PRN at ½ initial dose, up to a total dose of 3mcg/kg
 -  Dilaudid 0.25 mg – 1 mg IV/IO/IM/IN every 5 minutes as necessary. Maximum 4 mg.

 **AEMT STOP**

PARAMEDIC

- Toradol 30mg IV/IO/IN or 60mg IM (Geriatric 65+ is ½ the dose).
 - May be used in conjunction with narcotic.

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Additional doses of pain medication

Key Points/Considerations

- For patients with:
 - Severe burns without hemodynamic compromise.
 - Isolated extremity injuries such as fractures or dislocations with severe pain.
 - Suspected hip fractures should be treated as extremity injuries.
 - Shoulder injuries should be treated as extremity injuries.
 - Kidney Stones
 - Abdominal pains- stable and unstable.
 - Back Pains
 - Multi-systems trauma (without suspected head injury)
- For all other painful conditions, providers must consult medical control for orders.
- Contraindications to standing order pain management: altered mental status secondary to a suspected head injury, hypoventilation, hypoperfusion.
- Use caution and risk/benefits ratio with severely intoxicated/obtunded patients with traumatic injuries.
- SPO2 and cardiac monitoring required with any narcotic administration.
- **Acetaminophen:** Ensure pt has not had Tylenol, acetaminophen-containing medications such as cold medicines or narcotics within the last 4 hours

Adult Medical: Acute Agitation/ Excited Delirium Syndrome

EMR

EMT

- CAB and Vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check Blood Glucose. If not within normal limits treat per Diabetic Protocol.
- Physical restraints as needed (If possible, strongly consider deferring until chemical restraint is applied)
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation)



EMR AND EMT STOP

AEMT

- Cardiac Monitor
- IV/IO access Normal Saline normal saline only after chemical and physical restraints.
 - 1000 CC normal saline if excited delirium is suspected.



AEMT STOP

PARAMEDIC

- Haloperidol 5mg IM (if older than 65y/o give 2.5mg IM)
 - Utilize one of the following in conjunction with Haloperidol:
- Midazolam 5 mg IM Repeat once as necessary every 5-10 minutes up to 10 mg.
- Lorazepam 2 mg IM Repeat once as necessary in 10 minutes to maximum of 4 mg.
 - IM do not dilute, give injection deep into muscle mass.



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Additional Lorazepam
- Additional Midazolam
- Additional IV normal saline bolus

Key Points/Considerations

- If patient is physically restrained, chemical restraints should be strongly considered.
- Be aware for malignant hyperthermia in excited delirium patients, use aggressive external cooling measures.
- Repeat vitals every 5 minutes.
- Under no circumstances are patients to be restrained in the prone position.
- See restraint procedures.
- SPO2, cardiac monitoring, and end tidal CO2 monitoring is required with any chemical restraint administration.
- Consider all possible medical/trauma causes for patient behavior.
- If patient in excited delirium suffers cardiac arrest, consider fluid bolus and sodium bicarbonate early.

Adult Respiratory Emergencies: Asthma/Bronchospasms/COPD

EMR

- CAB and vital signs.
- Monitor SPO2 (obtain room air saturation if possible, prior to oxygen)
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose. If out of normal limits treat per Diabetic Protocol.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).

 **EMR STOP**

EMT

- Cardiac monitor: Acquire 12-lead (no interpretation)
- Determine if patient has been given their own asthma medications.
 - Albuterol 2.5 mg in 3 mL (unit dose) + Ipratropium 0.5 mg in 2.5 mL (unit dose) mixed, via nebulizer. Repeat, as necessary.

 **EMT STOP**

AEMT

- Cardiac Monitor: Acquire 12-lead.
- IV/IO access Normal Saline @ TKO.
- Medication options:

 **AEMT STOP**

PARAMEDIC

- Magnesium Sulfate 2 gms IV/IO over 20 minutes. Dilute into a 10% solution
 - **MED MATH: Draw up 2Gm/4mL Mag w/ 16mL NS into 60mL syringe for 10% solution; Administer 1mL/min**
- Methylprednisolone (Solu-Medrol) 125 mg IV/IO
- Consider 0.5-1 mg Lorazepam IV/IO/IM/IN **OR** 2mg Midazolam IV/IO/IM/IN for extreme anxiety to aid patient in accepting the CPAP mask.
- Epinephrine 1:1,000 dose 0.3 mg IM; Use with EXTREME caution in patients with cardiac history

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

- Repeat Epinephrine 1:1,000 dose 0.3 mg IM or for order for 1:10,000 dose 0.5 mg IV/IO.

Key Points/Considerations

- Position patient sitting or in position of comfort.
- Absence of breath sounds can be indicative of status asthmaticus. Be prepared for imminent respiratory arrest.
- Presence of wheezing is not only an indicator of Asthma/COPD. Consider pulmonary edema, pneumonia, upper airway obstruction, anaphylaxis, and acute allergic bronchospasm.
- Epinephrine should be reserved for those who are unable to generate adequate tidal volume to deliver aerosolized drug to the bronchial tree.
- Draft nebulizer with BVM
- CPAP: follow CPAP procedure.
- End-tidal CO₂ nasal cannula.
- Medication intervention other than oxygen - patient must be transported.
- Monitor heart rate closely. When administering Albuterol, contact a physician if heart rate increases to >150 bpm.

Adult Respiratory Emergencies: Pulmonary Edema/CHF

EMR

- CAB and vital signs.
- Monitor SPO2 (obtain room air saturation if possible prior to oxygen).
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Place patient in fowler's position.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).

 **EMR STOP**

EMT

- CPAP if tolerated and meets indications.
- Acquire 12-lead (no interpretation)

 **EMT STOP**

AEMT

- Cardiac Monitor.
- Acquire 12-lead.
- IV/IO access Normal Saline @ TKO.
- Nitroglycerin 0.4 mg sublingual (prior to CPAP application) if B/P is > 100 mmHg systolic. Repeat every 3 – 5 minutes
- Nitro-Paste 1-inch of 2% ointment topically.
- Morphine Sulfate (for chest pain) at 2 mg IV/IO over 1 minutes or 5mg IM/IN. Repeat in 5 minutes as necessary to a total dose of 10 mg.
 - Initial systolic B/P must be above 120 mmHg.
 - Patient must be conscious with adequate ventilatory effort.

 **AEMT STOP**

PARAMEDIC

- Furosemide (Lasix) 40 mg IVSP/IO max given over at least two minutes
 - 1mg/kg if already prescribed Lasix.
- Consider administering Lorazepam 0.5-1 mg IV/IO/IN/IM **OR** 2mg Midazolam IV/IO/IM/IN to assist relieving anxiety of the CPAP mask.

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Additional medications.

Key Points/Considerations

- Position patient in fowler's position.
- CPAP: follow CPAP procedure.
- Capnography can be monitored while using CPAP; Ensure applying prior to placement of CPAP mask.
- Monitor B/P, pulse oximetry and signs of respiratory depression.

Adult Medical: Seizures

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose. If out of normal limits treat per Diabetic Protocol.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation)



EMR STOP

EMT

- Acetaminophen 640mg/20mL PO (160mg/5mL or 2/ 325mg tablets total of 650mg) **WHEN FEVER IS PRESENT (TEMP >100.4°F OR 38°C and is alert and can manage own airway)**



EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- Administer **one** of the following within 3 minutes of seizure activity.
 - Lorazepam 1 – 4 IVSP/IO or 2 mg deep IM/IN as necessary up to a maximum dose 4 mg.
 - IV/IO not to exceed 2 mg/minute
 - Must be diluted in an equal amount of Normal Saline to give IV/IO.
 - IM do not dilute, give injection deep into muscle mass.
 - Midazolam 2-5 mg IV/IO/IM/IN maximum single dose of 5 mg. May repeat one time.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional benzodiazepine doses

Key Points/Considerations

- Do not delay medication administration. Consider IN or IM prior to IV access.
- Protect the patient and EMS crew from injury during the seizure.
- If seizure d/t Eclampsia, see OB/GYN Pre-eclampsia/Eclampsia Protocol
- Acetaminophen: Ensure pt has not had total dose of >650mg within the last 4 hours (e.g. Tylenol, acetaminophen-containing medications such as cold medicines, narcotics)

Adult Medical: Sepsis

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose level if equipped.
- Acquire 4-lead (No interpretation).
- Obtain temperature.

 **EMR STOP**

EMT

- Acquire 12-lead (No interpretation).
- Apply/monitor EtCO₂ capnography.
 - Acetaminophen 640mg/20mL oral soliton PO (160mg/5mL) or 2/325 mg (650mg) tablets PO **WHEN FEVER IS PRESENT (TEMP >100.4°F OR 38°C)**

 **EMT STOP**

AEMT

- Acquire/Interpret 12-lead ECG.
- Large-bore IV access or IO if unable to establish IV access AS SOON AS POSSIBLE.
- Normal Saline infusion at rate of 30mL/kg/hr in boluses using pressure bag.
 - May repeat x4 if no pulmonary edema up to max of 2000mL.
 - Boluses may be given in rapid succession if SBP <90mmHg.

 **AEMT STOP**

PARAMEDIC

BEFORE STARTING PRESSOR AGENTS, MUST HAVE ENTIRE FLUID BOLUS OF 30ML/KG INFUSED

- Dopamine infusion 2-20 mcg/kg/min IV/IO if hypotensive refractory to normal saline infusions. Titrate to effect.
- **MED MATH:** Add one vial (200mg/5mL) Dopamine to 120mL NS in a buretrol for a total of 200mg/125mL = 1600mcg/mL; Drip rate is figured same as with pre-mixed bag
- Epinephrine push-dose pressor 10-20mcg every 2 min, titrated to MAP>65mmHg
- **MED MATH:** Prepare 10mcg/mL (1:100,000) Epi by adding 1mL of 1:10,000 Epi (0.1mg/mL) to 9mL NS; Administer 10-20mcg boluses (1-2mL) every 2 minutes

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- Focus on rapid intervention, IV hydration, and early notification of concern for potential sepsis pt.
- Concern for any new or worsening infection can include: reported fever, shaking chills, sweateness, new cough, difficulty or less than usual urination, unexplained or new AMS, flushed skin, pallor, new rash or mottling.
- Vital signs should be obtained every 5 minutes.
- Acetaminophen: Ensure pt has not had total dose of >650mg within the last 4 hours (e.g. Tylenol, acetaminophen-containing medications such as cold medicines, narcotics)

Adult Medical: Stroke

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose. If out of normal limits treat per Diabetic Protocol.
- Determine at what point signs and symptoms appeared.
- If onset of symptoms is less than 6 hours or a “wake up stroke,” notify appropriate stroke center (Via Christi St Francis or Wesley Medical Center). Transport immediately.
- If onset is within 24 hours patient should be taken to a stroke center (Via Christi St. Francis or Wesley Medical Center) as a non-stroke alert patient.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

PARAMEDIC

- Monitor EKG.
- Acquire 12-lead.
- IV/IO access Normal Saline @ TKO.
 - If tachycardic and blood pressure is < 90 systolic, fluid bolus up to 1000 cc Normal Saline IV to maintain blood pressure above 90.
- Secure airway as needed.
- Treat seizure per seizure protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Contact Medical Control as soon as possible to give Stroke Alert.
- Limit scene time.
- Repeat vitals every 5 minutes if patient meets stroke criteria.
- All strokes that are within the 6 hour or “wake up stroke” time limit are Code Red. Transport regular traffic unless patient is hemodynamically unstable.
- Cincinnati Stroke Scale:
 - Speech – repeat: “You can’t teach an old dog new tricks.” Correct words and no slurring.
 - Facial Droop – Have patient smile.
 - Arm Drift – Have patient close eyes and hold arms out straight for 10 seconds without unequal movement or drifting.

Environmental

Adult Environmental: Carbon Monoxide Poisoning

EMR

- Remove patient from the source.
- CAB and vital signs.
- Airway management and high flow oxygen 15 LPM via non-rebreather.
- Consider ventilatory assistance for severe dyspnea or hypoventilation.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).

 **EMR STOP**

EMT

- Obtain CO reading by using the RAD 57 (Rescue 410, #420/Duty Officer vehicle, or the X-series monitor).

 **EMT STOP**

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- Treat seizures and cardiac dysrhythmias per protocols.

 **AEMT AND PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- All CO poisoning patients will be transported to Wesley Med Center for hyperbaric oxygen treatment.
- All patients with suspected CO poisoning are required to have high flow O2 initiated and maintained.
- S/S: dull headache, weakness, dizziness, N/V, shortness of breath, confusion, blurred vision, loss of consciousness
- SpO2/Pulse oximeter will be inaccurate.
- *100 percent oxygen reduces the elimination half-life of CO to 74 minutes, i.e. a SPCO 10% in theory would decrease to 5% after 74 minutes of oxygen @ 15 lpm via NRB.
- **In pregnant patients the CO levels of the fetus is double of the mother's

Level	Treatment
0-5%	Normal in Non-smokers
5-10%	Normal for smokers Non-smokers, assess for s/s • Treat w/ high flow O2 for 20 min * • If s/s still present, consider transport
10-15%	(In any patient) assess for s/s • Treat w/ high flow O2 • Transport to WMC
>10%	Pregnant patients** • Treat w/ high flow O2, immediate transport to WMC
>15%	(In any patient) • Treat w/ high flow O2, immediate transport to WMC

Adult Environmental: Hyperthermia

EMR

EMT

- Remove from heat source.
- CAB and vital signs.
- Obtain temperature.
- > 103 degrees - cool patient with ice packs to the groin, neck, and axillae. Cool water and fanning of the patient is also effective. (Remove when body temperature is 100 degrees.)
- Remove clothing.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac monitor (4-lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

- Monitor EKG if indicated.
- IV/IO access.
 - Normal Saline 500 mL up to 2000 mL's with hypovolemia and hemodynamic compromise. Assess breath sounds for tolerance.
 - Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes.



AEMT STOP

PARAMEDIC

- Treat seizures per seizure protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Use caution with cooling as it can cause shivering or hypothermia.
- Vigorous fluid administration may produce pulmonary edema.

Adult Environmental: Hypothermia

EMR

EMT

- Remove from cold environment and begin warming with blankets.
- CAB and vital signs (assess for 30-45 seconds)
- Obtain temperature.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose. If out of normal limits treat per Diabetic Protocol.



EMR AND EMT STOP

AEMT

PARAMEDIC

- IV/IO access.
 - Warm IV fluids up to 2000 mL
- Monitor EKG if indicated.
- If temperature is below 86 degrees F, CPR and one defibrillation at 200 joules if indicated until temp is above 86 degrees.
- Follow CPR protocol if patient is pulseless, apneic, and temperature is above 86 degrees F, but lengthen intervals of IV/IO medications until temperature is above 93 degrees F.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Prevent further heat loss in patients < 86 degrees; temperatures > 93 degrees may be re-warmed.
- Intubation should be avoided in the severely hypothermic patient with spontaneous respirations unless urgently needed using extreme caution.
- All hypothermic patients triaged code blue will be transported.

Trauma

Adult Trauma: Burns

EMR

EMT

- Stop the burning process.
 - Remove necessary clothing leaving all stuck clothing.
 - Remove rings, bracelets, and constricting bands.
 - Flush briefly with cool water if still burning (chemical burns only)
- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Maintain body temperature.
- Chemical burns: irrigate with copious amounts of water or saline for 20 minutes. EXCEPTION: Phosphorus and Lime Powder -> Brush off dry powder chemicals before flushing with water.
- Decontaminate all patients exposed to hazardous materials before entering the ambulance.
- Elevate extremity in position of function to minimize edema, monitoring neurovascular status.
- Treat for shock and hypothermia.
- Cover burns with dry sterile dressings.
- Cover patient with burn sheet.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

PARAMEDIC

- Monitor EKG if indicated.
- IV/IO access Normal Saline
 - ☞ Per ABLS Guidelines: patients age 14+ administer 500 ml/hour.
 - ☞ Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes.
- ☞ Morphine Sulfate 5 mg IV/IO/IN/IM. Repeat every 3-5 minutes as necessary to a maximum of 20 mg.
- ☞ Morphine allergy/unavailable: Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose 100mcg every 5 minutes PRN at ½ initial dose, up to total dose of 3mcg/kg
- ☞ Morphine allergy/unavailable: Dilaudid 0.25-1 mg IV/IO/IM/IN every 3-5 minutes to a maximum of 8 mg



AEMT STOP AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional narcotic pain relief

Key Points/Considerations

- Remove patient from exposure without compromising crew safety.
- Electrical Burns: consider spinal immobilization and assess for extremity fractures.
- Electrical, Flame, or scald burns should be covered with dry sterile sheets and dressings.

Adult Trauma: Chest / Abdominal Injuries

EMR

EMT

- CAB and vital signs.
 - Record vital signs every 5 minutes on critical patients.
- Stabilize life-threatening wounds. (Control hemorrhage, cover sucking chest wounds, etc.)
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if indicated.
- Moist sterile dressing to abdominal evisceration.
- Penetrating objects should be stabilized – DO NOT REMOVE THEM.
- Cardiac Monitor 4-Lead only maybe applied and printed (No interpretation)

EMR AND EMT STOP

AEMT

-  Cardiac monitor.
- IV/IO access.
 -  Normal Saline @ TKO or up to 2000 mL bolus to maintain systolic BP > 90. Assess lung sounds for tolerance.
 -  Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes.
 -  Morphine Sulfate for pain 2 mg IV/IO or 5mg IM/IN. Repeat as necessary every 3-5 minutes maximum 10 mg
 -  Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose of 100mcg; Repeat in 5 minutes PRN at ½ initial dose, up to total dose of 3mcg/kg

AEMT STOP

PARAMEDIC

- Chest injury with tension pneumothorax.
 - Chest decompression with available pneumothorax kit.
 - Needle decompression with 14 ga catheter if pneumothorax kit is unavailable
 - Code Blue Trauma – (bilateral) prior to termination of arrest.
- Consider sedation in conscious patient prior to decompression.
 -  Midazolam 2 mg IV/IO or 5 mg IM/IN.
- Crush injury for an extended time > 45 minutes.
 -  Consider Sodium Bicarbonate 1 mEq/kg IV/IO.

PARAMEDIC STOP

PHYSICIAN OPTIONS

-  Relief of non-tension pneumothorax.
-  Additional Normal Saline bolus.

Key Points/Considerations

- Target scene time is <10 minutes if trauma alert.
- Criteria for chest decompression
 - Absent or decreased breath sounds
 - Tracheal deviation (late sign)
 - Hypotension
 - Respiratory distress
 - Jugular venous distension (late sign)
 - Difficult BVM ventilations after ET tube confirmation ; O₂ saturation remains < 93% on 100% oxygen.

Adult Trauma: Drowning/Near Drowning

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Remove wet clothing and cover with dry blankets.
- Spinal motion restriction if indicated preferably before removal from water.
- Cold water – treat per hypothermia protocol.
- Cardiac arrest – treat per indicated cardiac, seizure, and trauma protocols.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

- Cardiac monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- Treat cardiac dysrhythmias per protocol.



AEMT STOP

PARAMEDIC

- Treat seizures per protocol
- Sodium Bicarbonate 1 mEq/kg IV/IO up to 100 mEq for suspected acidosis.



PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Resuscitation should be considered if submersion has been less than 2 hours, especially if water temperature is below 70 degrees.
- Obtain the following history:
 - Submersion time.
 - Water temperature and depth.
 - Clean or dirty water and any known pollutants.
 - Mechanism of injury.
 - Type of clothing and flotation device.
 - Location last seen.

Adult Trauma: Head Injuries

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if indicated.
- Penetrating objects should be stabilized – DO NOT REMOVE THEM.
- Cardiac Monitor (4-lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

- Monitor if indicated.
- IV/IO access Normal Saline @ TKO.
 - Systolic blood pressure < 90 and tachycardia present administer 500 ml Normal Saline bolus. Repeat as necessary to maintain > 90 systolic blood pressure. Auscultate to ensure tolerance.
- Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes as needed.



AEMT STOP

PARAMEDIC

- Combative head injuries and/or seizures utilize one of the following:
 - Midazolam 2 mg IV/IO or 5 mg IM/IN. Repeat as necessary to maximum of 10 mg.
 - Lorazepam 1-2 mg IV/IO/IM/IN; repeat as necessary up to maximum of 6 mg in 5 minutes.
 - IV/IO do not exceed 2 mg/minute.
 - Must be diluted in equal amounts of Normal Saline to give IV/IO.
 - IM do not dilute, give injection deep into muscle mass.



PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Scene time target is 10 minutes.
- Record vital signs every 5 minutes if patient is critical.

Adult Trauma: Hemorrhage Control Management

EMR

EMT

- Correct circulation life threats with correct treatment procedures.
 - If hemorrhaging cannot be controlled by direct pressure, elevation or pressure points, apply tourniquet superior to hemorrhaging site. Distal pulses should be absent after application. Consider hemostatic agent application.
- CAB and vital signs, (vitals taken every five minutes).
- Spinal motion restriction if indicated.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor 4-Lead only may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor
- 12-lead for suspected cardiac contusions.
- IV/IO access
 - Normal Saline @ TKO or up to 2000 ml bolus to maintain systolic BP >90. Assess lung sounds for tolerance.
- Pain control - see pain management protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

- Additional Normal Saline bolus.

Key Points/Considerations

- Contact Medical Control as soon as possible.
- Transport immediately to an appropriate Level 1 Trauma Center.
- See patient transport policy to determine appropriate Trauma Center.
- Tourniquets are only used for life threatening peripheral hemorrhages when all other means of hemorrhage control have been unsuccessful.
- Tourniquets should only be removed by a physician in an appropriate hospital setting, preferably a level 1 trauma center.

Adult Trauma: *Isolated*

EMR

- CAB and vital signs.
- Stabilize fractures and stop bleeding.
- Spinal motion restriction if indicated.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac monitor (4-lead only) may be applied and printed (no interpretation).



EMR STOP

EMT



See pain management protocol for pain management.



EMT STOP

AEMT

PARAMEDIC

- Monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- See pain management protocol for pain management.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- All fractures should be stabilized.
- Ensure that pulses, motor, and sensations are checked in extremities before, after, and whenever the patient is moved.
- Traction splints are to be used only on mid shaft femur fractures.

Adult Trauma: Multi-Systems

EMR

EMT

- Correct any life threats with correct treatment procedures.
- CAB and vital signs (vitals taken every five minutes).
- Spinal motion restriction if indicated.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac monitor (4-Lead only) may be applied and printed (no interpretation).



EMR AND EMT STOP

AEMT

- Cardiac monitor.
- 12-lead for suspected cardiac contusions.
- IV/IO access.
 - Normal Saline @ TKO or up to 2000 ml bolus to maintain systolic BP > 90. Assess lung sounds for tolerance.
- Pain control - see pain management protocol.



AEMT STOP

PARAMEDIC

- Tranexamic Acid- Adult (>40kg): Mix 1000mg (1 gram) of TXA in 100 mL of 0.9% Normal Saline; Infuse over 10 minutes (5gtt/3seconds OR 25gtt/15seconds)
 - **Any patient at high risk for ongoing massive traumatic hemorrhage and meeting one or more of the following criteria:**
 - Systolic BP < 90mmHg, and/or physical shock appearance (confusion, AMS, cool extremities, etc.)
 - Heart rate >110bpm



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Additional Normal Saline bolus.

Key Points/Considerations

- Contact Medical Control as soon as possible.
- Transport immediately to an appropriate Level 1 Trauma Center.
- **Ensure receiving facility is informed PTA of TXA administration**
- See patient transport policy to determine appropriate Trauma Center.

Adult Trauma: Shock/Hypoperfusion

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac monitor (4-lead only) may be applied and printed (no interpretation).

 **EMR AND EMT STOP**

AEMT

- Cardiac monitor.
- IV/IO access.
 - Normal Saline @ TKO or up to 2000 mL bolus to maintain systolic BP >90. Assess lung sounds for tolerance.
 - Ondansetron (Zofran) for nausea 4 mg IV/IO. Repeat once in 5 minutes.

 **AEMT STOP**

PARAMEDIC

- Dopamine 2-20 mcg/kg/minute IV/IO infusion.
 - **MED MATH:** DOPAMINE: Add one vial (200mg/5mL) Dopamine to 120mL NS in a buretrol for a total of 200mg/125mL = 1600mcg/mL; Drip rate is figured same as with pre-mixed bag

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Additional Normal Saline bolus.

Key Points/Considerations

- For patients with hypoperfusion due to trauma, bleeding, vomiting, diarrhea, or sepsis, fluid bolus must be administered **first**.
- Contact a physician if you suspect cardiogenic shock.
- Contact receiving hospital early.

Adult Trauma: Spinal Motion Restriction

EMR
EMT
AEMT
PARAMEDIC

- Perform an appropriate trauma assessment.
- Determine mechanism:
 - **Low Risk:** (Requires **all** of the following)
 - Patient reliable and competent?
 - Normal level of alertness (GCS 15)
 - No evidence of clinical intoxication.
 - No distracting injury.
 - Able to communicate adequately.
 - Lack of point tenderness on palpation of entire length of spinous processes.
 - Lack of focal deficit (motor or sensory) on neuro exam.
- **No** to any of the above -> Spinal Motion Restriction is required.
 - **High Risk:** (Requires at least one of the following)
 - Ejection from a vehicle
 - Motorcycle crash >20 mph
 - Auto vs. pedestrian or bike at > 20 mph
 - Axial load to head
 - Fall from 3x patient's height
- **Yes** to any of the above Spinal Motion Restriction is required.
- Patient documentation must specifically include all of the above findings.

 **EMR, EMT, AEMT, and PARAMEDIC STOP**

PHYSICIAN OPTIONS



Key Points/Considerations
• Transport as soon as possible.
• Penetrating trauma patients without neuro deficits do NOT require spinal motion restriction.
• Long backboard or KED may still be used in patient extrication even if spinal motion restriction is not required by protocol, but should be removed as soon as possible.

Pediatric Emergencies

- IO access should be attempted if no peripheral access can be obtained after 1 attempt or 90 seconds in patients with cardiovascular collapse.
- Protocols may be used fluidly.
- Buretrol is required for all IVs that have fluids running on patients < 9 years of age.
- Have a Broselow Pediatric Tape or similar device available to accurately determine the correct medication dosage. Deviate from color codes to match actual weight. At no time should the pediatric dosage exceed adult dosage.
- Normal Vital Signs for Infants and Children:

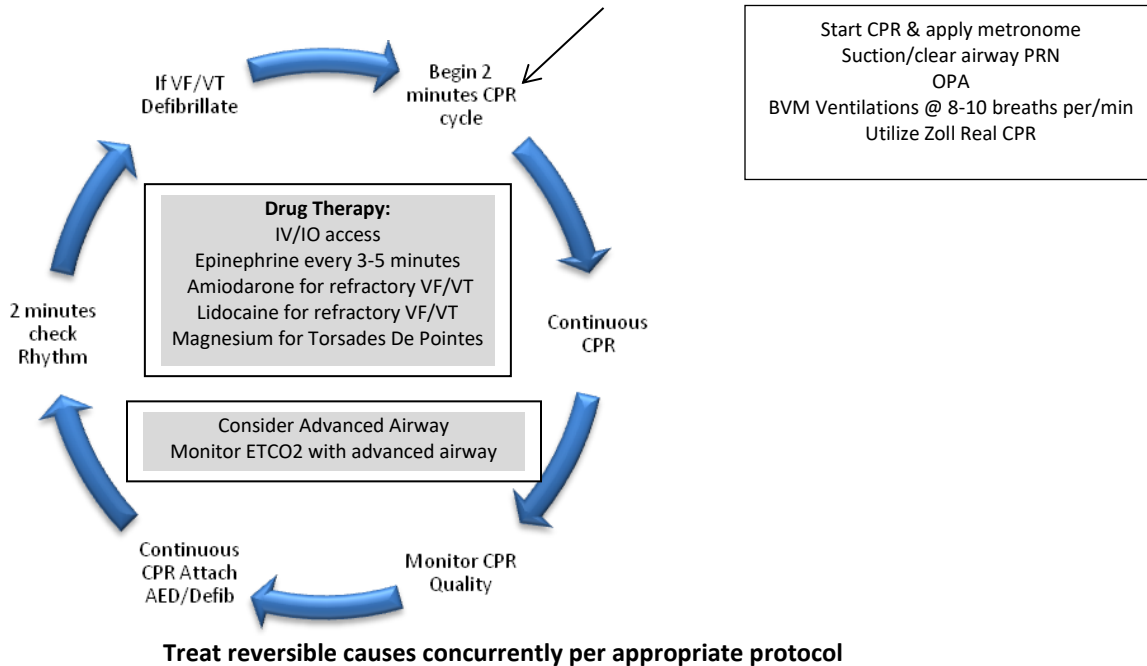
Age	Respirations	Pulse	Systolic BP
Newborn	30-60	100-180	>60
Infant (<1 year)	30-60	100-160	>60
Toddler (1-3 years)	24-40	90-150	>70
Preschooler (3-5 years)	22-34	80-140	>75
School-aged (6-8 years)	18-30	70-120	>80

From: American Academy of Pediatrics, Pediatric Education for Pre-hospital Professionals.

Pediatric Cardiac

Pediatric Cardiac Arrest: *Cardiopulmonary Resuscitation (CPR)*

Indications: All pediatric cardiac arrest situations.



CPR Quality

- Push fast (110/min), allow complete recoil.
- Pediatric/Neonate chest compressions should be $>1/3^{\text{rd}}$ depth of the anterior-posterior diameter of chest.
- Minimize interruptions in chest compressions; interruptions should be infrequent and for no longer than 10 seconds.
- Avoid excessive ventilation. Ventilate at a rate of 8-10 breaths per minute.
- 1-rescuer Pediatric/Neonate CPR: 30:2 compression-ventilation ratio (continuous compressions upon adequate resources)
- 2-rescuer Pediatric/Neonate CPR: 15:2 compression-ventilation ratio (continuous compressions upon adequate resources)
- After advanced airway placement, perform continuous chest compressions and ventilate at 8-10 breaths per minute.
- Rotate compressor every 220 or when compression quality decreases as indicated by Real CPR or capnography trending.

Electrical Therapy

- Biphasic defibrillation: Pediatric/Neonate 1st dose 2j/kg. All repeat doses 4j/kg.
- AED settings are preset by manufacturer.

Medication Therapy

- Normal Saline IV/IO :
 - Pediatric 20 ml/kg. May repeat once if needed
- Pediatric/Neonate: Epinephrine 0.01 mg/kg IV/IO every 3-5 minutes
- Pediatric/Neonate Pulseless VF/VT options:
 - Magnesium 25-50 mg/kg IV/IO diluted into 10% solution; Max single dose 2 gm (Torsades De Pointes Only)
 - Amiodarone 5 mg/kg IV/IO repeat once in 3-5 minutes.
 - Lidocaine 1 mg/kg IV/IO may repeat once in 3-5 minutes.
- Consider Sodium BiCarb 1mEq/kg IV/IO after 10 minutes. Repeat once in 5 minutes 0.5 mEq/kg (Asystole, PEA, VF, VT)

Pediatric Cardiac Arrest

EMR

EMT

- Begin CPR and attach AED
- Airway management and appropriate oxygen therapy via BVM.

EMR AND EMT STOP

AEMT

- Determine and document rhythm in two leads.
- Biphasec defibrillation:
 - First dose: 2j/kg.
 - Repeat doses: 4j/kg.
- Normal Saline IV/IO at 20mL/kg; May repeat once PRN
- Epinephrine 1:10,000 0.01 mg/kg IV/IO every 3-5 minutes; **No more than 5 Epi total regardless of rhythm changes**
- Pediatric/Neonate Pulseless VF/VT options:
 - Amiodarone 5 mg/kg IV/IO; repeat once in 3-5 minutes.
 - Lidocaine 1 mg/kg IV/IO; repeat once in 3-5 minutes

AEMT STOP

PARAMEDIC

- Magnesium 25-50 mg/kg (50% solution) IV/IO diluted into 10% solution; Max single dose 2 gm (Torsades)
 - **MED MATH: Draw up 1Gm/2mL into 8mL NS = 1Gm/10mL = 1000mg/10mL; Example: If you have 20kg pt, dose is 500mg -> 5mL**
- Consider Sodium BiCarb 1mEq/kg IV/IO after 10 minutes; Repeat once in 5 minutes 0.5 mEq/kg (Asystole/PEA/VT/VF).

PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Early physician contact.
- Confirm asystole in more than 1 lead
- Perform CPR for up to 3 minutes between medication doses.
- IO access should be attempted if no peripheral access can be obtained after 1 attempt or 90 seconds in patients with cardiovascular collapse
- **DEXTROSE MED MATH:** For patients <40 kg: Dilute D50 to concentration of D25: Draw up 25 ml of D50 and dilute with normal saline to total volume of 50 ml (Equals 1 gm/2 ml)
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- Do not interrupt compressions for placement of an advanced airway. Avoid excessive ventilation.
- Insert NG tube for gastric distension if needed.
- Capnography to confirm and monitor ET tube placement and quality of CPR.
- Consider and treat causes of PEA:

➤ Hypoxemia	(hyperventilate)
➤ Hypovolemia	(fluid challenge 20 ml/kg bolus)
➤ Hypoglycemia (<60 mg/dL)	(Dextrose)
➤ Hypothermia	(warm)
➤ Hyperkalemia/Hydrogen Ion (acidosis)	(NaHCO3 1mEq/kg with a max 100 mEq)
➤ Pulmonary Embolism	(ventilate)
➤ Tension Pneumothorax	(Needle decompression)
➤ Cardiac Tamponade	(fluid challenge)
➤ Overdose (Opiate)	(Narcan 0.1 mg/kg with a max single dose 2 mg)

Pediatric Cardiac Arrest: Management/Post Resuscitation

EMR

EMT

- CAB and vital signs.
- Continue high flow oxygen via non-rebreather or BVM as indicated.
- Check blood glucose level and treat per diabetic protocol if indicated.
- Acquire 4-lead (No interpretation).



EMR, EMT STOP

AEMT

- Acquire 12-lead
- IV/IO access:
 - Normal saline bolus 20 ml/kg repeat as necessary to maintain a B/P of at least 90 systolic. Repeat if indicated.



AEMT STOP

PARAMEDIC

- Capnography monitoring if intubated.
- Post resuscitation ventricular ectopy: watchful waiting is usually the most appropriate action while improved oxygenation takes effect and catecholamine levels return to normal.



PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional fluids.

Key Points/Considerations

- Contact Medical Control to tailor the treatment to the patient's specific needs.
- Vital signs should be recorded every 5 minutes on critical patients.
- Even mild hypotension should be avoided because it can impair recovery of cerebral function.
- Symptomatic Bradycardia – Reassess ventilations and airway. If profound bradycardia persists after reassessment, consider Cardiac Dysrhythmia/Bradycardia Protocol.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- Monitor lung sounds when administering fluid. Do **not** fluid overload the patient.

Pediatric Cardiac Arrest: Termination of Resuscitation

EMR

EMT

- Resuscitative efforts for patients in cardiac arrest should not be initiated if:
 - Patient presents with significant dependent lividity, rigor mortis, decomposition and/or injuries incompatible with life (such as decapitation).
 - There is a signed DNR order.
- For all other patients in respiratory or cardiac arrest, the EMS provider MUST start BLS care, including defibrillation, and request a Type I assist if Type II.
- All pediatric cardiac arrests from birth to 17 years that do not fall under the above code black criteria will be resuscitated for a minimum of 20 minutes on scene unless ROSC occurs. After 20 minutes of resuscitation and no ROSC the patient will be transported code blue to the nearest appropriate facility.



EMR AND EMT STOP

AEMT

PARAMEDIC

- Complete standing orders appropriate to presenting rhythm.
- **EXCEPTION:** Parent(s) or legal guardian understand and agree to termination of efforts.
- All **“TRAUMA”** pediatric cardiac arrest from birth to 17 y/o that do not meet Code Black criteria:
 - Will be assessed BLS/ALS **AND**
 - Basic airway maneuvers performed **AND**
 - Bilateral chest decompression(s) performed **AND**
 - Ages 0-1years: 22g IV needle, 2nd ICS mid-clavicular line
 - Ages 1-4years: 20g IV needle, 2nd ICS mid-clavicular line
 - Confirmation of asystole in 2 or more leads.
 - **NO RESPONSE:** Triage Code Black in the field and **DO NOT** transport.
 - **RESPONSE:** If the patient responds to efforts, continue with Code Blue Trauma protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Attempt to obtain a copy of the DNR or advanced directive to be attached to the chart.

Pediatric Cardiac: Chest Pain

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead (No interpretation).



EMR STOP

EMT

- Acquire 12-lead (No interpretation).



EMT STOP

AEMT

PARAMEDIC

- Acquire 12-lead.
- IV/IO access.
- Normal Saline TKO.
 - Consider 20 ml/kg normal saline bolus if BP < 90mmHg (or hypotensive for age)
- Follow appropriate protocol for rhythm changes.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional medications

Key Points/Considerations

- Notify receiving hospital as soon as possible if patient is a STEMI alert.

Pediatric Cardiac: Cardiac Dysrhythmia/Bradycardia

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- If heart rate is < 60 bpm with poor perfusion despite effective ventilation with oxygen, initiate chest compressions w/ compression to ventilation ratio 15:2 or 30:2 if single provider.
- Check blood glucose and treat hypoglycemia per protocol
- Acquire 4-lead (No interpretation).

EMR STOP

EMT

- Acquire 12-lead (no Interpretation)

EMT STOP

AEMT

- Acquire 12-lead EKG.
- IV/IO access.
- Normal Saline:
 - Consider 20 ml/kg bolus

AEMT Stop

PARAMEDIC

- Bradycardia:
 - Persistent or responds only transiently:
 - Epinephrine push-dose pressor: 0.01mg/kg (0.1mL/kg) every 3 minutes; **TITRATE TO MAP >65mmHg**; max single dose 10mg or 1mL; EXAMPLE: if you have 10kg pt, give 1mL
 - **MED MATH:** Prepare 10mcg/mL (1:100,000) Epi by adding 1mL of 1:10,000 Epi (0.1mg/mL) to 9mL NS;
 - Atropine 0.02 mg/kg IV/IO; minimum initial dose 0.1 mg, max initial dose 0.5mg; max total dose of 1 mg.
 - Give for increased vagal tone or cholinergic drug toxicity
 - Medication failure: Transcutaneous pacing starting @ 100 bpm or call physician for further orders.
 - Consider sedation and/or pain control prior to pacing:
 - Midazolam IV/IO 0.05mg/kg infused slowly over 1-2 minutes **OR** IM/IN 0.1mg/kg; Max single dose 5mg. Repeat as necessary up to a maximum of 10 mg.
 - Morphine 0.1 mg/kg IV/IO/IM/IN, repeat in 3-5 minutes as necessary up to maximum 10 mg.
 - Fentanyl 0.5-1 mcg/kg IV/IO/IM/IN with max initial dose 100mcg; Repeat in 5 min PRN at ½ initial dose, up to total dose 3mcg/kg
 - Dilaudid 0.01-0.02 mg/kg IV/IO/IM/IN as necessary. Maximum of 0.5 mg.

PARAMEDIC STOP

PHYSICIAN OPTIONS

Additional Fluids or Medications

Key Points/Considerations

- Early physician contact.
- Newborn/Infant bradycardic if pulse less than 80 bpm; child over 1 year of age bradycardic if pulse less than 60 bpm.
- Symptomatic includes poor systemic perfusion, hypotension, hypoxia, respiratory difficulty or altered level of consciousness
- IO access should be attempted if no peripheral access can be obtained after 1 attempt or 90 seconds in patients with cardiovascular collapse
- If bradycardia is due to increased vagal tone or primary AV block give atropine before giving epinephrine.
- Do not treat asymptomatic bradycardia. Contact medical control.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric Cardiac: Dysrhythmia / Tachycardia with a Pulse

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor 4-Lead only may be applied and printed (No interpretation of ECG rhythm)



EMR STOP

EMT

- 12-lead (no interpretation)



EMT STOP

AEMT

- Cardiac Monitor.
- Acquire 12-lead.
- IV/IO access Normal Saline TKO.
 - Normal Saline 20mL/kg bolus if systolic BP is <90 mmHg and if no signs and symptoms of fluid overload exist.
- Identify and treat possible causes.
- Unstable rhythms require synchronized cardioversion:
 - Narrow, Regular QRS: 1st dose 1J/kg; Repeat doses 2J/kg
 - Wide, Regular QRS: 1st dose 1J/kg; Repeat doses 2J/kg



AEMT STOP

PARAMEDIC

- **If patient is stable use medications first, then perform synchronized cardioversion if refractory to medications.**
- Narrow, Regular QRS (e.g. SVT):
 - Adenosine 0.1mg/kg IVFP/IO w/ NS flush, max 6mg
 - May repeat in 3 minutes at 0.2mg/kg IVFP/IO w/ NS flush, max 12mg
- Narrow, Irregular QRS (e.g. A-Fib, A-Flutter, MAT):
 - Amiodarone 5mg/kg IV/IO over 10 minutes, max 150mg/dose
 - **MED MATH:** Draw up 150mg Amiodarone into 10mL NS for 15mg/mL concentration; Example: if you have a 3kg pt, give 15mg/1mL at 0.1mL/min; 10kg pt give 50mg OR 3mL at 0.3mL/min. **Hold onto the syringe plunger while infusing to prevent backflow; will need to be given with fluids to allow for flushing between doses.**
- Wide, Regular QRS (e.g. V-Tach, SVT/A-Fib/A-Flutter w/ aberrancy):
 - Amiodarone 5mg/kg IV/IO over 10 min, max 150mg/dose
- Wide, Irregular QRS (e.g. A-Fib w/ aberrancy, WPW w/ A-Fib, MAT, Polymorphic VT/Torsades de Pointes)
 - Amiodarone 5mg/kg IV/IO over 10 min, max 150mg/dose
- Consider sedation prior to cardioversion:
 - Midazolam 0.2 mg/kg IV/IO/IM/IN infused slowly over 1-2 minutes; Max singled dose 5mg. Repeat as necessary up to a maximum of 10 mg.



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Synchronized cardioversion of irregular rhythms
- Additional or alternative medication options
 - Adenosine for SVT w/ aberrancy

Key Points/Considerations

- Contact Medical Control as soon as possible.

- Wide, regular QRS?
 - If **YES**, is there a RBBB/LBBB morphology?
 - **YES:**
 - Consider **SVT** w/aberrancy. Ask about hx + adenosine
 - If V1 has Rsr' (with left rabbit ear taller than right) then rhythm is more likely **ventricular tachycardia**.
 - **NO:**
 - Positive QRS in aVR and/or negative QRS in aVF?
 - If **YES**, is there all positive or all negative QRS complexes in V1-6?
 - If **YES**, then rhythm is most likely **ventricular tachycardia**.
 - If **NO**, consider **antidromic AVRT**.
 - If wide, regular with **NO** RBBB/LBBB morphology:
 - Consider **ventricular tachycardia** until proven otherwise.
- Wide, irregular QRS?
 - If **YES**, is there a RBBB/LBBB morphology?
 - **YES:**
 - Consider **BBB + afib**. Ask about hx.
 - If RBBB morphology, V1 **must** have rSR' morphology (right rabbit ear taller than left rabbit ear)
 - If **BBB + afib**: Fluids & treat underlying cause of tachycardia (sepsis, dehydration, etc.)
 - **NO:**
 - Consider **WPW + afib**. Do **not** treat with *adenosine*.
 - If **WPW + afib**, *cardiovert* PRN as rate can be very fast.
- Narrow, regular QRS?
 - If **YES**, does the rate fluctuate or does it stay the same constantly?
 - *Fluctuates:*
 - Consider **sinus tachycardia**
 - *Constant rate:*
 - Consider **PSVT**
 - **AVNRT** most common (tx: *adenosine*)
 - **AVRT** requires accessory pathway (*WPW*) (tx: *adenosine*)
 - **Aflutter** (look at II for flutter waves, V1 for individual P waves)
- Narrow, irregular QRS?
 - Consider **atrial fibrillation** or (less likely) **multifocal atrial tachycardia (MAT)**.
 - Afib + RVR is usually due to secondary etiology (dehydration, sepsis, dyspnea).
Treat underlying cause before administering rate control medication.

Pediatric: Newborn Resuscitation

EMR

EMT

- CAB and vital signs
- APGAR
- CPR if indicated
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead EKG (No interpretation).

EMR AND EMT STOP

AEMT

PARAMEDIC

- Monitor EKG.
- IV/IO access.
-  Normal Saline bolus 20 ml/kg IV/IO as needed
- Check blood glucose level and treat per diabetic protocol if indicated.
- Strong cry, regular respiratory effort, good tone, term gestation -> skin-to-skin w/ mom and cover w/ dry linens
- Weak cry, signs of respiratory distress, poor tone, preterm -> Position airway to sniffing position and clear as needed
 - Suction meconium with meconium aspirator and advanced airway.
- HR >100bpm:
 - Monitor for central cyanosis; Provide blow-by oxygen as needed
 - Monitor signs of respiratory distress
 - Apneic or significant distress: BVM w/ **room air** at 40-60 breaths/min
 - Consider intubation
- HR <100bpm:
 - BVM ventilations w/ **room air** at 40-60 breaths/min
 - If no improvement after 90 seconds: change to oxygen administration until HR normalizes
 - Consider ETI if ventilations are ineffective
- HR <60bpm:
 - Ensure effective ventilations w/ supplementary oxygen and adequate chest rise
 - If no improvement after 30 seconds:
 - Initiate chest compressions w/ two-thumb-encircling-hands technique
 - Coordinate compressions w/ PPV (3:1 ratio; 90 compressions and 30 breaths every minute)
 - Consider ETI
 -  Epinephrine 1:10,000 dose 0.01mg/kg IV/IO

AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

-  Additional fluid bolus.
-  Access of umbilical vein if unable to obtain venous or intraosseous access

Key Points/Considerations

- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- **Keep warm.**

Pediatric Medical

Pediatric: Abdominal Pain

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac Monitor 4-Lead only may be applied and printed (No interpretation of ECG rhythm)

 **EMR STOP**

EMT

- 12-lead (no interpretation)
 - Ondansetron (Zofran) for nausea 4mg PO may repeat x1 in 5 minutes.

 **EMT STOP**

AEMT

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO.
- Ondansetron (Zofran) for nausea 1 mg IV/IO/IM. May repeat once in 5 minutes.
- Morphine 0.1 mg/kg IV/IO/IM/IN, repeat in 3-5 minutes as necessary up to maximum 10 mg.
- Fentanyl 0.5-1 mcg/kg IV/IO/IM/IN with max initial dose 100mcg; Repeat in 5 min PRN at ½ initial dose, up to total dose 3mcg/kg
- Dilaudid 0.01-0.02 mg/kg IV/IO/IM/IN as necessary. Maximum of 0.5 mg.

 **AEMT STOP**

PARAMEDIC

- Toradol: <2yrs – 0.25mg/kg; >2yrs – 0.5mg/kg; Max dose 30mg

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS



Key Points/Considerations

- Begin transport as soon as possible.
- Transport patient in a position of comfort if possible.
- Use caution palpating patients with possible cardiac involvement, i.e. aortic aneurysms.
- **Pediatrics must be > 4 years to administer Ondansetron.**
- SPO2 and cardiac monitoring required with any narcotic administration.

Pediatric: Abuse / Neglect

EMR
EMT
AEMT
PARAMEDIC

- CAB and vital signs.
- Initiate treatment as necessary.
- If possible remove child from scene, transporting to hospital even if there is no medical reason for transport.
- Allow a parent to be with the child during transport unless a suspicion of further trauma exists or the severity of injuries warrants the separation.
- If parents refuse permission to transport, notify law enforcement to respond to scene. If patient is in immediate danger, let law enforcement handle scene.
- **AVOID ACCUSATIONS** as this may delay transport. Adult with child may not be the abuser.
- **Never** communicate via radio abuse suspicions.
- Carefully document findings and report to physicians at the hospital.
- Immediately contact the appropriate law enforcement agency to file a report. Make sure to include this in your documentation of the call.



EMR, EMT, AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- EMS care providers are MANDATORY REPORTERS by law with the duty to notify authorities of suspected child abuse. Reports must also be submitted via DCF website.
- Child abuse/neglect are widespread enough that nearly all EMTs and Paramedics will see these problems at some time. The first step in recognizing abuse or neglect is to accept that they exist and to learn the signs and symptoms:
 - Noticeable injuries not mentioned in the history.
 - History does not explain physical findings of the injury.
 - Injuries with a pattern:
 - Bite marks
 - Cigarette Burns
 - Rope (circumferential or linear marks at the ankles or wrists)
 - Loop scars (doubled-over electrical wire or telephone cord)
 - Belt marks
 - Etc.
 - Injuries to genital or perineal areas.
 - Multiple repetitive unexplained injuries.
 - Refer to Broselow Pediatric Emergency Tape.

Pediatric: Altered Mental Status/Unresponsive

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if trauma suspected.
- Check blood glucose level and treat per diabetic protocol if indicated.
- Acquire 4-lead (no interpretation)

 **EMR AND EMT STOP**

AEMT

PARAMEDIC

- Acquire 12-lead.
- IV/IO access.
-  Normal Saline TKO.
-  D25 Dextrose 0.5 gm/kg IV/IO per Pediatric Diabetic protocol followed with 20 ml/kg normal saline infusion up to 100 ml.
 -  **DEXTROSE MED MATH:** For patients <40 kg: Dilute D50 to concentration of D25: Draw up 25 ml of D50 and dilute with normal saline to total volume of 50 ml (Equals 1 gm/2 ml)
-  Narcan 0.1mg/kg IV/IO/IM/IN per the Overdose protocol.

 **AEMT AND PARAMEDIC STOP**

PHYSICIAN OPTIONS



Key Points/Considerations

- Consult Medical Control as soon as possible.
- IO access should be attempted if no peripheral access can be obtained after 1 attempts or 90 seconds in patients with cardiovascular collapse.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- Potential causes of altered mental status:
 - A – Alcohol
 - E – Epilepsy, Endocrine, Electrolytes
 - I – Insulin
 - O – Opiates and other drugs
 - U – Uremia
 - T – Temperature, Trauma
 - I – Infection
 - P – Psychogenic, Poisoning
 - S – Shock, Stroke, Space-occupying lesion, Subarachnoid hemorrhage
- Questions to consider:
 - Is patient newborn of narcotic using mother?
 - What was patient doing prior to this?
 - Has it happened before?

Pediatric Medical: Anaphylaxis/ Allergic Reactions

EMR

-  CAB and vital signs.
-  Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
 - Consider ventilator assistance for severe dyspnea or hypoventilation.
- Determine cause of reaction, time of exposure, and amount of substance patient was exposed to if possible.
 - Remove patient from source or source from patient if still present.
 - For Moderate Allergic Reaction (e.g., hives, itching, redness, stable vitals) administer:
 -  Diphenhydramine 9kg(22lbs)-20kg (43lbs)- 12.5mg PO (1 tablet)
 -  Diphenhydramine 21kg(55lbs) 36kg(80lbs) – 25mg PO (2 tablets)

For Anaphylaxis administer:

-  Epinephrine 25kg (55lbs) 1:1,000 dose 0.15 mg IM used first for shock states; Repeat x1 in 5 minutes, as necessary.

EMR STOP

EMT

-  Albuterol 2.5 mg in 3 mL (unit dose) + Ipratropium 0.5 mg in 2.5 mL (unit dose) mixed together via nebulizer for wheezing. May repeat, as necessary.

EMT STOP

AEMT

- Acquire 12-lead.
- IV/IO access.
- Normal Saline TKO.
 -  If hypotensive 20 mL/kg fluid bolus. Repeat one time to maintain BP > 90.
 -  Diphenhydramine 1 mg/kg IV/IO/IM

AEMT STOP

PARAMEDIC

-  Epinephrine 1:10,000 dose 0.01 mg/kg IV/IO; max 0.3 mg.
-  Methylprednisolone (Solu-Medrol) 1 mg/kg IV/IO push.
-  Epinephrine Push Dose Pressor 0.01mg/kg = 0.1mL/kg 1:100,000 every 3-5 minutes, max single dose 10mcg
 - **MED MATH:** Prepare 10mcg/mL (1:100,000) Epi by adding 1mL of 1:10,000 Epi (0.1mg/mL) to 9mL NS;
Example: 10kg pt, give 1mL

PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- S/S of allergic reaction: gradual onset, mild flushing, rash, or hives, mild bronchoconstriction, and normal mental status.
- S/S of anaphylaxis: sudden onset, severe flushing, rash or hives, severe bronchoconstriction, laryngospasm, anxiety, sense of impending doom, may decrease to confusion, then unconsciousness.
- Severe reactions may develop within 2-20 minutes following exposure.
- Consider concurrent use of Pediatric Respiratory Distress protocol.

Pediatric: Dehydration

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Acquire 12-lead
- IV/IO access
- With signs and symptoms of dehydration present:
 - Patient must have tachycardia > 140 bpm and at least one additional sign and symptom:
 - Dry mucous membranes
 - Skin tinting
 - Hypotension
 - Decreased LOC related to suspected dehydration
 - High Fever
 - Severe vomiting or diarrhea
 - Heat related emergency
 - Pointed and peaked T-waves
- Normal Saline 20 ml/kg bolus; may repeat x1.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Monitor lung sounds closely throughout bolus.
- Monitor SPO2.
- Excessive rapid fluid replacement can cause an imbalance in electrolytes and other lab values. This may increase the patient's stay at the hospital. With this said minor dehydration will not be treated in the field. The above criteria must be met to treat dehydration in the field.

Pediatric: ***Diabetic Emergencies***

EMR

EMT

- Airway management and appropriate oxygen therapy.
- Check blood glucose level and treat per diabetic protocol if indicated.
- Cardiac monitor (4-lead only) may be applied and printed if indicated (no interpretation)
 - If blood glucose is known to be low and patient is able to self-administer and swallow on command, give oral glucose one unit dose (24 gms).

 **EMR AND EMT STOP**

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO
- Hyperglycemia: Blood glucose above 250 mg/dl and ONLY if signs of dehydration are present:
 - Normal Saline bolus 20 ml/kg IV/IO.
- Hypoglycemia:
 - Blood glucose <60 administer Dextrose D25 at 0.5mg/kg IV/IO followed with 20 ml/kg normal saline infusion up to 100 ml
 - **DEXTROSE MED MATH:** Dilute D50 to a concentration of D25: draw up 25 ml of D50 and dilute with normal saline to total volume of 50 ml. Equals 12.5gms/50mL or 0.25gm/mL
 - If unable to obtain IV, administer Glucagon:
 - 0.5 mg IM for Broselow colors gray-white.
 - 1 mg IM for Broselow colors blue-green.
 - If IM Glucagon unsuccessful establish IO for Dextrose.

 **AEMT AND PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Additional Dextrose
-  Additional Normal Saline bolus

Key Points/Considerations

- DKA has slow onset lasting 12-24 hours. S/S include polyuria, polydipsia, polyphagia, warm, dry skin and mucous membranes, nausea/vomiting abdominal pain, tachycardia, Kussmaul's respirations, fruity breath odor, decreased mental function, or coma.
- If the patient's parent/guardian wishes to refuse transport:
 - Must be present.
 - Patient is alert.
 - Blood Glucose greater than 60 mg/dl.
 - No other complaints or complications.
- Instructed to give complex carbohydrates, contact physician, and to recheck blood glucose regularly.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Nausea and/or Vomiting

EMR

- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead. (no interpretation)



EMR STOP

EMT

- Ondansetron (Zofran) ODT 4mg PO. Repeat once in 10 minutes if needed.



EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO.
- Ondansetron (Zofran) 1 mg IV/IO/IM for nausea as necessary. Repeat one time in 5 minutes.
- Consider dehydration protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Pediatric patient must be > 4 years to receive Ondansetron.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Overdose or Toxic Exposure

EMR

- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Determine what was taken, when, and how much.
- Contact Poison Control Center at 1-800-222-1222.
- Check blood glucose level. Treat per Diabetic protocol if indicated.
- Acquire 4-lead. (no interpretation)
- Suspected Opiate Overdose -
 - ☞ If symptomatic (unresponsive with respirations less than 10/min or signs of inadequate perfusion, pinpoint pupils): Naloxone (Narcan) 1mg IM repeat in 5 minutes as necessary to a maximum of 2 mg.
 - ☞ Cardiac Arrest Dose: 1mg IM repeat in 5 minutes as necessary to a maximum of 2 mg.

 **EMR STOP**

EMT

- ☞ Activated Charcoal 1-2 gms/kg PO (Maximum dose 50 gms).

 **EMT STOP**

AEMT

- Acquire 12-lead if indicated.
- IV/IO access.
- Normal Saline @ TKO.
- ☞ Symptomatic opiate overdose: Naloxone (Narcan) 0.1 mg/kg IV/IO/IN every 5 minutes as necessary; not to exceed 6 mg.

 **AEMT STOP**

PARAMEDIC

- ☞ Organophosphate poisoning: Atropine 0.05 mg/kg IV/IO every 3 – 5 minutes until secretions dry.
- ☞ Dystonic reaction: Diphenhydramine (Benadryl) 1 mg/kg IV/IO/IM.
- ☞ TCA overdose: Sodium Bicarbonate 1 mEq/kg IV/IO over 2 minutes.
- ☞ Beta blocker OD: Glucagon
 - 0.5 mg IM for Broselow colors gray-white
 - 1 mg IM for Broselow colors blue-green.
- ☞ Calcium channel blocker OD: Glucagon
 - 0.5 mg IM for Broselow colors gray-white
 - 1 mg IM for Broselow colors blue-green.

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Repeat Activated Charcoal.
-  Additional antidotes as needed.



Key Points/Considerations

- Dystonic reaction is uncontrolled contractions of face, neck, or tongue.
- Monitor patients with TCA overdose closely for rapid changes and deterioration in changes in their condition.
- Obtain and give to ER staff all meds/containers found.
- Obtain what meds were taken (pill bottles), number of pills taken, time of ingestion and if vomiting occurred.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- See following page for further information.
- S/S of Phenothiazine overdose:
 - Tardive Dyskinesia (involuntary movements of the facial muscles and tongue) and acute dystonic reactions.
- S/S of TCA overdose:
 - Widened QRS >0.10
 - Tachycardia >120 bpm
 - Hypotension
- S/S of Organophosphate poisoning:
 - Muscle fasciculation
 - Diarrhea
 - Wheezing
 - Abdominal cramping
 - Salivation
 - Seizures
 - Altered mental status
 - Pinpoint pupils
 - DUMBELS: Diarrhea, urination, miosis, bradycardia, bronchospasm, bronchorrhea, emesis, lacrimation, salivation, sweating, secretions, seizures.
 - SLUDGE: Salivation, lacrimation, urination, diarrhea, GI cramps, emesis.
- S/S of atropinization:
 - Papillary dilation
 - Tachycardia
 - Flushing
 - Diminished diaphoresis
 - Drying of secretions
- S/S of Opiate overdose:
 - Respiratory <10
 - Pinpoint pupils
 - Loss of consciousness

Calcium Blockers	TCA	Beta Blockers	Organophosphates
Verapamil	Amitriptyline	Inderal	Disulfoton
Cardiazem	Amoxapine	Lopressor	Tetraethyl Pyrophosphate
Nifedipine	Clomipramine	Tenormin	Parathion
Nicardipine	Desipramine	Corgard	Phosdrin
Nimodipine	Dothiepin	Blocadren	Phorate
Felodipine	Imipramine	Cartrol	Doumaphos
Amlodipine	Lofepamine	Visken	Chlorpytigos
Bepidil	Maprotiline	Sectral	Trichlorfon
Flunarizine	Nortriptyline	Normodyne	Ronnel
Isradipine	Protriptyline	Betapace	Diazinon
Nitrendipine	Trimipramine	Zebeta	Malathion

List is not all inclusive.

Pediatric: Pain Management

EMR

- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Monitor SPO₂.
- Acquire 4-lead (no interpretation)

 **EMR STOP**

EMT

- 🔧 Acetaminophen Oral Solution 15mg/kg PO or: (**ONLY FOR PTS >60 DAYS OLD**)

Weight (lbs)	Weight (kg)	mL of Acetaminophen
<6	<2.7	Not indicated
6 - 11	2.7 - 5.0	1.25
12 - 17	5.1 - 7.7	2.5
18 - 24	7.8 - 10.9	3.75
25 - 49	11.0 - 22.3	5
≥50	≥22.3	10

 **EMT STOP**

AEMT

- Acquire 12-lead.
- IV/IO access.
- 🔧 Normal Saline @ TKO.
- 🔧 Ondansetron (Zofran) 1 mg IV/IO/IM for nausea (> 4 years). Repeat one time in 5 minutes.
- 🔧 Morphine 0.1 mg/kg IV/IO/IM/IN up to 5mg; repeat as necessary one time in 5 – 10 minutes; maximum 10 mg
- 🔧 Fentanyl 0.5-1 mcg/kg IV/IO/IN/IM w/ max initial dose 100mcg; Repeat in 5 min PRN at ½ initial dose, up to total dose of 3mcg/kg
- 🔧 Dilaudid 0.01 – 0.02 mg/kg IV/IO/IM/IN; repeat as necessary once up to a max of 1 mg.

 **AEMT STOP**

PARAMEDIC

- 🔧 Toradol: <2yrs – 0.25mg/kg IV; >2yrs – 0.50mg/kg IV; Max dose 30mg
 - May be used alone or in conjunction with narcotic

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS

-  Additional pain medication doses

Key Points/Considerations

- For patients with:
 - Severe burns without hemodynamic compromise.
 - Isolated extremity injuries such as fractures or dislocations with severe pain.
 - Suspected hip fractures should be treated as extremity injuries.
 - Shoulder injuries should be treated as extremity injuries.
- Contraindications to standing order pain management: altered mental status secondary to a head injury, hypoventilation, hypoperfusion.
- **Pediatrics must be > 6 months to administer Dilaudid.**

• Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
• SPO2 and cardiac monitoring required with any narcotic administration.
• Acetaminophen: Ensure pt has not had total dose of >15mg/kg within the last 4 hours (e.g. Tylenol, acetaminophen-containing medications such as cold medicines or narcotics)

Pediatric: Respiratory Emergencies

EMR

- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Monitor SPO₂.
- Determine if patient has been given their own asthma medications.
- Acquire 4-lead (no interpretation)

 **EMR STOP**

EMT

- Albuterol 2.5 mg in 3 ml (unit dose) + Ipratropium 0.5 mg in 2.5 ml (unit dose) mixed together, via nebulizer; repeat as necessary.

 **EMT STOP**

AEMT

- Cardiac Monitor if indicated.
- IV/IO access.
- Normal Saline TKO.

 **AEMT STOP**

PARAMEDIC

- Methylprednisolone (Solu-Medrol) 1 mg/kg IV/IO.
- Epinephrine 1:1000 dose 0.01 mg/kg IM max 0.3 mg. May repeat x 1.
- Magnesium Sulfate 25 - 50 mg/kg IV IO diluted to a 10% solution over 1 – 2 minutes
 - **MED MATH: Using 10mL syringe, draw up 1gm/2mL Mag and then add 8mL NS for 100mg/mL and then use 3-way stopcock to transfer desired dose from this syringe; Example: 5kg pt, draw/give 125mg = 1.25mL; 20kg pt, draw/give 5mL = 500mg**
- Severe Croup Treatment:
 - < 4 years: 2.5 mg (2.5mL) Epinephrine 1:1000 diluted to 3 ml with normal saline.
 - > 4 years: 4 mg (4mL) Epinephrine 1:1000 nebulized.

 **PARAMEDIC STOP**

PHYSICIAN OPTIONS



Key Points/Considerations

- Position patient sitting or in position of comfort.
- Absence of breath sounds can be indicative of status asthmaticus. Be prepared for imminent respiratory arrest.
- Presence of wheezing is not only an indicator of Asthma/COPD. Consider pulmonary edema, pneumonia, upper airway obstruction, anaphylaxis, and acute allergic bronchospasm.
- Epinephrine should be reserved for those who are unable to generate adequate tidal volume to deliver aerosolized drug to the bronchial tree.
- Draft nebulizer with BVM
- Humidified O₂ for croup/epiglottitis
- Invasive airway maneuvers with upper airway disorders (croup, foreign body, epiglottitis) should only be attempted when in respiratory arrest.
- Monitor heart rate closely. When administering Albuterol contact a physician if heart rate increases to >180 bpm.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Seizures

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose level refer to Pediatric Diabetic protocol.
- Cardiac monitor (4-lead only) may be applied and printed (no interpretation)

EMR STOP

EMT

-  **FEVER >100.4°F/38°C:** Acetaminophen Oral Solution 15mg/kg PO or: (**ONLY FOR PTS >60 DAYS OLD**)

Weight (lbs)	Weight (kg)	mL of Acetaminophen
<6	<2.7	Not indicated
6 - 11	2.7 - 5.0	1.25
12 - 17	5.1 - 7.7	2.5
18 - 24	7.8 - 10.9	3.75
25 - 49	11.0 - 22.3	5
≥50	≥22.3	10

EMT STOP

AEMT

PARAMEDIC

- IV/IO access.
-  Normal Saline @ TKO.
- Administer one of the following within 3 minutes of seizure activity:
 -  Lorazepam 0.1 mg/kg IVSP/IO/IN/IM. Repeat x 1 to a maximum of 4 mg.
 - IV/IO do not exceed 2 mg/min; Must be diluted w/ equal amount of normal saline
 - IM do not dilute, give injection deep into muscle mass.
 -  Midazolam IV/IO 0.05mg/kg infused slowly over 1-2 minutes **OR** IM/IN 0.1mg/kg; Max single dose 5mg.
Repeat as necessary up to a maximum of 10 mg.

AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

-  Additional Midazolam.
-  Additional Lorazepam.

Key Points/Considerations

- Do not delay medication administration. Consider IN or IM prior to IV access.
- Protect the patient and EMS crew from injury during the seizure.
- Paramedic may assist the patient's family or caregivers with administration of rectal Valium (Diastat) if available.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- Fever in infants <60 days is NOT to be treated w/ Acetaminophen by EMS
- Acetaminophen: Ensure pt has not had total dose of >15mg/kg within the last 4 hours (e.g. Tylenol, acetaminophen-containing medications such as cold medicines or narcotics)

Pediatric Medical: Sepsis

EMR

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose level if equipped.
- Acquire 4-lead (No interpretation).
- Obtain temperature



EMR STOP

EMT

- Acquire 12-lead (No interpretation).
- Apply/monitor EtCO₂ capnography.
- **FEVER >100.4°F/38°C:** Acetaminophen oral solution 15mg/kg PO or: (**ONLY FOR PTS >60 DAYS OLD**)

Weight (lbs)	Weight (kg)	mL of Acetaminophen
<6	<2.7	Not indicated
6 - 11	2.7 - 5.0	1.25
12 - 17	5.1 - 7.7	2.5
18 - 24	7.8 - 10.9	3.75
25 - 49	11.0 - 22.3	5
≥50	≥22.3	10



EMT STOP

AEMT

PARAMEDIC

- Acquire/Interpret 12-lead ECG.
- Large-bore IV access or IO if unable to establish IV access w/in 90 seconds.
- Normal Saline infusion at rate of 30mL/kg/hr in boluses using pressure bag.
 - May repeat 3x up to 60mL/kg, ensuring no pulmonary edema.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Focus on rapid intervention, IV hydration, and early notification of concern for potential sepsis pt.
- Concern for any new or worsening infection can include: reported fever, shaking chills, sweatiness, new cough, difficulty or less than usual urination, unexplained or new AMS, flushed skin, pallor, new rash or mottling.
- Vital signs should be obtained every 5 minutes.
- Fever in infants <60 days is NOT to be treated w/ Acetaminophen by EMS
- Acetaminophen: Ensure pt has not had total dose of >15mg/kg within the last 4 hours (e.g. Tylenol, acetaminophen-containing medications such as cold medicines or narcotics)

Pediatric: Stroke

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Determine at what point signs and symptoms appeared.
- If onset of symptoms is less than 5 hours, notify appropriate stroke center and transport immediately.
- If onset is within 24 hours patient should be taken to a stroke center as a non-stroke alert patient.
- Check blood glucose. If out of normal limits treat per Diabetic Protocol.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Secure airway as needed.
- Acquire 12-lead.
- IV/IO access.
- Normal Saline @ TKO
- If tachycardic and blood pressure is < 90 systolic, 10 ml/kg normal saline IV/IO, to maintain blood pressure above 90.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS

Key Points/Considerations

- Contact hospital as soon as possible to give Stroke Alert.
- Limit scene time.
- Repeat vitals every 5 minutes if patient meets stroke criteria.
- All strokes that are within the 6 hour or “wake-up stroke” time limit are Code Red. Transport regular traffic unless hemodynamically unstable.
- Cincinnati Stroke Scale
 - Speech – repeat: “You can’t teach an old dog new tricks”. Correct words and no slurring.
 - Facial Droop – Have patient smile.
 - Arm Drift – Have patient close their eyes and have them hold their arms out straight for 10 seconds without unequal movement or drifting.

Environmental

Pediatric: Carbon Monoxide Poisoning

EMR

- Remove patient from the source.
- CAB and vital signs.
- Airway management and high flow oxygen at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Consider ventilatory assistance for severe dyspnea or hypoventilation.
- Acquire 4-lead (no interpretation)



EMR STOP

EMT

- Obtain carbon monoxide reading.



EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO
- Treat seizures and cardiac dysrhythmias per protocols.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Pulse oximeter reading will be inaccurate.
- All CO poisoning patients will be transported to Wesley Medical Center for hyperbaric oxygen treatment.
- S/S: dull headache, weakness, dizziness, N/V, shortness of breath, confusion, blurred vision, loss of consciousness
- **All patients with suspected CO poisoning are required to have high flow oxygen initiated and maintained.**
- ***100 percent oxygen reduces the elimination half-life of CO to 74 minutes, i.e. a SPCO 10% in theory would decrease to 5% after 74 minutes of oxygen @ 15 lpm via NRB. **In pregnant patients the CO levels of the fetus is double of the mother's**

Level	Treatment
0-5%	Normal in Non-smokers
5-10%	Normal for smokers Non-smokers, assess for s/s • Treat w/ high flow O2 for 20 min * • If s/s still present, consider transport
10-15%	(In any patient) assess for s/s • Treat w/ high flow O2 • Transport to WMC
>10%	Pregnant patients** • Treat w/ high flow O2, immediate transport to WMC
>15%	(In any patient) • Treat w/ high flow O2, immediate transport to WMC

Pediatric: Drowning

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Remove wet clothing and cover with dry blankets.
- Spinal motion restriction if indicated preferably before removal from water.
- Cold water – treat per hypothermia protocol.
- Cardiac arrest – treat per indicated Cardiac, Seizure, and Trauma protocols.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor if indicated.
- IV/IO access Normal Saline @ TKO.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- No longer requires a trauma alert unless there is associated traumatic MOI involved (e.g. boating accident, diving, significant fall, etc.)
- Resuscitation should be considered if submersion has been less than 2 hours, especially if water temperature below 70 degrees.
- Refer to Broselow Pediatric Emergency Tape.
- Obtain the following history:
 - Submersion time.
 - Water temperature and depth.
 - Clean or dirty water and any known pollutants.
 - Mechanism of injury.
 - Type of clothing and flotation device.
 - Location last seen.

Pediatric: Hyperthermia

EMR

EMT

- Remove from heat source.
- CAB and vital signs.
- Obtain temperature.
 - > 103 degrees cool patient with ice packs to the groin, neck, and axilla. Cool water and fanning of the patient is also effective. Remove when body temperature is 100 degrees.
- Remove clothing.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO or
 - 20 ml/kg to maintain BP > 90, with hypovolemia and hemodynamic compromise. Assess breath sounds for tolerance.
- Treat seizures per Seizure protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Use caution with cooling - this can cause shivering or hypothermia.
- Vigorous fluid administration may produce pulmonary edema.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Hypothermia

EMR

EMT

- CAB and vital signs. (Assess for 30-45 seconds)
- CPR if indicated.
- AED if indicated – one attempt only.
- If not in cardiac arrest remove patient from environment avoiding any rough handling. Remove any wet clothing, cover with blankets, and place the heater on high. Patient must be kept warm.
- Obtain temperature.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Check blood glucose level.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO.
 - Warm IV fluids at 20 ml/kg.
- If temperature is below 86 degrees F, CPR and one defibrillation @ 2 j/kg if indicated until temp is above 86 degrees.
- Follow Cardiac Arrest protocol if patient is pulseless, apneic, and temperature is above 86 degrees F, but lengthen intervals of IV medications until temperature is above 93 degrees F.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Prevent further heat loss in patients < 86 degrees; temperatures > 93 degrees may be re-warmed.
- Intubation should be avoided in the severely hypothermic patient with spontaneous respirations unless urgently needed using extreme caution.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric Trauma

Pediatric: Burns

EMR

EMT

- Stop the burning process.
 - Remove necessary clothing leaving all stuck clothing.
 - Remove rings, bracelets, and constricting bands.
 - Flush briefly with cool water if still burning. (chemical burns only)
- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Maintain body temperature.
- Chemical burns:
 - Irrigate with copious amounts of water or saline for 20 minutes. **EXCEPTION:** Phosphorus and Lime Powder.
 - Brush off dry powder chemicals before flushing with water.
- Decontaminate all patients exposed to hazardous materials before entering the ambulance.
- Elevate extremity in position of function to minimize edema monitoring neuro/vascular status.
- Treat for shock and hypothermia.
- Acquire 4-lead (no interpretation)

 **EMR AND EMT STOP**

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access.
 - ☞ Normal Saline:
 - 5 year old and younger – 125 ml/hour
 - 6-13 year old – 250 ml/hour
 - 14+ year old – 500 ml/hour
 - ☞ Morphine 0.1 mg/kg IV/IO/IM/IN up to 5 mg. Repeat as necessary every 3-5 minutes to a maximum of 20 mg.
 - ☞ Fentanyl 0.5-1mcg/kg IV/IO/IM/IN with max initial dose 100mcg; Repeat in 5min PRN at ½ initial dose, up to total dose of 3mcg/kg
 - ☞ Dilaudid 0.01-0.02 mg/kg IV/IO; Max single dose 0.25mg; Max total dose 4mg

 **AEMT AND PARAMEDIC STOP**

PHYSICIAN OPTIONS

Key Points/Considerations

- Begin transport as soon as possible to Via Christi on St. Francis.
- Consider burn alert criteria.
- Use Rule of Nine's when determining total body surface.
- Remove patient from exposure without compromising crew safety.
- Infuse warm IV fluids if available.
- Electrical Burns: consider spinal immobilization and assess for extremity fractures.
- Electrical, Flame, or scald burns should be covered with dry sterile sheets and dressings.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric Trauma: Chest / Abdominal Injuries

EMR

EMT

- CAB and vital signs.
 - Record vital signs every 5 minutes on critical patients.
- Stabilize life-threatening wounds (Control hemorrhage, cover sucking chest wounds, etc.)
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if indicated.
- Moist sterile dressing to abdominal evisceration.
- Penetrating objects should be stabilized – DO NOT REMOVE THEM.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO or:
 - ☞ 20 ml/kg bolus IV/IO maintaining a systolic B/P > 90.
- Treat pain per pain management protocol.



AEMT STOP

PARAMEDIC

- Chest injury with tension pneumothorax.
 - Chest decompression with available pneumothorax kit.
 - Needle decompression with 14 ga IV catheter if pneumothorax kit is unavailable.
 - Age 0-1years: 22ga IV catheter
 - Age 1-4years: 20ga IV catheter
 - Code Blue Trauma – (bilateral) prior to termination of arrest.
- Consider sedation in conscious patient prior to decompression.
 - ☞ Midazolam IV/IO 0.05mg/kg infused slowly over 1-2 minutes **OR** IM/IN 0.1mg/kg; Max single dose 5mg. Repeat as necessary up to a maximum of 10 mg.
- Crush injury for an extended time > 45 minutes.
 - ☞ Consider Sodium Bicarbonate 1 mEq/kg. IV/IO.



PARAMEDIC STOP

PHYSICIAN OPTIONS

- Relief of non-tension pneumothorax.
- Additional Normal Saline bolus.

Key Points/Considerations

- Begin transport as soon as possible.
- Target scene time is <10 minutes if trauma alert.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.
- Criteria for chest decompression.
 - Absent or decreased breath sounds
 - Tracheal deviation
 - Hypotension
 - Severe respiratory distress
 - Jugular venous distension
 - Difficult BVM ventilations after ET tube confirmation. O₂ saturation remains >93% on 100% oxygen.

Pediatric Trauma: Head Injuries

EMR

EMT

- CAB and vital signs.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Spinal motion restriction if indicated.
- Penetrating objects should be stabilized – DO NOT REMOVE THEM.
- Consider soft restraints for combative patients.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

- Cardiac monitor if indicated.
- IV/IO access.
- Normal Saline @ TKO:
 - 20 ml/kg bolus maintaining a systolic B/P of 90.



AEMT STOP

PARAMEDIC

- Combative head injury:
 - Lorazepam 0.1 mg/kg IV/IO/IM/IN up to 2 mg repeat X 1 in 5 minutes. IV/IO dilute in equal amount normal saline.
 - IV/IO do not exceed 2mg/min
 - Must be diluted in equal amount of normal saline to give IV/IO; IM do not dilute, give deep into muscle mass
 - Midazolam IV/IO 0.05mg/kg infused slowly over 1-2 minutes **OR** IM/IN 0.1mg/kg; Max single dose 5mg. Repeat as necessary up to a maximum of 10 mg.



PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Scene time target is 10 minutes.
- Record vital signs every 5 minutes if patient is critical.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Isolated Trauma

EMR

EMT

- CAB and vital signs.
- Stabilize fractures and stop bleeding.
- Spinal motion restriction if indicated.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor if indicated.
- IV/IO access Normal Saline @ TKO.
- See pain management protocol for pain management.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- All fractures should be stabilized.
- Ensure that pulses, motor, and sensations are checked in extremities before, after, and whenever the patient is moved.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Multi-Systems Trauma

EMR

EMT

- Correct any life threats with correct treatment procedures.
- CAB and vital signs taken every five minutes.
- Spinal motion restriction if indicated.
- Airway management and oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Acquire 4-lead (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac monitor.
- 12-lead for suspected cardiac contusions.
- IV/IO access.
- Normal saline @ TKO or:
 - 20 ml/kg bolus maintaining a systolic B/P of 90.
- Pain control: see pain management protocol.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Additional normal saline bolus.

Key Points/Considerations

- Contact Medical Control as soon as possible.
- Transport immediately to Wesley Medical Center.
- Tourniquets are only used for life threatening peripheral hemorrhages when all other means of hemorrhage control have been unsuccessful.
- Tourniquets should only be removed by a physician in an appropriate hospital setting, preferably a level 1 trauma center.

Pediatric: Shock/Hypoperfusion

EMR

EMT

- CAB and vital signs.
- Airway management and appropriate oxygen therapy at 2-6 LPM via nasal cannula or when appropriate 12-15 LPM via non-rebreather.
- Cardiac monitor (4-lead only) may be applied and printed (no interpretation)



EMR AND EMT STOP

AEMT

PARAMEDIC

- Cardiac Monitor if indicated.
- IV/IO access.
- Normal Saline:
 - 20 ml/kg bolus maintaining a systolic B/P > 90.



AEMT AND PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Contact a physician if you suspect cardiogenic shock.
- Diagnostic criteria for hypoperfusion includes: capillary refill time > 2 seconds, cool, clammy or mottled skin, inability to recognize parents, restlessness, listlessness, tachycardia, tachypnea, systolic BP less than 70 mmHg (2 years and older) or systolic BP less than 60 mmHg (less than 2 years old).
- Contact receiving hospital early.
- Heart rate is more important than blood pressure if less than 3 years old.
- Refer to Broselow Pediatric Emergency Tape/Pediatric Reference Cards.

Pediatric: Spinal Motion Restriction

EMR

EMT

AEMT

PARAMEDIC

- Perform an appropriate trauma assessment.
- Determine mechanism:
 - **Low Risk:** (Requires **all** of the following)
 - Patient reliable and competent?
 - Normal level of alertness (GCS 15)
 - No evidence of clinical intoxication.
 - No distracting injury.
 - Able to communicate adequately.
 - Lack of point tenderness on palpation of entire length of spinous processes.
 - Lack of focal deficit (motor or sensory) on neuro exam.
- **No** to any of the above Spinal Motion Restriction is required.
 - **High Risk:** (Requires at least one of the following)
 - Ejection from a vehicle
 - Motorcycle crash >20 mph
 - Auto vs. pedestrian or bike at > 20 mph
 - Axial load to head
 - Fall from 3x patient's height
- **Yes** to any of the above Spinal Motion Restriction is required.
- Patient documentation must specifically include all of the above findings.



EMR, EMT, AEMT, and PARAMEDIC STOP

PHYSICIAN OPTIONS



Key Points/Considerations

- Transport as soon as possible.
- Penetrating trauma patients without neuro deficits do NOT require spinal motion restriction.
- Long backboard, pediatric board, or KED may still be used in patient extrication even if spinal motion restriction is not required by protocol but **should be removed as soon as possible.**